**DOMAIN 1 ADVANCED APPLICATION OF SAFETY PRINCIPLES**

821. Maintenance grounds-keepers often use self-propelled machines to assist them with their work. The most protective type of guard used for this equipment is the:

1) Self-adjusting
2) Adjustable
3) Interlocked
4) Fixed

**Ans.3** Self-propelled lawn mowers and snow blowers are interlocked with the source of power so that the machine cannot be operated without all guards in place. The interlocking guard prevents operation of the control that sets the machine in motion until the guard is moved into position, so that the operator cannot reach the point of operation or the point of danger.

822. The most common guard for an automatic guillotine paper cutter is:

1) Barrier
2) Remote controls
3) Sensing
4) Mechanical

**Ans.2** Remote controls are the most common guards on these devices.

823. Some common causes of eye injury involve exposure to:

1) High energy arcs, compressed air, dilute boric acid
2) High energy arcs, methanol, compressed air
3) Compressed air, metal filings, de-ionized water
4) De-ionized water, dilute boric acid, compressed air

**Ans.2** High energy arcs, methanol, and compressed air have all been causes of eye injuries. Dilute boric acid is used to treat eye inflammation; de-ionized water is ultrapure water.

824. Acetylene can be described as a:

1) Colorless gas with a slight garlic-like odor
2) Pungent, irritating greenish-yellow gas
3) Faint yellow liquid
4) White powder with a chlorine-like odor

**Ans.1** Acetylene is a colorless gas with a slight garlic-like odor.

825. Which of the following is NOT an example of protective footwear?

1) Steel-toed boots
2) Fitted orthotics
3) Molders (foundry) shoes
4) Conductive footwear

**Ans.2** Fitted orthotics is not a type of safety protective footwear.

Safety shoes have impact-resistant toes and heat-resistant soles that protect the feet against hot work surfaces common in roofing, paving and hot metal industries. The metal insoles of some safety shoes protect against puncture wounds. Safety shoes may also be designed to be electrically conductive to prevent the buildup of static electricity in areas with the potential for explosive atmospheres or nonconductive to protect workers from workplace electrical hazards.

Electrically conductive shoes provide protection against the buildup of static electricity. Employees working in explosive and hazardous locations such as explosives manufacturing facilities or grain elevators must wear conductive shoes to reduce the risk of static electricity buildup on the body that could produce a spark and cause an explosion or fire. Foot powder should not be used in conjunction with protective conductive footwear because it provides insulation, reducing the conductive ability of the shoes. Silk, wool and nylon socks can produce static electricity and should not be worn with conductive footwear. Conductive shoes must be removed when the task requiring their use is completed. Note: Employees exposed to electrical hazards must never wear conductive shoes.

Electrical hazard, safety-toe shoes are nonconductive and will prevent the wearers' feet from completing an electrical circuit to the ground. These shoes can protect against open circuits of up to 600 volts in dry conditions and should be used in conjunction with other insulating

equipment and additional precautions to reduce the risk of a worker becoming a path for hazardous electrical energy. The insulating protection of electrical hazard, safety-toe shoes may be compromised if the shoes become wet, the soles are worn through, metal particles become embedded in the sole or heel, or workers touch conductive, grounded items. Note: Nonconductive footwear must not be used in explosive or hazardous locations.

Foundry Shoes, in addition to insulating the feet from the extreme heat of molten metal, foundry shoes keep hot metal from lodging in shoe eyelets, tongues or other shoe parts. These snug-fitting leather or leather-substitute shoes have leather or rubber soles and rubber heels. All foundry shoes must have built-in safety toes.

826. The minimum concentration of gasoline in air that can be ignited is 7.6%. This minimum concentration is called the:

1) Lower explosive limit
2) Upper flammability limit
3) Lower ignitable limit
4) Both 1 and 3

**Ans.1**The minimum concentration of a material in air that can be ignited is commonly called the LEL, or lower explosive limit. The maximum concentration of a material in air that can be ignited is called the upper explosive limit or UEL. The terms lower explosive limit and lower flammability limit (LFL) are used interchangeably as are upper explosive limit and upper flammability limit (UFL).

827. A rapid reaction during which heat is transferred progressively from a reacting material to another nearby material whose temperature is then raised to the point at which it also reacts is called:

1) Reaction terminal
2) Deflagration
3) Detonation
4) Explosion

**Ans.2** Deflagration is a rapid reaction during which heat is transferred progressively from a reacting material to another nearby whose temperature is then raised to a point at which it also reacts.

828. The tendency for flame to spread from the source of ignition through a flammable mixture is known as:

1) Flame spread
2) Flammability of the mixture
3) Propagation of flame
4) Rate of diffusion

**Ans.3** Propagation of flame is the spread of the flame away from the source of ignition.

829. One of the most common causes of fire in building renovation is from:

1) Electrical fires
2) Temporary heating equipment
3) Welding and cutting operations
4) Trash burning

**Ans.3** Welding and cutting operations are one of the leading sources of fires during remodeling and renovation work.

830. The most common source of ignition for fires in industrial operations is:

1) Smoking
2) Burner flames
3) Friction
4) Electrical sources

**Ans.4** Electrical sources of ignition account for most industrial fires.

831. Compared to local exhaust ventilation, dilution ventilation:

1) Is more suitable for highly toxic substances
2) Is not adequate for ventilating hazardous point source emissions
3) Must be used as make-up air
4) Uses less air than local exhaust ventilation

**Ans.2** General dilution ventilation is not appropriate for point source emissions or for use with highly toxic materials. In these situations local exhaust ventilation should be used. Because of these limitations, short circuiting of exhaust is less critical than in local exhaust ventilation. The air from general ventilation may or may not be used for local exhaust ventilation make-up air, depending upon the type of local exhaust ventilation device.

832. Dilution ventilation is:

1) Used to control a contaminant at its source
2) Used for controlling vapors with low toxicity
3) More economical than local exhaust ventilation
4) Is the first consideration for hazard control

**Ans.2** Dilution ventilation lowers the concentration of a contaminant by adding air to the general work area. Since the air is added to the general work area it will not effectively control exposure to a toxic substance used in a specific location (point source).

833. You are a safety professional in a large corporation with a high-rise corporate office building. The building is occupied totally by employees of your company except for a few service contractors and first floor space leased by store tenants. Life safety is one of your principle concerns. Strategies for evacuation that you may consider include all the following EXCEPT:

1) Relocating occupants
2) Constructing a vertical exit
3) Constructing a horizontal exit
4) Staging, or phasing, the evacuation

**Ans.2** Relocation means to move people away from the fire floor, generally two floors up or down from the fire. A horizontal exit on the same floor is possible when fire walls, ceilings, and floors exist that allow people to evacuate through a fully protected enclosure before arriving at a stairwell. Staged evacuation can take several forms but essentially means to evacuate people in stages and to perhaps use a series of safe areas in the staging. There is no such term as a vertical exit.

834. For a chemical to be required to be listed on an MSDS by the manufacturer, the chemical must meet all of the following criteria EXCEPT:

1) Be present in a concentration greater than 1%
2) Be listed as a hazardous material on selected lists
3) If a carcinogen, be present in a concentration greater than 0.1%
4) Have an LEL greater than 10%

**Ans.4** There are a variety of criteria for determining if a material must be listed on the ingredient list of an MSDS. It must be listed on one of the identified lists as a hazardous material and must be present in some minimum quantity. The LEL is not one of the criteria.

835. In conducting a safety and health audit at a plant owned by your company, you discover that an extremely hazardous solvent is being used which has a high probability of injury. What is your first course of action?
1) Stop the operation
2) Recommend that the solvent be substituted
3) Take air samples for analysis
4) Initiate actions to construct engineering controls

**Ans.3** The first course of action is to take air samples for analysis. Then, you can determine whether further action is warranted.

With no indication of negative impacts occurring, the most appropriate action is to conduct further assessment of the solvent use to verify that no imminent hazard exists. If an imminent hazard is determined to exist, immediate action is required, such as stopping the operations and requiring appropriate PPE. If no imminent hazard exits, then a longer term action can be considered, such as material substitution or engineering controls.

836. In fire safe designs, the acceptable levels of risk and the focus of fire safety analysis and design process is concentrated in three areas. The three areas are in the list below. Which is NOT one of the three principle areas of study, analysis, and design?

1) Life safety
2) Contents protection
3) Property protection
4) Continuity of building operations

**Ans.2** All but contents protection are major areas of concern and study in fire safety design.

837. All of the following practices can affect the success of fire prevention EXCEPT:

1) Housekeeping
2) Security
3) Education of occupants
4) Exposure

**Ans.4** Exposure is a factor affecting bringing heat and ignitable material together.

838. If the concentration of flammable gases in a work environment is high enough to be a fire hazard, which of the following hazards also probably exists?

1) A product liability exposure

2) An environmental hazard
3) A health hazard
4) A chemical contamination hazard

**Ans.3** Generally, if a fire or explosion hazard is created by a concentration of gas, that concentration exceeds the safe levels of human exposure as well.

839. Anticipating and controlling hazards are best accomplished:

1) Twenty-four hours following a workplace inspection
2) Any time before a pilot process is closed to production
3) During all design stages of an activitt
4) After an accident has occurred

**Ans.3** Anticipating and controlling hazards at the design stages of an activity is the cornerstone of a safety program. The ultimate goal for any safety program is to design environments, equipment, and job procedures so that employee exposures to injuries will be either eliminated or controlled as completely as possible.

840. Arrange the following hazard control steps into the proper sequence:

A. Guard the hazard
B. Engineer the hazard out if possible
C. Educate personnel

1) B, A, C
2) C, A, B
3) A, B, C
4) C, B, A

**Ans.1** The proper sequence in hazard control is to engineer out, guard, and educate (B, A, C).

841. In general, personal protective equipment tends to _____ worker efficiency and _____ personal injury.

1) Increase, Increase
2) Decrease, Decrease
3) Increase, Decrease
4) Decrease, Increase

**Ans.2** Personal protective equipment will decrease the injuries/illnesses incurred by the workers but their efficiencies will probably also decrease due to sensory impairment and restricted motion.

842. Which of these circumstances would you expect to have the HIGHEST likelihood of producing a severe injury?

1) Routine, planned work
2) Production work
3) Low energy sources
4) Non-routine work

**Ans.4** Most accidents resulting in severe injuries occur in non-routine work, non- production work, high energy work, or construction.

843. Hot working of metals refers to:

1) Power presses
2) Foundries
3) Shears
4) Punches

**Ans.2** The term "hot working" refers to foundry operations where metals are heated to their melting points and are then poured.

844. The most common accident in an office setting is:

1) Falling

2) VDT over-exposure
3) Paper cuts
4) Alcoholism

**Ans.1** Falling is the primary accident type in office settings.

845. Experience has indicated that the most effective way to reduce accidents is:

1) Training employees to be more safety conscious
2) Behavior modification
3) Designing safety features into systems
4) Job rotation

**Ans.3** Hazards can be more effectively reduced or eliminated through system design, whereas hazard reduction through changing human behavior is limited and requires continuous reinforcement.

846. The concentration of flammable vapors in a confined space should be less than _____ upon entry.

1) 0.1% of LEL
2) 1% of LEL
3) 10% of LEL
4) 100% of LEL

**Ans.3** The acceptable entry concentration of flammable vapor should be less than 10% of the Lower Explosive Limit (LEL).

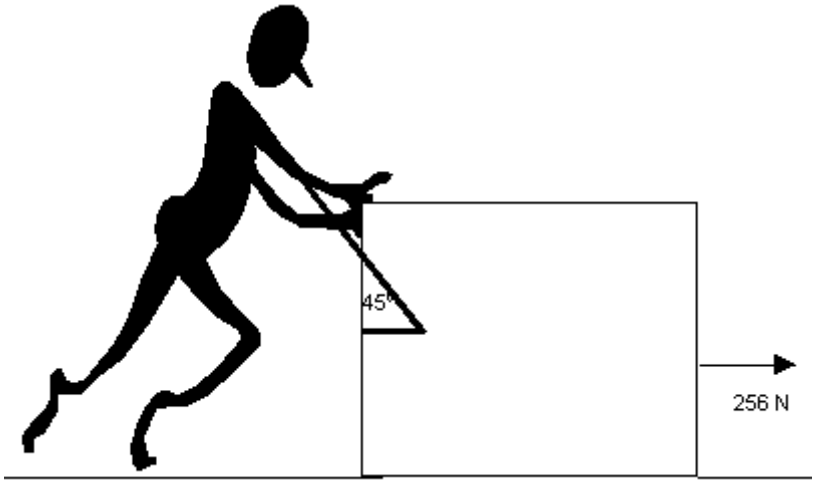
847. Hazardous locations are areas where flammable liquids, gases, vapors, combustible dusts, or other easily ignitable materials exist in sufficient quantities to

produce an explosion or fire. Hazardous locations are classified as Class I, Class II, or Class III. Which one of the following statements is NOT true?

1) Class II locations are those in which combustible liquids may be founClass I locations are those in which flammable vapors and gases may be present
2) Class II locations are those in which combustible liquids may be found
3) Class III locations are those in which there are ignitable fibers or flyings
4) Class I and Class II locations are subdivided into groups of gases, vapors, and dusts having similar properties

**Ans.2**Class II locations are those in which combustible dust may be found.

848.




A worker's muscles act on an angle of 45º with respect to the direction of motion. The horizontal force necessary to start moving the item is 256 N and is measured standing in front of the item to be moved. What is the actual muscle force needed to start moving the item?

1) 181 N
2) 256 N
3) 362 N
4) 551 N

**Ans.3** Using the cosine trigonometric identity and vector mechanics:
Cosine of Angle = Adjacent Component / Hypotenuse Resultant
cos 45º = 256 N / Hypotenuse Resultant
Hypotenuse Resultant = 256 N / cos 45º
= 256 N / 0.7071 = 362 N

849. A 3/4in manila rope has a safety factor of 5 and a tensile strength of 1,000 lb. What is the maximum safe working load?

1) 100 lb
2) 200 lb
3) 750 lb
4) 1,000 lb

**Ans.2** To determine safe working load, divide the tensile strength by the safety factor. 1,000 lb / 5 = 200 lb

850. What device is used to measure traction on walking surfaces?

1) Friction-meter
2) Slip-meter
3) Drag-meter
4) Shoe-sole meter

**Ans.2** A slip-meter is used to make these measurements.

851. What does a "bird cage" in a wire rope usually signify?

1) Repeated bending over sheaves
2) Jumping a sheave
3) Fatigue
4) Sudden release of tension and rebound

**Ans.4** A "bird cage" signifies that the wire rope was subjected to a sudden release of tension.

852. The boom indicator on a mobile crane:

1) Provides the angle of the boom
2) Shows the safe load with the boom at any angle
3) Indicates load capacity
4) Signals when the crane is out of balance

**Ans.1** A boom indicator is a device that shows the angle of the boom. The boom angle is used in conjunction with the crane load chart to determine safe loads. Modern load measuring systems for cranes allow for real-time direct reading of actual crane load and can provide output warnings indicating that the rated capacity is approaching. Such devices can even be integrated with the crane controls to stop the crane operation.

853.

| Rope Diameter (inches) (improved plow steel) | Number of clips | | Minimum Spacing |
|---|---|---|---|
| | Drop Forged | Other material | |
| 1/2 | 3 | 4 | 3 |
| 5/8 | 3 | 4 | 3 3/4 |
| 3/4 | 4 | 5 | 4 1/2 |
| 7/8 | 4 | 5 | 5 1/4 |
| 1 | 5 | 6 | 6 |
| 1 1/8 | 6 | 6 | 6 3/4 |
| 1 1/4 | 6 | 7 | 7 1/2 |
| 1 3/8 | 7 | 7 | 8 1/4 |
| 1 1/2 | 7 | 8 | 9 |

When using u-bolt drop forged steel wire rope clips to make an eye in a 1 in. improved plow steel wire rope, you should use:

1) 4 clips
2) 5 clips
3) 6 clips
4) 7 clips

**Ans.2** The Table is from Table H-20 from 29 CFR 1926.251. The reference requiring the use of this table is from 29 CFR 1926.251 (c)(5).

854. The electromagnetic force (emf) produced by a generator, when operating at constant speed, chiefly depends on:

1) The length of time the generator operates
2) The wire thickness of the armature
3) The wire thickness of the field magnet
4) The strength of the magnetic field

**Ans.4** The electromagnetic force (emf) induced in a wire depends on three things:
a) The length of wire cutting the magnetic field,
b) The strength of the magnetic field being cut, and
c) The speed with which the field is being cut at right angles.

855. Electronic circuits designed to produce high-frequency alternating currents are usually known as:

1) Rectifiers
2) Oscillators
3) Amplifiers
4) Diodes

**Ans.2** Oscillators capable of producing high-frequency alternating current may include crystals capable of producing particular frequencies or they may involve electronic components, like capacitors and inductors, capable of being tuned to varying frequencies.

856. Heat energy is converted directly into electrical energy by which of the following:

1) A photoelectric cell
2) A thermocouple
3) A piezoelectrical crystal
4) A calorimeter

**Ans.2** Thermocouples usually consist of connections between wires of two dissimilar metals. They are frequently calibrated so that the amount of voltage produced can be directly related to the temperature and the capability of measuring it.

857. You discover chemicals stored several years past their expiration date. Which chemical presents the highest hazard?

1) Nitric acid
2) Trichloroethylene
3) Perchloric acid
4) Perchloroethylene

**Ans.3** Perchloric acid forms chlorate salts with age, which are highly explosive upon contact.

858. Electromagnetic energy:

1) Is not a factor in heat stress calculations
2) Is not a factor in heat transfer calculations
3) Is best controlled by forced air ventilation
4) Insignificantly heats the air in the room as it passes through

**Ans.4** Radiant heat is a form of electromagnetic energy. It is a major input in heat stress and heat transfer equations. It is best controlled by shielding and heats the air it passes through only insignificantly.

859. Which of the following is NOT true about pitot tubes?

1) Not a direct reading instrument
2) May be susceptible to plugging
3) Needs periodic calibration
4) Measures velocity and static pressures

**Ans.3** A pitot tube is a metallic, precision made instrument that is made to specific dimensions. The tube itself cannot be calibrated. The calibration occurs with the manometer that it is used with, to determine velocity and static pressures and air volume flowing through a duct. Tiny holes in the side for measuring static pressure could become plugged in a particulate airstream.

860. The aspect ratio of fiber is defined as L/W. The aspect ratio of an exhaust hood is defined as:

1) Ratio of length to width
2) Ratio of length squared to width
3) Ratio of width to length
4) Ratio of width squared to length

**Ans.3** The formula for the aspect ratio of an exhaust hood is AR = W/L where AR is Aspect ratio, W is width, and L is length. This is the reciprocal of the aspect ratio that would be used when discussing the aspect ratio of an asbestos fiber.

861. When assessing ventilation systems, the pitot tube is limited to:

1) Use at diffusers
2) In-duct measurements
3) Use at inlets
4) Use at outlets

**Ans.2** Velometers and thermoanemometers can be used at diffusers and inlets. Pitot tubes are used to make in-duct measurements.

862. Velocity pressure can be measured in a ventilation system using a:

1) Pitot tube
2) Thermoanemometer
3) Smoke tube
4) Piezo-electric anemometer

**Ans.1** Of all devices listed, only the pitot tube can be used to measure velocity pressure in ventilation systems. Smoke tubes can measure direction of flow, thermoanemometers

measure velocity, and Piezo-electric anemometer
is fictitious.

863. What device is commonly used to measure the capture velocity of the fume hood?

1) Velometer
2) Pitot tube
3) Cascade impactor
4) Air pump

**Ans.1** Velometers are commonly used to conduct those types of measurements.

864. A good indicator gas for determination of adequacy of outdoor air exchange in buildings is:

1) Carbon monoxide
2) Oxygen
3) Carbon dioxide
4) Nitrogen

**Ans.3** Carbon dioxide is an indoor air pollutant that is given off by occupants of the building and generally is a good indicator of the degree of indoor air exchanged with outdoor air. Levels of carbon dioxide below 1,000 ppm indicate, in most cases, an adequate supply of outdoor air.

865. For ventilation measurements, a pitot transverse refers to:

1) Systematic equal area velocity measurements
2) A one point method of establishing flow rate
3) Measurements of velocity along the length of duct
4) Random velocity measurements to determine flow rate

**Ans.1** Since the airflow in the cross section of a duct is not uniform, it is necessary to obtain the average by measuring velocity pressure at points in a number of equal areas of a cross

section. The pitot tube transverse is a classic measurement method for obtaining duct velocities.

866. Some of the following six statements VIOLATE hoisting equipment and crane regulations. Indicate which statements violate these regulations.

I. Use of equipment is restricted to qualified employees.
II. Load ratings must be marked on each side of the hoist.
III. Loads may be only traversed over the heads of the actual materials handling crew performing the lift.
IV. Holding and self-setting brakes are optional for loads under 10,000 lbs.
V. Hoist rope or chain must be free from kinks or twists.
VI. Hoisting or traveling is permissible only when a qualified person is on the load or hook.

1) I, III, IV, and VI
2) III, III, IV, and V
3) III, IV, V, and VI
4) III, IV, and VI

**Ans.4** OSHA regulations for cranes, hoists, chains, cables, ropes, and hooks are detailed in 29 CFR 1910.179. Spelled out are that loads must not be carried over the heads of people; hoists must be equipped with a self-setting brake that is applied to the motor shaft or gear train; holding brakes must be applied automatically when the power is off for powered hoists; and hoisting, lowering, or traveling is not permitted while anyone is on the load or hook.

867. The AAOO-AMA "Guide for the Evaluation of Hearing Impairment" has a dB "window" at selected frequencies below which no impairment in hearing is accepted. What is this "window"?

1) 10 dB
2) 15 dB
3) 20 dB
4) 25 dB

**Ans.4** A worker must exhibit at least a 25 dB threshold of hearing at selected frequencies to be considered impaired.

868. The loss of life from buildings equipped with sprinklers has been:

1) Almost negligible
2) Used as an excuse by architects to avoid installing sprinklers in new construction
3) Statistically the same as buildings not equipped with sprinklers
4) Statistically higher than buildings not equipped with sprinklers

**Ans.1** Automatic sprinkler protection is a particularly important factor in safeguarding life from fire. NFPA records show that the loss of life by fire in buildings equipped with automatic sprinklers has been almost negligible. The few deaths that have occurred have been unusual circumstances where the sprinkler system was inappropriately shut off or where the fire was confined to clothing and did not generate enough heat to fuse sprinklers.

869. The 1947 Texas City, Texas disaster killed 200 people and injured over 600 people. The explosion was caused by:

1) Nitrate mixing with water
2) Ammonium nitrate mixing with oil
3) Ionizing radiation
4) Molasses

**Ans.2** One of the worst industrial accidents of all time occurred in the port of Texas City, Texas when ammonium nitrate cargo mixed with oil and caused a series of huge explosions and fires.

870. The catastrophic hotel fire which occurred in Las Vegas, Nevada in 1980 was in the:

1) MGM Grand Hotel
2) Caesar's Palace
3) Bally's Hotel
4) Flamingo Hilton

**Ans.1** The MGM Grand Hotel fire occurred in Las Vegas, Nevada in 1980. This fire has been extensively studied. Building codes were subsequently modified in many municipalities as a result of the MGM fire.

871. The Triangle Shirtwaist fire in New York led to the development of what standard?

1) Sprinkler standards for public occupancies
2) Standards on interior finish
3) Life safety standards
4) Occupant loading standards

**Ans.3** This fire resulted in a major loss of life and was one of the reasons life safety codes were developed. The "Committee on Safety to Life" was formed after this fire.

872. Two of the top three deadliest fires in US history have been associated with:

1) Forest fires
2) Industrial plant fires
3) Aircraft fires
4) Passenger ship fires

**Ans.4** Steamship fires were two of the top three deadliest fires in US history. The SS Sultana fire on the Mississippi River claimed 1,547 lives and the General Slocum fire in New York claimed 1,030 lives.

873. Occupational fatalities due to workplace violence (homicide) is the _____ leading cause of occupational fatalities in the United States.

1) Highest
2) Second highest
3) Fifth highest
4) Eighth highest

**Ans.2** Workplace violence has emerged as an important safety and health issue in today's workplace. Its most extreme form, homicide, is the second leading cause of fatal occupational injuries in the United States. Nearly 1,000 workers are murdered, and 1.5 million are assaulted in the workplace each year. According to the BLS Census of Fatal Occupational Injuries (CFOI), there were 709 workplace homicides in 1998, accounting for 12% of the total 6,026 fatal work injuries in the United States. Environmental conditions associated with workplace assaults have been identified and control strategies implemented in a number of work settings. OSHA has developed guidelines and recommendations to reduce worker exposures to this hazard but is not initiating rulemaking at this time.

874. The primary value of accident and injury/illness records is to:

1) Locate and prevent recurrences of the same accident
2) Meet reporting requirements
3) Provide information to OSHA
4) Union management progress reports

**Ans.1** Records of all accidents, property damage, and personal injury/illnesses are essential to efficient and successful safety programs. Records supply the information necessary for implementing controls to prevent the accident(s) from happening again.

875. In liquid systems, one ppm is equivalent to:

1) Milligram per liter of solution
2) Microgram per liter of solution
3) Milligram per cubic meter
4) Both 1 and 3

**Ans.1** In liquid systems, the expression "parts per million" represents one part by weight of solute per one million parts by weight of solvent.

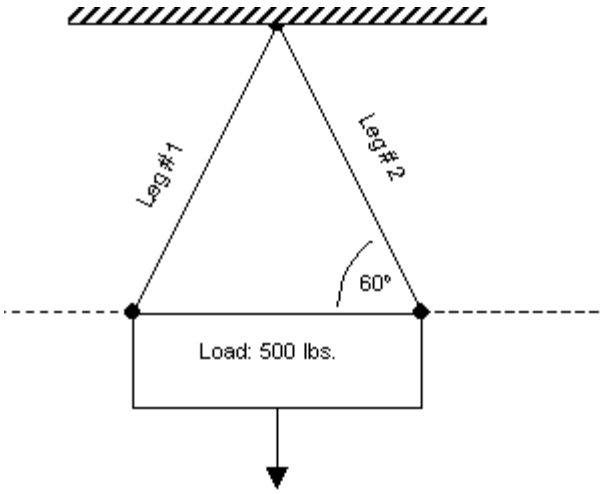
876. Pick the appropriate formula to calculate safety margin:

1) Minimum strength / 2

2) Minimum strength / (minimum strength - maximum strength)
3) (Minimum strength - maximum stress) / minimum strength
4) Minimum strength / 4

**Ans.3** The appropriate formula to calculate safety margin is: K = (minimum strength - maximum stress) / minimum strength

877.




A load of 500 lb is supported by a two-leg sling. If the legs of the sling make an angle of 60º with the load, what force is exerted on each leg of the sling?

1) 144.3 lb
2) 288.7 lb
3) 433.0 lb
4) 500.0 lb

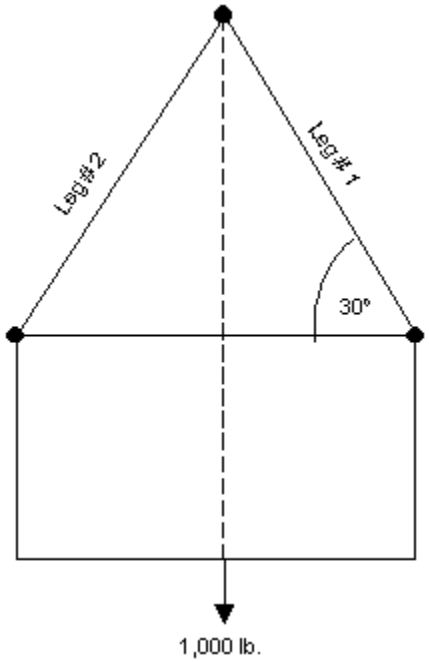
**Ans.2** Use the sine trigonometric identity.Total load is divided by 2 because we have a two-leg sling.sine of angle = opposite leg length / hypotenuse leg length
sin 60º = 250 lb / hypotenuse leg length
0.8660 = 250 lb / hypotenuse leg length
hypotenuse leg length = 250 lb / 0.8660
= 288.68 lb

878. A pulley system consisting of 3 fixed and 2 movable pulleys is being used to raise a 500 lb load to a height of 20 ft. Neglecting friction, how much effort must be exerted?

1) 100 lb
2) 125 lb
3) 250 lb
4) 500 lb

**Ans.1** As there are five ropes holding up the load, the mechanical advantage is 5.
NOTE: The distance is irrelevant to the effort needed.
F (effort) = W(weight) / MA(mechanical advantage)
F = W/MA
F = 500 lb / 5 = 100 lb

879.




A 1,000 lb load is lifted with a two-leg sling whose legs are at a 30º angle with respect to the load. The stress on each leg of the sling is:

1) 500 lb
2) 577 lb
3) 1,000 lb
4) 2,000 lb

**Ans.3** Use the sine trigonometric identity.The vertical component of each sling supports 500 lb (1,000 lb / 2).sine of angle = opposite leg length / hypotenuse leg length
sin 30 º = 500 lb / hypotenuse leg length
0.5000 = 500 lb / hypotenuse leg length
hypotenuse leg length = 1,000 lb

880. Given a two-leg sling (each leg makes a 60º angle with respect to the load) supporting 20,000 lb, what is the tension on the left sling leg?

1) 8,754 lb
2) 9,967 lb
3) 11,547 lb
4) 12,333 lb

**Ans.3** Remember that you must divide the load by 2 because this is a two-leg sling. Your sketch should include a figure with a triangle with 10,000 lb as the side opposite the 60 º angle. The hypotenuse leg (the tension in the sling leg itself) is calculated using the sine trigonometric identity.
sine of angle = opposite leg length / hypotenuse leg length
10,000 lb / sin 60 º = 10,000 lb /0.8660 = 11,547 lb

881. A crane is lowering a 1,500 lb load to the ground. The operator applies the brakes causing the cable to break. What is the maximum tension on the cable if the breaking acceleration is 50 ft/sec$^2$?

1) 2,344 lbs
2) 3,486 lbs
3) 3,844 lbs
4) 6,224 lbs

**Ans.3** F = force
W = weight
G = gravity constant
A = acceleration

Use the formula:
$F = (W/G)A$
$F = (1500 \text{ lbs} / 32 \text{ ft/sec}^2)(50 \text{ ft/sec}^2)$
F = 2344 lbs + 1500 lbs (Load) = 3844 lbs

882. The main source of fire standards/codes used in the United States is promulgated by:

1) ANSI
2) NFPA
3) OSHA
4) UL

**Ans.2** The National Fire Protection Association (NFPA) is the main source of fire standards/codes in the United States.

883. Explosion proof apparatus is intended for what class of locations?

1) I
2) II
3) III
4) IV

**Ans.1** Explosion proof apparatus is intended for Class I locations, while dust- ignition-proof equipment is primarily intended for Class II and III locations. NFPA 70 Article 500-2 (National Electric Code)

884. What frequency range of hearing loss is generally considered compensable under workers' compensation?

1) 500 - 2,000 Hz
2) 10,000 - 20,000 Hz
3) 3,000 - 4,000 Hz
4) 0 - 500 Hz

**Ans.1** The range most important for speech is 500 to 2,000 Hz. Significant losses in this frequency range often result in compensation eligibility.

885. What is the limit of hearing damage, currently suggested by the American Medical Association (AMA), before compensation should be awarded?

1) 15 dB loss at 2,000, 3,000, and 4,000 Hz octave bands
2) 25 dB loss at 2,000, 3,000, and 4,000 Hz octave bands
3) 25 dB loss at 500, 1,000, and 2,000 Hz octave bands
4) 15 dB loss at 500, 1,000, and 2,000 Hz octave bands

**Ans.3** If the average monaural hearing level at 500, 1,000, and 2,000 Hz is 25 dB or less, the AMA guide states that usually no impairment exists in the ability to hear everyday speech under everyday conditions. If there is more than 25 dB in loss, then compensation is suggested.

886. Boilers and unfired pressure vessels are designed, constructed, tested, and installed in accordance with which code:

1) NFPA
2) ANSI
3) ASME
4) ASTM

**Ans.3** Design, construction, testing, and installation of boilers and unfired pressure vessels is covered by applicable sections of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel code. Boilers and pressure vessels are not covered by present OSHA Standards.

887. Hazardous locations are classified as Class I, Class II or Class III, depending on what type of hazardous substance is or may be present. Each of these classes is divided into two hazard categories, Division 1 and Division 2, depending on the likelihood of the presence of a flammable or ignitable concentration of a substance. Which one of the following statements is NOT true?

1) Division 1 locations are designated as such because a flammable gas, vapor, dust, or easily ignitable material is normally present in hazardous quantities
2) Division 2 locations are designated as such because a flammable gas, vapor, dust, or easily ignitable material is normally present in hazardous quantities
3) In Division 2 locations, the existence of hazardous quantities of these materials is not normal, but they may occasionally exist either accidentally or when material in storage is handled
4) In general, the installation requirements for Division 1 locations are more stringent than for Division 2 locations

**Ans.2** Division 1 locations are designated as such because a flammable gas, vapor, dust, or easily ignitable material is normally present in hazardous quantities. NFPA 70 Article 500-3 (National Electric Code)

888. If the mean and median diastolic blood pressure screened on a large population of men are both 83 mm Hg, and the standard deviation of the screening is 12 mm Hg, which is TRUE?

1) The mode is 83 mm Hg
2) About 95% of the men have a diastolic blood pressure between 59 and 107 mm Hg
3) About 80% of the men have a diastolic blood pressure between 71 and 95 mm Hg
4) About 50% of the men have a diastolic blood pressure between 71 and 94 mm Hg

**Ans.2** When the mean and median numbers of a set are the same, one may assume that the population is normally distributed. In a normal distribution, approximately 95% of the results are within the mean plus or minus 2 standard deviations. This means that the diastolic blood pressure of 95% of the population of men falls between the mean minus 24 mm Hg and the mean plus 24 mm Hg (1 standard deviation is 12 mm Hg, 2 is 24 mm Hg).

889. A correlation between two variables measures the degree to which they are:

1) Statistically associated
2) Causally related
3) Statistically significant
4) Cause one another

**Ans.1** Correlation measures the statistical association between two variables. A determination of causation requires consideration of other factors in addition to statistics.

890. What major code(s) are utilized in the design and construction of pressure vessels in the United States?

1) ASME Code
2) API Standard 620
3) Code of Federal Regulations
4) Both 1 and 2

**Ans.4** The two major codes in use today are ASME (American Society of Mechanical Engineers) Code and the API (American Petroleum Institute) Standard 620. An earlier code, discontinued in 1956, was the API-ASME Code.

891. The organization which has been the most influential in researching temperature-humidity comfort is:

1) NIOSH
2) ACGIH
3) ASHRAE
4) OSHA

**Ans.3** The American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE), has been the most influential organization in researching temperature- humidity comfort. ASHRAE publishes ventilation standards that specify minimum ventilation rates and indoor air quality which are acceptable to human occupants and are intended to avoid adverse health effects. ASHRAE 62-1989, "Ventilation for Acceptable Indoor Air Quality" is its most recent standard.

892. Which organization authors the "National Fire Codes" and is recognized as the worldwide clearinghouse for fire safety information?

1) ANSI
2) FM
3) NFPA
4) UL

**Ans.3** The National Fire Protection Association (NFPA) is the clearinghouse for information on the subject of fire protection and fire prevention. It is a non-profit technical and educational organization whose technical standards are widely accepted and used as the basis for good practice. Many are incorporated by reference in OSHA regulations.

893. Which organization is responsible for establishing the codes governing electric hazards as a source of ignition?

1) ANSI
2) NFPA
3) IEEE
4) OSHA

**Ans.2** The National Fire Protection Association (NFPA) governs electrical hazards as a source of ignition through the National Electric Code (NEC), also called NFPA 70. This organization is responsible for reviewing, revising, and setting the standards that control electrical wiring and hardware in Class I, II, and III hazardous locations.

894. The Fire Protection Handbook is published by which of the following organizations?

1) OSHA
2) NIOSH
3) NSC
4) NFPA

**Ans.4** The National Fire Protection Association (NFPA) publishes the Fire Protection Handbook.

895. For a TLV change listed in the notice of intended changes to be considered for adoption, the ACGIH waits:

1) At least one year
2) 2 years
3) 3 years
4) Adoption requires specific action

**Ans.1** At least one year must pass for comments before it can be considered for adoption.

896. The primary organization dealing with compressed gas cylinder safety is:

1) ANSI
2) CGA
3) ASTM
4) NFPA

**Ans.2** The CGA (Compressed Gas Association) is the primary organization dealing with compressed gas safety. Some of the CGA documents are incorporated by reference into OSHA standards.

897. Who accredits industrial hygiene laboratories?

1) OSHA
2) NIOSH
3) ABIH
4) AIHA

**Ans.4** Only the AIHA accredits industrial hygiene laboratories.

898. The administrative sponsor of the National Electric Code is:

1) OHSA
2) ANSI
3) NFPA
4) IEEE

**Ans.3** In 1911, the National Fire Protection Association (NFPA) assumed sponsorship and control of the National Electric Code. The 1984 Edition of the Code (NFPA 70- 1984) was adopted by the Fire Protection Association (FPA) and approved by the American National Standards Institute (ANSI) as ANSI/NFA 70-1984.

899. What is the name of the journal published by the National Safety Council?

1) Professional Safety
2) Occupational Safety & Health
3) Occupational Health & Safety
4) Safety + Health

**Ans.4** Safety + Health is the journal of the National Safety Council.

900. What is the name of the journal published by the American Society of Safety Engineers?

1) Professional Safety
2) ASSE Journal
3) Safety + Health
4) Safety Management

**Ans.1** The ASSE journal is entitled "Professional Safety."

901. Of the following, which occupational illness occurs at the highest rate?

1) Cumulative trauma
2) Pneumoconioses
3) Poisoning
4) Heat-related

**Ans.1** According to OSHA statistics, cumulative trauma disorders have the highest incidence rate.

902. A pallet of material in a freight elevator which goes up with constant upward acceleration exerts a push on the floor of the elevator whose value is:

1) Equal to the weight of the pallet and its load
2) Greater than the weight of the pallet and its load
3) Less than the weight of the pallet and its load
4) Less than the weight of the pallet but greater than its load

**Ans.2** Applying Newton's third law of motion, "Every action (force) is accompanied by an equal and opposite reaction (force)", the push of the floor of the elevator must be enough to support the weight of the pallet and its load and in addition, must accelerate the combined weights upward. Hence, the push of the elevator is greater than the weight of the pallet and load by an amount needed to give it acceleration [F = W + W (a/g)].

903. The worker/working surface coupling has been cited as a causative factor in a large percentage of slips and falls accidents. The major mediating variable in this type of accident is the:

1) Shoe sole/heel composition and contact area
2) Floor surface and wet/greasy conditions
3) Coefficient of friction between the shoe sole and the working surface
4) Floor surface coefficient of friction

**Ans.3** The general recommendation for the control of slip and fall accidents is to adjust shoes and working surfaces to give a static coefficient of friction (SCOF) of at least 0.50. This is particularly important in manual materials handling where forward movement requires increased frictional forces to prevent foot slippage (NIOSH publication No. 81-122).

904. Factors that effect the stability of a trench include all of the following EXCEPT:

1) Moisture content of the soil
2) Freezing and thawing
3) Location of spoil pile and nearby equipment
4) Resonance

**Ans.4** The stability of a trench can be affected by the moisture content, freezing and thawing, and location of the spoil pile. Other factors include the depth of the trench and external vibration sources.

905. A ground fault circuit interrupter functions by:

1) Breaking the circuit
2) Control switches
3) Monitoring voltage
4) Monitoring current

**Ans.4** A ground fault circuit interrupter functions by monitoring current.

906. Storage of lead in the human body takes place primarily in the:

1) Brain
2) Bones
3) Lymph nodes
4) Lungs

**Ans.2** The primary reservoir for lead in the human body is the bones.

907. The result of acute exposure to beryllium inhalation is:

1) Chemical pneumonitis
2) Kidney failure
3) Collapse of arteries
4) High blood pressure

**Ans.1** Beryllium is a suspect carcinogen that causes chemical pneumonitis upon acute inhalation exposure. Chronic exposure can, among other things, cause berylliosis, a cumulative lung disorder.

908. Bladder cancer is sometimes associated with:

1) Benzo(a) pyrene
2) Benzo(e) pyrene
3) Vinylchloride monomer (VCM)

4) Aniline

**Ans.1** Benzo(a) pyrene is a suspect human carcinogen associated with bladder cancer. Benzo(e) pyrene is also a volatile benzene soluble derivative of coal tar pitch; however, it is not currently classified as a suspect human carcinogen. VCM is associated with human liver cancers.

909. Shaver's Disease is associated with exposure to:

1) Asbestos
2) Silica and aluminum dust/fume
3) Beryllium
4) Rickettsia

**Ans.2** Shaver's Disease is a bauxite pneumoconiosis associated with silica. Rickettsia is a human pathogen often carried by arthropods.

910. An occupational disease which is characterized by wrist drop, anemia, and abdominal cramps is:

1) Mercury poisoning
2) Lead poisoning
3) Asbestosis
4) Insecticide poisoning

**Ans.2** Other symptoms of lead overexposure include: Lassitude, insomnia, pallor, eye irritation, anorexia, low-weight, malnutrition, gingival lead line, and tremors.

911. A cigarette contains 6 mg of tar. Calculate the cumulative daily dose of tar received by a 160 lb adult male who smokes 2 packs per day. Each pack contains 20 cigarettes.

1) 1.5 mg tar/kg body weight per day

2) 3.3 mg tar/kg body weight per day
3) 4.4 mg tar/kg body weight per day
4) 6.6 mg tar/kg body weight per day

**Ans.2** Dose = (6 mg tar/cig * 20 cig/pack * 2 packs/day) / [(160 lbs * (1 kg/2.2 lbs)] =3.3 mg tar/kg body weight per day

912. The effective temperature is a determined index of the degree of warmth perceived on exposure to different combinations of:

1) Radiant heat, humidity, and air movement
2) Temperature, humidity, and air movement
3) Clothing, humidity, and air movement
4) Temperature, humidity, and radiant heat

**Ans.2** The effective temperature is an empirically determined index of different combinations of temperature, measured as the dry bulb, humidity, measured as the wet bulb, and air movement.

913. A worker in a warehouse on a hot summer day suffers from heat stroke. Her body temperature is:

1) Lowered
2) Elevated
3) Normal
4) Irregular

**Ans.2**Elevated body temperature in heat stoke is a major factor in causing the most important other clinical symptom, that being lack of sweating. Lack of sweating causes an increase in body temperature.

914. The most accurate measure of heat stress to the body is:

1) Blood pressure
2) Skin temperature
3) Rectal temperature
4) Oral temperature

**Ans.3** Rectal temperature measures the body's core temperature. Body core temperature rise is used as the limiting parameter in heat stress work.

915. If the air temperature is above the skin temperature one should _____ the air velocity to reduce the convective load:

1) Increase
2) Decrease
3) Stop
4) Not change

**Ans.2** If the air temperature is ABOVE the skin temperature, increasing the air velocity will increase the convective load.

916. A person can lose his or her heat acclimatization in as little as:

1) 12 hours
2) 3 days
3) 7 days
4) 14 days

**Ans.2** Significant loss of acclimitization may be seen after 3 days off.

917. The most serious form of heat injury is:

1) Heat stress
2) Heat syncope
3) Prickly heat
4) Heat stroke

**Ans.4** Heat stroke is truly life threatening and therefore the most serious. The body temperature rises to dangerous levels causing potential irreversible damage and possibly death.

918. Thermal stress is associated with all of the following jobs EXCEPT:

1) Freezer workers
2) Life guards
3) Foundry workers
4) Teachers

**Ans.4** Thermal stress refers to both heat stress and cold stress.

919. Of the following, which is NOT taken into account when figuring the effective temperature index?

1) Ambient temperature
2) Wall temperature
3) Moisture
4) Air movement

**Ans.2** Effective temperature is figured by measuring ambient temperature, air movement, and humidity.

920. Of the factors listed, which LEAST influences the interchange of heat between a person and their environment?

1) Air velocity
2) Air density
3) Air temperature
4) Radiant temperature

**Ans.2** Air velocity, air temperature, and radiant temperature appear as variables in the heat stress equations.

921. The condition that results when the body's heat regulating mechanism fails to function is:

1) Heat cramps
2) Heat exhaustion
3) Heat stroke
4) Heat syncope

**Ans.3** With heat syncope, heat cramps, and heat exhaustion, the body is still capable of producing sweat and is therefore able to achieve some cooling.

922. A primary sign of heat stroke on a worker is:

1) Cyanosis
2) Pale, clammy skin
3) Hot, dry skin
4) Nausea

**Ans.3** Pale, clammy skin and nausea are associated with heat exhaustion.

923. A risk-specific dose (RSD) applies to:

1) Non-carcinogens
2) Carcinogens
3) Mutagens
4) Teratogens

**Ans.2** The RSD is a dose that corresponds to a specific level of risk to an individual of contracting cancer over a 70-year lifetime due to the intake of contaminated drinking water.

924. A reference dose (RfD) applies to:

1) Non-carcinogens
2) Carcinogens
3) Mutagens
4) Teratogens

**Ans.1** An RfD is an estimate of daily exposure of the human population, including sensitive subgroups, to a substance that would not represent an appreciable risk of deleterious effects during a lifetime. RfD applies to non-carcinogens, while RSD applies to carcinogens.

925. If the ambient temperature rises above the dew point:

1) Precipitation will occur
2) Dew will form
3) Evaporation is likely to occur
4) The relative humidity will reach 10%

**Ans.3** At the dew point, the air is saturated with water vapor. When the temperature of the air rises, it can hold more water vapor. Its relative humidity will be less than 100%, and evaporation will occur if water is present.

926. What happens to a normal gas when it is compressed rapidly?

1) Nothing
2) The temperature increases
3) The temperature decreases
4) Work is done by the gas

**Ans.2** Work is done ON the gas when it is compressed, and the gas's temperature will rise.

927. What happens to the heat energy of a 2 qt pan of water when it is heated from 30ºC to 100ºC?

1) The energy remains the same
2) The energy is doubled
3) The energy is less than doubled
4) The energy is more than doubled

**Ans.3** The heat energy varies directly with the absolute temperature of the substance (i.e., Kelvin), not the relative temperature (i.e., degrees Celsius). Water heated from 30ºC to 100ºC is being heated from 303° K to 373° K (add 273 to the Celsius temperatures to get the absolute Kelvin temperatures). This is less than a doubling of temperature resulting in less than a doubling of heat energy.

928. The temperature of a block of ice while melting remains constant. In the course of changing state, the block of ice is:

1) Giving up energy
2) Gaining energy but using energy to change state
3) Converting energy to mass
4) It neither gives up nor gains energy

**Ans.2** In the process of melting, the block of ice gains heat energy. To illustrate, if a block of ice is heated, its temperature will reach 32ºF. The temperature then remains constant and the

ice begins to melt. As more heat is added, more ice will melt until it all turns to 32ºF water. The energy gained by the ice is used to melt it. The energy necessary to "melt", or to change from a solid state to liquid state, is called the latent heat of fusion.

929. The amount of heat required to change 100 g of water at 15ºC to 75ºC is:

1) 3,000 cal
2) 6,000 cal
3) 9,000 cal
4) 12,000 cal

**Ans.2** Heat gained/lost = (m * c) * (□T) where:
m = mass of the substance
c = specific heat of the substance
□T = change in temperature
1 cal is required to heat 1 g of water 1ºC
Heat required = 100 g * [1 cal/(g ºC)] * (75º C - 15ºC)
= 100 g * 60 cal/g
= 6,000 cal

930. The amount of heat needed to raise the temperature of 1 lb of water 1ºF is called:

1) The specific heat
2) A calorie
3) A Btu
4) The Kelvin scale

**Ans.3** The British thermal unit (Btu) is the amount of heat that is required to raise the temperature of 1 lb of water 1ºF.

931. The quantity of heat needed to raise the temperature of 1 g of a material by 1ºC is a characteristic property of the material. It is commonly called:

1) A calorie
2) The specific heat
3) A Btu
4) Temperature rise

**Ans.2** The specific heat is the heat energy that is required to raise the temperature of a substance 1ºC.

932. The ability of a material to be drawn out into the form of wire is called:

1) Tenacity
2) Malleability
3) Ductility
4) Hardness

**Ans.3** Copper, gold, silver, brass, and other substances can be drawn out in the form of wires. A material which can be drawn out this way is said to be ductile.

933. The specific gravity of cast iron is 7. A 1 $ft^3$ bar of it will weigh:

1) 874 lb
2) 437 lb
3) 218 lb
4) 125 lb

**Ans.2** The number of times a substance is as heavy as an equal volume of water is called specific gravity. The specific gravity (SG) of a substance may be found from the relation:
SG = Weight of substance/Weight of an equal volume of water

1 $ft^3$ of water weighs 62.4 lbs and SG of cast iron is 7
Then:
SG = (Weight of 1 $ft^3$ of iron)/(Weight of 1 $ft^3$of water)
7 = (Weight of 1 $ft^3$ of iron)/(62.4 lbs)
(7)(62.4 lbs) = (weight of 1 $ft^3$ of iron)
436.8 lbs = (Weight of 1 $ft^3$ of iron)

934. Find the density of a block of iron 12 cm long, 8 cm wide, and 2 cm high, if the block has a mass of 700 g:

1) 1.8 $g/cm^3$
2) 3.7 $g/cm^3$
3) 5.5 $g/cm^3$
4) 7.3 $g/cm^3$

**Ans.2** Apply the following formula.
Density = Mass / Volume
Density = 700 g / (12 cm * 8 cm * 2 cm)
= 700 g / 192 $cm^3$
= 3.65 $g/cm^3$

935. A quantity of air exerts a pressure of 720 mm Hg when it occupies a volume of 50 $cm^3$. This air is compressed until it exerts a pressure of 810 mm Hg. If the temperature is kept constant, what would be the volume of the gas after compression?

1) 22 $cm^3$
2) 44 $cm^3$
3) 66 $cm^3$
4) 88 $cm^3$

**Ans.2** If the temperature of a confined gas is kept constant, its volume will vary inversely as its pressure. This may be expressed by the following formula:
$P_1 V_1 = P_2 V_2$
720 mm Hg * 50 $cm^3$ = 810 mm Hg * $V_2$
$V_2$ = 44.4 $cm^3$

936. A rectangular barge, 10 ft wide and 40 ft long sinks 6 in. deeper in the water after a truck is loaded on it. What is the weight of the truck?

1) 1.6 ton
2) 3.1 ton
3) 6.3 ton
4) 12 ton

**Ans.3**A floating body will displace its own weight of the fluid in which it is immersed. The additional volume of water displaced by the barge after the truck is loaded on it is equal to:
10 ft * 40 ft * 0.5 ft = 200 $ft^3$
Since:
1 $ft^3$ of water weights 62.4 lb
Then:
200 $ft^3$ * 62.4 lb/$ft^3$ = 12,500 lb
12,500 lb / 2,000 (lb/ton)
= 6.25 ton

937. At a depth of 1,000 ft, what is the pressure on the deck of a submarine? (The density of salt water is 64.4 $lb/ft^3$)

1) 32,200 $lb/ft^2$
2) 16,100 $lb/ft^2$
3) 447 $lb/in^2$
4) 671 $lb/in^3$

**Ans.3** Pressure is caused by the weight of the fluid acting on a surface. A dense liquid will exert more pressure on bodies immersed in it than will a lighter liquid. Pressure on the deck of the submarine is equal to the height of the column of salt water acting on it multiplied by the density of the salt water.
Pressure = 1,000 ft * 64.4 $lb/ft^3$
= 64,400 $lb/ft^2$ 1 $ft^2$
= 144 $in^2$
Pressure = (64,400 $lb/ft^2$) / (144 $in^2/ft^2$)
= 447.2 $lb/in^2$
= 447.2 psi

938. When atmospheric water vapor condenses in the air, it always leaves the surrounding air temperature:

1) Unchanged
2) Cooler
3) Warmer
4) Either cooler or warmer, depending on time of the day

**Ans.3** When a liquid is vaporized, it must be heated. Whenever it reverts back to a liquid state through condensation, it must give up this heat. The heat that is given off must therefore warm the surrounding air.

939. What is the ratio of elemental Pb atoms (non-oxidized) to molecules of $H_2S0_4$ after balancing the equation?

$Pb + H_2SO_4 + PbO_2 \rightarrow PbSO_4 + H_2O$

1) 1 to 1

2) 1 to 2
3) 2 to 1
4) 2 to 3

**Ans.2** The balanced equation consisting of lead, sulfuric acid and lead oxide is:

$Pb + 2\ (H_2SO_4) + PbO_2 \rightarrow 2\ (PbSO_4) + 2\ (H_2O)$

940. Which of the following would react with a strong oxidizing agent?

1) Iron
2) Freon 22
3) Oxygen
4) Silicon dioxide

**Ans.1** Iron (Fe) reacts readily with oxidizing agents like oxygen to form rust. Oxygen ($O_2$) is an oxidizer itself, and silicon dioxide ($SiO_2$) is unreactive to oxidizing agents. Freons are noted for their relative unreactive nature.

941. What is the weight of a hollow concrete cylinder with an outside diameter of 5 feet, an inside diameter of 3 feet, and a length of 6 feet?(Concrete weighs 150 $lb/ft^3$).

1) 45,000 lb
2) 18,000 lb
3) 15,000 lb
4) 11,000 lb

**Ans.4** Find the volume of the cylinder as though it were solid, and then subtract the volume of the empty space. Finally, multiply the difference by the density of concrete.
Volume is □ * $radius^2$ * length
Volume (solid) = □ * $(2.5\ ft)^2$ * 6 ft = 117.75 $ft^3$
Volume (empty space) = □ * $(1.5\ ft)^2$ * 6 ft = 42.39 $ft^3$
Volume concrete = 117.75 $ft^3$ - 42.39 $ft^3$ = 75.36 $ft^3$
Weight = 75.36 $ft^3$ * 150 $lb/ft^3$ = 11,304 lb

942. If the two legs of a right triangle have lengths of 20 feet and 30 feet, what is the angle formed by the longer leg and the hypotenuse?

1) 49º
2) 30º
3) 34º
4) 52º

**Ans.3** Draw the triangle. The angle between the longer leg and the hypotenuse is determined by finding the tangent.
tangent = opposite side /adjacent side
= 20 ft /30 ft
= 0.6667 $\tan^{-1}$
(0.6667) = 33.7º

943. If the two legs of a right triangle have lengths of 10 feet and 15 feet, how long is the hypotenuse?

1) 12 ft
2) 18 ft
3) 20 ft
4) 25 ft

**Ans.2** In mathematics, the sum of the sides of a triangle squared equals the sum of the hypotenuse squared or:
$c^2 = a^2 + b^2$
Thus:
$c^2 = (100\ ft^2 + 225\ ft^2) = 325\ ft^2$
$c = \sqrt{325\ ft^2}$
= 18 ft

944. If the diagonal of a rectangle has a length of 20 ft and forms a 30º angle with the base, what are the dimensions of the rectangle?

1) 10 ft * 17.3 ft
2) 10 ft * 15.5 ft
3) 8 ft * 12 ft
4) 8 ft * 10 ft

**Ans.1** Draw a picture. The length of the sides can be found by using trigonometry.
opposite side = hypotenuse side * sine of the angle the hypotenuse makes with the other side

of the triangle
opposite side = sin 30 / 20 ft
= 10 ft
The length of the base can be found by dividing the length of the opposite side by the tangent of the angle the unknown side makes with the hypotenuse.
base = 10 ft / tan 30
= 10 ft / 0.5774
= 17.3 ft

945. What is the wavelength of an electromagnetic wave with a frequency of $10^8$ Hz? (The speed of light is 3 x $10^8$ m/sec.)

1) 1 m
2) 1/3 m
3) 3 m
4) 1 m

**Ans.3** Speed of light = wavelength * frequency:
3 x $10^8$ m/sec = wavelength * ($10^8$ Hz)
wavelength = 3 m

946. The term "Hertz" refers to:

1) A car rental company
2) The wavelength of an electromagnetic wave, in meters
3) 10% of the speed of light
4) Frequency, in cycles per second

**Ans.4** Hertz is defined as a cycle per second.

947. Two forces of 50 lb each act on a bolt at angles from the horizontal of 30º and 60º. What is the magnitude of the resultant force on the bolt?

1) 100 lb
2) 97 lb
3) 75 lb
4) 67 lb

**Ans.** 2

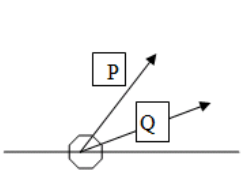


Figure 1

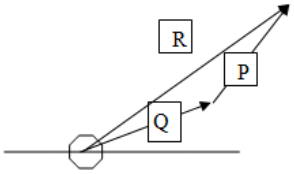


Figure 2

948. Two forces of 50 lb each act on a bolt at angles from the horizontal of 30º and 60º. What is the direction of the resultant force?

1) 30º
2) 60º
3) 45º
4) 57º

**Ans.3** Draw this problem. This problem can be solved intuitively or by using the Law of Cosines followed by the Law of Sines.
$c^2 = a^2 + b^2 - 2ab \cos C$
$c^2 = 50^2 + 50^2 - 2(50)(50) \cos 150°$
$c^2 = 9330$
$c = 96.6$
$a/\sin A = b/\sin B = c/\sin C$
$a/\sin A = c/\sin C$
$50/\sin A = 96.6/\sin 150°$
$\sin A = .26$
$A = 15°$

Direction of resultant force = 30º + 15º = 45º

949. A unit used to describe sound pressure is:

1) Newtons per square meter
2) Joules
3) Kilograms per meter
4) Newtons per square foot

**Ans.1** Sound pressure is measured in newtons per square meter.

950. Another unit used to describe sound pressure is:

1) Pascals
2) Footpounds
3) Ergs
4) Slugs

**Ans.1** One pascal equals one newton per square meter.

951. Effective temperature is defined as:

1) A non-comfort index which combines the effect of a body of temperature, humidity, and air movement
2) Any specific temperature, the vapor pressure, and absolute humidity
3) The rate in which temperature effects the body
4) Comfort index which combines the effects of a body temperature, humidity, and air movement

**Ans.4** Effective temperature is defined as the comfort index which combines the effects of a body temperature, humidity, and air movement.

952. Hypothermia is defined as a condition when:

1) The body experiences extremely high blood pressure
2) The body experiences extremely low blood pressure
3) The body gains heat faster than it can lose it
4) The body loses heat faster than it can produce it

**Ans.4** Hypothermia is defined as a condition when the body loses heat faster than it can produce it.
NOTE: Hypotension is very low blood pressure, and hypertension is very high blood pressure.

953. The combined effects of all or some of physical climate factors can be expressed in the form of:

1) Combined temperature index
2) Non-effective temperature
3) Climate index
4) Temperature index

**Ans.3** The combined effects of all or some physical climate factors can be expressed in the form of climate index.

954. Acclimatization to a hot environment includes controlled blood flow to the skin, facilitated sweating, and increased stroke volume of the heart without an increase in heart rate. It can be accomplished in:

1) One or two weeks
2) One month
3) 7 days
4) 12 days

**Ans.1** Acclimatization to a hot environment includes controlled blood flow to the skin, facilitated sweating, and increased stroke volume of the heart without an increase in heart rate. It can be accomplished in one or two weeks.

955. Identify the major avenue(s) by which the human body controls heat transfer between the core and the skin.

1) Sweat production
2) Efferent motor pathways, sweat production, and vasomotor
3) Sudomotor pathways only
4) Motor, vasomotor pathways

**Ans.2** The efferent motor pathways, sweat production, and vasomotor mechanisms are the systems by which the body transfers heat from the core to the skin.

956. Which is the most vulnerable to injury induced by laser energy?

1) Skin
2) Eye
3) Internal organs
4) Glands

**Ans.2** The eyes are the most vulnerable to injury by laser energy.

957. The most common cause of ultraviolet radiation exposure is from:

1) Television/computer monitors
2) Ultraviolet lights (tanning lights, black lights, etc.)
3) The sun
4) Special devices

**Ans.3** Exposure to direct and indirect sunlight is the most common cause of ultraviolet radiation exposure.

958. An important element of a production machine shop safety program is:

1) Safety training schedule
2) Organizational reporting protocols
3) Lockout/tagout procedures
4) Personnel policies for discipline

**Ans.3** A specific lockout/tagout program facilitates protecting workers from life threatening hazards. This is more important than the other choices.

959. What may be a negative consequence of wearing personal protective equipment?

1) Uncomfortable
2) Inconvenient
3) Requires more effort
4) All of the above

**Ans.4** Negative consequences of wearing personal protective equipment may include less comfort, more inconvenience, and more effort.

E. Scott Geller, The Psychology of Safety

960. Byssinosis is an occupational disease related to exposure to:

1) Cotton fiber dust
2) Droppings of infected cattle
3) Rickettsial organism
4) Fumes of metals

**Ans.1** Byssinosis is a disease occurring in those who experience prolonged exposure to heavy air concentrations of cotton or flax dust.

961. The most frequent occupational disease is attributed to:

1) Pneumoconiosis
2) Dermatitis
3) Chemical poisoning
4) Radioactive exposure

**Ans.2** Industrial skin diseases account for more than half of all compensation claims for occupational disease. These claims are scattered throughout all types of industry and sometimes appear where occupational skin disease may be least expected. There are two general types of dermatitis, primary irritation and sensitization. Cause can be classified under the headings of: chemical agents, mechanical agents, physical agents, plant poisons, and biological agents.

962. Occupational arsenic exposure is highly unlikely in which of the following industries?

1) Agricultural pesticide workers
2) Electronics industry workers
3) Tire manufacturing
4) Copper smelting

**Ans.3** Some arsenates are used as pesticides; various arsenic compounds are used as doping agents in the semiconductor industry, and arsenic trioxide is produced in large quantities in the smelting of copper.

963. Of all occupational diseases occurring in industry, the most common is?

1) Lead poisoning
2) Silicosis
3) Mercurialism
4) Dermatitis

**Ans.4** Dermatitis is the most common of all occupational diseases.

964. Dermatitis tends to be most severe in which of the following industries?

1) Silver
2) Meat packing
3) Lead battery
4) Concrete products

**Ans.1** Using lost work day rates as a measure of severity, dermatitis is the most severe in the silver industry, followed by the meat packing industry, then the lead battery industry and finally the concrete production industry.

965. Overexertion injuries associated with manual materials handing jobs account for at least _____ of all occupational injuries:

1) 15%
2) 20%
3) 25%
4) 30%

**Ans.3** The National Safety Council reports that at least 25% of all occupational injuries resulted from overexertion associated with manual materials handling work consisting of lifting, lowering, pushing, and pulling operations.

966. Carpal tunnel syndrome occurs more frequently in:

1) Men
2) Women
3) Approximate equal distribution
4) Depends on body weight

**Ans.2** Statistics indicate that more women get carpal tunnel syndrome than men.

967. When is the highest incidence of sharp object injuries from a medical device?

1) During use
2) Pre-use
3) During disposal
4) After use, pre-disposal

**Ans.4** The highest incidence of sharp object injuries from medical devices occur after use, prior to disposal. This is the most dangerous time, since the object is contaminated with potential infectious agents from blood.

968. Which of the following chemicals is not combustible?

1) Carbon black
2) Sulfur
3) Lamp black
4) Sodium chloride

**Ans.4** Most organic chemicals (those with carbon atoms) are combustible, while most inorganic chemicals (those without carbon atoms) are non-combustible.

969. The ability of a material to be beaten or rolled into thin sheets is called:

1) Hardness
2) Ductility
3) Malleability
4) Tenacity

**Ans.3** The ability of a material to be beaten or rolled into thin sheets is called malleability. Gold is among the most malleable substances. Others include iron, copper, tin and aluminum.

970. 10 ml of 10 molar HCl contains _____ moles of HCl. This amount of HCl will neutralize _____ ml of 0.1 normal $HNO_3$.

1) 0.1, 0.1
2) 0.1, 0.05
3) 0.05, 0.1
4) HCl will not neutralize $HNO_3$

**Ans.4** Since both are strong acids, no neutralization will take place. Neutralization is a reaction between an acid and a base. Since both are very strong acids they will be nearly 100% ionized in solution; both will use water as their proton acceptor.

971. A radioisotope has a half life of 1 year. How many years will it take to reduce the initial activity to less than 10%?

1) 1 year
2) 2 years
3) 3 years
4) 4 years

**Ans.4** After 1 year, the activity will be 50%. After two years, the activity will be half of 50%, or 25%. The activity after three years will be 12.5% of the initial activity. After 4 years the activity will be 6.25% of the initial activity. This is less than 10%.

972. A solution contains $10^{-1}$ moles of HYDROGEN ions per liter. The solution is:

1) A strong base
2) A strong acid
3) A weak base
4) A weak acid

**Ans.2** $10^{-1}$ moles of H+ ions is equal to a pH of 1. This is a strong acid solution.
pH = -log (concentration of hydrogen ions/liter) = -log ($10^{-1}$) = 1

973. A tank of helium contains 150 l of gas at a pressure of 110 atm. What volume would the gas occupy at 1 atm assuming the temperature remains unchanged?

1) 16,500 liters
2) 1.36 liters
3) 0.73 liters
4) 6.06 x $10^{-5}$ liters

**Ans.1** Use Boyle's law ($P_1 V_1 = P_2 V_2$) and solve for $V_2$
$V_2 = (P_1 V_1) / P_2$
= (110 atm * 150 liters) / 1 atm
= 16,500 liters

974. Tuberculosis is spread by:

1) Intimate sexual contact
2) Contaminated blood
3) Airborne droplet nuclei
4) Fomites

**Ans.3** Tuberculosis (TB) is a bacterial disease which is spread by airborne droplet nuclei containing the infectious bacteria in closed air spaces. TB is not spread by sexual contact or contaminated blood or blood products.

975. A person infected with tuberculosis:

1) Always shows symptoms
2) Not always tests positive
3) Can spread disease with no symptoms
4) May never show symptoms

**Ans.4** People infected with tuberculosis may never develop active TB. 5-15% develop active disease within the first 2 years of primary infection. Therefore, the symptoms may never be evident. When the infected person is not showing symptoms, they are not infectious and cannot spread the disease. TB infected people always test positive for TB.

976. Tuberculosis outbreaks have increased in recent years because of all of the following EXCEPT:

1) Non-compliance with drug therapy
2) Contaminated blood supply
3) Multi-drug resistant bacteria
4) Human immunodeficiency virus

**Ans.2** Tuberculosis is not transmitted by contaminated blood. Infected people who do not comply with the proper drug therapy cause an increase in drug resistant bacteria. Many HIV infected patients develop TB due to their reduced immunocompetence.

977. Disinfection means:

1) Killing/removing all organisms
2) Reducing the number of organisms so there is no threat of disease
3) Removing microorganisms from skin or living tissus
4) Stopping the growth of microorganisms

**Ans.2** Disinfection is used on inanimate objects to reduce the number of pathogens to such a low concentration that there is no threat of disease. Sterilization is the killing and removing of all organisms from a surface. Antisepsis is the removal of microbes from the skin or living tissue. Bacteriostatic agents stop bacterial growth.

978. The phenol coefficient is:

1) A measure of microbial resistance to phenol disinfectant
2) A measure of microbial resistance to disinfectants
3) A measure of phenol in a disinfectant
4) A measure of the effectiveness of a disinfectant

**Ans.4** The phenol coefficient is the ratio of the smallest concentration of a disinfectant that will kill a certain percentage of organisms in 5-10 minutes to the smallest concentration of phenol that will kill in the same time period. It measures the effectiveness of the disinfectant.

979. Bacterial and viral disinfectants work in all of the following ways EXCEPT:

1) Alteration of protein structure
2) Disruption of lipids in cell membrane
3) Destruction of nuclear membrane
4) Nucleic acid destruction

**Ans.3** Viruses and bacteria have no nucleus or nuclear membrane to disrupt. Disinfectants work through action on proteins, cell membranes, and viral nucleic acid.

980. What biosafety level should be used when working with blood and blood products?

1) 1
2) 2
3) 3
4) 4

**Ans.2** Biosafety level 2 is recommended by the Centers for Disease Control for working with human-derived blood, body fluids, or tissues where the presence of an infectious agent may be unknown.

981. What is the function of the ear's eustachian tube?

1) Pressure equalization
2) Transmission of sound waves
3) Collect dirt
4) Prevent infection

**Ans.1** The eustachian tube serves to equalize pressure in the middle ear with the external atmospheric pressure. It opens during swallowing and yawning.

982. Which of the following chemicals is least likely to produce kidney damage in humans?

1) Chlorinated hydrocarbons
2) Chromium
3) Benzene
4) Carbon disulfide

**Ans.3** Chlorinated hydrocarbons, carbon disulfide, chromium, and other heavy metals such as uranium, lead, and mercury are nephrotoxins - chemicals that produce kidney damage in humans. Benzene is toxic primarily to the hematopoietic systems.

983. Which of the following is the most hazardous to your health if ingested?

1) Wood alcohol
2) Isopropyl alcohol
3) Glyceryl alcohol
4) Grain alcohol

**Ans.1** Wood alcohol is methyl alcohol (methanol), which is toxic when ingested; drinking it may cause blindness. Isopropyl alcohol is rubbing alcohol, glyceryl alcohol is an industrial solvent, and flavored diluted grain alcohol is a beverage.

984. Carpal tunnel syndrome involves an injury to the median nerve. The nerve is compressed due to inflammation in the carpal tunnel. Carpal tunnel syndrome will NOT result in:

1) Numbness in the little finger
2) Pain in the wrist upon exertion
3) Pain in the second and third fingers
4) Numbness in the thumb

**Ans.1** The little finger is not served by the median nerve, therefore the little finger is not affected by compression of the median nerve in the carpal tunnel.

985. In heat stroke, the:

1) Body temperature is lowered
2) Body temperature is elevated
3) Body temperature is normal
4) Patient sweats profusely

**Ans.2** In heat stroke the patient has an elevated temperature and does not sweat.

986. In which part of the ear is mechanical energy converted to electrical energy?

1) Outer ear
2) Middle ear
3) Inner ear
4) Outer ossicular area

**Ans.3** In the cochlea the energy is changed from mechanical (fluid waves) to electrical (nerve impulses).

987. Melanoma is a serious type of skin cancer. Which of the following is the most common occupational cause of melanoma?

1) Asphalt tar
2) Ultraviolet light from black lights
3) Sunlight
4) Nitrosoamines

**Ans.3** Sunlight is the most common occupational cause of melanoma skin cancer in caucasian workers.

988. Numbness of the fingers is associated with:

1) Siderosis
2) Raynaud's Phenomenon
3) Lead poisoning
4) Tetanus

**Ans.2** Raynaud's Phenomenon is a disorder associated with the use of hand held vibrating tools. It results in a loss of feeling and is characterized by white spots on the fingers and hands.

989. The cochlea is essentially a coiled tube. At one end of the tube is the oval window. The function of the oval window is to:

1) Receive the vibrations from the eardrum
2) Shield the semicircular canals from interference
3) Protect the auditory canal
4) Receive neural input from the auditory nerve

**Ans.1** The oval window is the entry point for the vibrations coming from the eardrum via the ossicles. The vibrations enter the cochlea, move through the fluid and stimulate hair-like cells. Stimulation of these cells sends an impulse to the brain for analysis.

990. What is the most serious heat related disease?

1) Heat stroke
2) Heat neurosis
3) Acute heat exhaustion
4) Heat syncope

**Ans.1** In heat stroke the body temperature rises to dangerous levels and the heat control mechanisms are shut down.

991. Which of the following is likely to increase proneness to skin problems?

1) Dry skin
2) Oily skin
3) Fair skin
4) All of the above

**Ans.4** Dry, oily, or fair skin makes the individual more susceptible to one form or another of skin disorders.

992. For indoor air quality investigations, carbon dioxide concentrations above _____ppm indicate inadequate ventilation.

1) 350 ppm
2) 600 ppm
3) 700 ppm
4) 1,000 ppm

**Ans.4** NIOSH guidelines specify that carbon dioxide concentrations of 1,000 ppm or greater indicate inadequate ventilation. The number may even be less than 1,000 ppm (e.g., in the 750-800 ppm range).

993. You are asked to inspect a 10 psi(g) unfired pressure vessel. What do you already know about it?

1) Safety professionals do not perform pressure vessel inspections
2) No formal inspection criteria exist
3) Unfired pressure vessel regulations do apply
4) Unfired pressure vessel regulations do not apply

**Ans4** Unfired pressure vessel regulations do not apply to unfired pressure vessels which operate at pressures below 15 psi(g).

994. A static coefficient of friction (SCOF) of ____ or higher is considered the minimum for pedestrian safety under most conditions.

1) 0.10
2) 0.30
3) 0.50
4) 0.70

**Ans.3** Although there is no regulation setting this as the minimum SCOF, most authorities agree that 0.50 is the minimum acceptable SCOF. A SCOF of 0.50 or greater is required by UL 410 for a floor product to be listed under that consensus standard.

995. When a crane is repaired or extensively overhauled, how much load must be placed on the crane to test it prior to returning it to service?

1) 100% of rated capacity
2) 125% of rated capacity
3) 150% of rated capacity
4) 200% of rated capacity

**Ans.2** The standard load test requirement is 125% of the rated capacity of the crane. This is the same load that is used when the unit is new from the manufacturer.

996. If an employee is exposed to continuous noise at the level of 90 dBA averaged over the eight hour work day, the employer must do all of the following EXCEPT:

1) Implement feasible engineering or administrative controls
2) Include the employee in a Hearing Conservation Program
3) Establish a Hearing Conservation Program
4) Isolate the noise source

**Ans.4** OSHA's Hearing Conservation Standard requires that feasible administrative or engineering controls be implemented if exposure is greater than or equal to 90 dBA. If the noise exposure is greater than or equal to 85 dBA, a hearing conservation program must be established and any exposed employees must be enrolled in it.

997. What is the total resistance of three 30 □ devices connected in series?

1) 10 O
2) 30 O
3) 90 O
4) 27,000 O

**Ans.3** For a series circuit: $R_T$ = 30 □ + 30 □ + 30 □
$R_T$ = 90 □

998. What is total resistance of three 30 □ devices connected in parallel?

1) 10 O
2) 30 O
3) 90 O
4) 27,000 O

**Ans.** For a parallel circuit: $1/R_T$ = 1/30 □ + 1/30 □ + 1/30 □
$R_T$ = 10 □

999. What is the total power of a parallel 110 V circuit where the current flow through the branches is 12.3 A, 8.5 A and 4.2 A?

1) 1,265 W
2) 2,750 W
3) 3,552 W
4) 4,321 W

**Ans.2** For a parallel circuit, the total current is the sum of the branch currents (opposite from calculating resistances).
$I_T$ = 12.3 A + 8.5 A + 4.2 A
$I_T$ = 25 A
P = E * I
P = 110 V * 25 A = 2,750 W

1000. What is the maximum number of lighting and appliance circuits permitted in any one circuit breaker cabinet?

1) 22
2) 42
3) 100
4) No limit, cabinets can be sized to hold additional circuits

**Ans.2** No more than 42 circuits of this type are permitted in one circuit breaker cabinet.
NFPA 70 Article 384-15 (National Electric Code)

1001. Equipment connected by cord and plug shall be grounded for any of the following EXCEPT when:

1) The equipment is used by persons standing on metal floors
2) The equipment operates at 220 V to ground
3) The equipment is used by persons working inside a metal tank
4) The equipment operates at 120 V to ground

**Ans.4** Equipment connected by cord and plug shall be grounded if it is used in a hazardous location, if it operates at more than 150 V to ground, if it is used in a damp or wet location, if it is used by persons standing on the ground, or on metal floors, or working inside metal tanks or boilers, or if it is likely to be used in wet and conductive locations.
29 CFR 1910.305 (g)

1002. The voltage on the secondary side of a step-down transformer is _____ the voltage on the primary side.

1) Lower than
2) Higher than
3) Equal to
4) Not related to

**Ans.1** The secondary side of a step-down transformer side features the "stepped down" voltages, therefore the secondary side is of lower voltage.

1003. As a safety professional, you are notified that some rigid metal conduit is warm to the touch. What is your suspicion?

1) Short in system
2) Transformer is undersized
3) Aluminum wire
4) Current flow is too high

**Ans.4** This would probably indicate that current flow is too high in the conduit. Further investigation and corrective action should also take place.

1004. Electric power tools which do not have separate grounding conductors must:

1) Never be used
2) Have three-wire systems
3) Have grounding straps attached to the chassis
4) Be double insulated

**Ans.4** Double insulated tools do not require separate grounding conductors.

1005. In an industrial transformer, the "primary" side is generally recognized by any of the following EXCEPT:

1) Direct feed from power lines
2) Last part of transformer to see current

3) Higher voltages than the secondary side
4) First part of transformer to see current

**Ans.2** The "primary" side of a conventional industrial transformer receives the highest voltage from the electrical service and then steps down the voltage.

1006. A megohm meter is used to:

1) Measure large voltages
2) Measure large resistances
3) Measure earth faults
4) Measure large currents

**Ans.2** A megohm meter measures large resistances.

1007. With what instrument can the continuity of a ground be tested in an electrical hand tool?

1) Voltmeter
2) X-Ray
3) Tester with light
4) Ohmmeter

**Ans.4** An ohmmeter can test the continuity of a ground in an electrical hand tool.

1008. When does the Corona Effect occur?

1) Upon electromagnetic radiation effects
2) Upon low voltage current from which a spark occurs
3) Upon a partial breakdown of a gap between two conductors in a high voltage circuit
4) Upon the breakdown of the insulation in a high voltage conductor

**Ans.3** Corona occurs in high voltage circuits when breakdown of a gap between two conductors is partial.

1009. Which is not an acceptable grounding means?

1) Open switch ground
2) Resistance ground
3) Reactance ground
4) Resonance ground

**Ans.1** Open switch ground is not an acceptable grounding means.

1010. For Class I locations Groups A, B, C and D, the classifications involve determinations of:

1) Minimum explosion pressure, minimum safe clearance between parts of a clamped joint in an enclosure, and the maximum auto-ignition temperature of the atmospheric mixture
2) Maximum explosion pressure, maximum safe clearance between parts of a clamped point in an enclosure, and the minimum auto-ignition temperature of the atmospheric mixture
3) The chemical mixture and explosive pressure
4) The ignition temperature of the gas only

**Ans.2** Maximum explosion pressure and maximum safe clearance between joints in the enclosure are the critical issues. NFPA 70 Article 500-3 (National Electric Code)

1011. A 10 part block and tackle is used to lift a load of 3 tons. Neglecting shear friction, what is the force required?

1) 100 lb
2) 200 lb
3) 300 lb
4) 600 lb

**Ans.4** 600 lb is required.
3 ton * 2,000 lb/ton = 6,000 lb
6,000 lb / 10 parts = 600 lb

1012. All circular table saws must have:

1) A hood which encloses that portion of a saw above the material being cut
2) A spreader to prevent material from squeezing the saw blade
3) Anti-kickback fingers to prevent material from being thrown back at the operator
4) Both 1 and 3

**Ans.1** OSHA requires that all circular table saws shall have a hood over the portion of the saw above the table, so mounted that the hood will automatically adjust itself to the thickness of, and remain in contact with, the material being cut. Spreaders and anti kickback fingers apply to special applications, such as ripping.
29 CFR 1910.213 (a) and (c)

1013. A punch press guard normally consists of:

1) Barrier
2) Remote controls
3) Sensing
4) Locks

**Ans.2** Remote controls are the most common guard on punch presses.

1014. A wheelbarrow contains a load of 216 lb at an average distance of 1.50 ft from the axle. If the shaft of the wheelbarrow is 4 ft long, what is the force exerted upward by the fulcrum?

1) 45.0 lb
2) 67.5 lb

3) 135 lb
4) 180 lb

**Ans.3** Since the wheelbarrow is in equilibrium, the forces pulling it upward are equal to the sum of the forces pulling it downward.
The downward force = load = 216 lb
The upward force = (216 lb * l.50 ft) / 4 ft= 81 lb
Therefore the fulcrum must exert an upward force of: 216 lb - 81 lb = 135 lb

1015. If water weighs 8.34 lb/gal, what work is required to move 300 gal of water from a mine 400 ft deep?

1) 1,000,200 ft-lb
2) 1,000,400 ft-lb
3) 1,000,600 ft-lb
4) 1,000,800 ft-lb

**Ans.4** Use the formula W = F D, where W = Work, F = Force, D = Distance
W = (8.34 lb/gal * 300 gal)(400 ft)
W = 1,000,800 ft-lb

1016. The ASME code requires pressure vessel hydrostatic testing to 150% of the:

1) Design pressure
2) Pressure relief limit
3) Maximum operating pressure (MOP)
4) Burst pressure

**Ans.3** The ASME code calls for 150% of the maximum operating pressure.

1017. To what percentage of the rated load must a crane be tested?

1) 75%
2) 100%
3) 125%
4) 150%

**Ans.3** Cranes must be proof-tested to 125% of the rated load.

1018. What predominant frequency can be expected from a fan which is direct driven at 3,600 rpm to deliver 2,400 cfm at 0.5 in. of water? The fan has 60 backward curved blades.

1) 1,200 Hz
2) 1,800 Hz
3) 3,600 Hz
4) 5,400 Hz

**Ans.3** The frequency is determined by how many times the blades pass a fixed point.
3,600 rpm = 60 revolutions per second (rps)
60 blades turning at 60 rps = 3,600 passes of a blade past the point
This is 3,600 Hz.

1019. A two piece block and tackle with 5 parts of rope to lift a weight of 3,000 pounds requires how much force for lift (neglecting friction)?

1) 500 lb
2) 600 lb
3) 750 lb
4) 1,000 lb

**Ans.2** To determine the lifting force without friction, divide the total load by the number of rope parts.
3,000 lb / 5 = 600 lb

1020. In any man-machine system, humans serve three functions: sensor, controller and:

1) Information seeker
2) Information processor
3) Stimulus responder
4) Information controller

**Ans.2** The purpose of human factors engineering is to minimize errors in using displays and controls by designing systems that will be compatible with both the human and machine components. It is assumed that humans will serve as one or all of their functions, namely, sensing, controlling, and information processing. In the role of an information processor, humans are required to make judgments.

1021. The Steiner Tunnel Test is used to determine:

1) Sprinkler head RTI values
2) Dust deflagration (explosion) venting values
3) Extinguishing agent effectiveness
4) Material flame spread/smoke development rating

**Ans.4** The Steiner Tunnel Test is the test for flame spread and smoke developed for interior finishes such as carpeting and wall coverings.

1022. Vapors of flammable and combustible liquids are:

1) Generally heavier than air
2) Generally lighter than air
3) Normally liquid at room temperature and have molecular weights lower than those which are gases
4) Unstable and unpredictable

**Ans.1** The vapors of flammable liquids are generally heavier than air. Examples are alcohol, naptha, gasoline, kerosene, and carbon disulfide. This is logical because gas or vapor density is proportional to molecular weight, and the compounds which are normally liquids at room temperature have higher molecular weights than those which are gases. Note that because vapors of flammable and combustible liquids are heavier than air, they tend to settle in lower areas and the trail of vapors can spread far from the liquid itself. If this vapor trail contacts a source of ignition, the resulting fire can FLASHBACK to the source of the vapors.

1023. Increasing the temperature or pressure will have the following effect on the LEL and UEL:

1) Raise the LEL and lower the UEL
2) Lower the LEL and raise the UEL
3) Raise the LEL and raise the UEL
4) Lower the LEL and lower the UEL

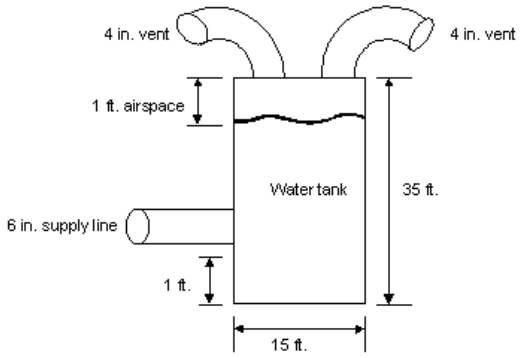
**Ans.2** Flammable or explosive limits are concentrations of a vapor or gas in air below or above which propagation of a flame does not occur on contact with a source of ignition. The lower explosive limit (LEL) is the minimum vapor concentration below which the vapor-air mixture is too lean to burn or explode. The upper explosive limit (UEL) is the maximum vapor concentration above which the mixture is too rich to burn or explode. Increasing the temperature or pressure lowers the LEL and raises the UEL.

1024. Decreasing the temperature or pressure will have the following affect on the LEL and UEL:

1) Raise the LEL and lower the UEL
2) Lower the LEL and raise the UEL
3) Raise the LEL and raise the UEL
4) Lower the LEL and lower the UEL

**Ans.1** Flammable or explosive limits are concentrations of a vapor or gas in air below or above which propagation of a flame does not occur on contact with a source of ignition. The lower explosive limit (LEL) is the minimum vapor concentration below which the vapor-air mixture is too lean to burn or explode. The upper explosive limit (UEL) is the maximum vapor concentration above which the vapor-air mixture is too rich to burn or explode. Decreasing the temperature or pressure raises the (LEL) and lowers the (UEL).

1025.




A water tank used as a secondary water supply in a small manufacturing plant is 15 ft in diameter and is 35 ft tall. The supply line has a diameter of 6 in.. There is 1 ft of air space at the top of the tank when full. Two 4 in. diameter vents are located on the top of the tank to supply make up air as the tank is drained. What is the gauge pressure in the supply line (located 1 ft from the bottom of the tank)?

1) 1.9 psi
2) 14.3 psi
3) 272 psi
4) 2,059 psi

**Ans.2** $P = 33 \text{ ft} * 62.4 \text{ lb/ft}^3$
$= 2{,}059 \text{ lb/ft}^2$
Convert $\text{lb/ft}^2$ to psi...
$P = 2{,}059 \text{ lb/ft}^2 (1 \text{ ft} / 12 \text{ in.})^2$
$= 14.3 \text{ lb/in}^2$
$= 14.3$ psi
NOTE: This is a gauge pressure

1026. The most common method for measuring flow from an open stream discharging from an orifice, nozzle, or open pipe is by direct measurement of the velocity head of the stream. This is measured by:

1) A pressure gauge
2) A pitot tube and pressure gauge
3) A pitot tube
4) A play pipe

**Ans.2** Pitot tubes in combination with pressure gauges are used to measure pressure which can then be converted into flow rates.

1027. A pitot tube reading of 30 psi was measured on an open 2.5 in. smooth rounded hydrant butt. The coefficient of friction is 0.90. The theoretical flow rate for a 30 psi velocity head is 990 gpm. What is the actual discharge in gpm?

1) 330 gpm
2) 891 gpm
3) 990 gpm
4) 1,089 gpm

**Ans.2** Q = flow in gpm = theoretical discharge * discharge coeff.
= 990 gpm * 0.90
= 891 gpm

1028. Why should acetylene cylinders be stored in an upright position?

1) To prevent liquid acetone from escaping
2) For proper securing
3) In order for the acetylene to dispense properly
4) Both 1 and 2

**Ans.4** Acetylene cylinders must always be stored in an upright position to allow proper securing and to prevent the liquid acetone solvent from escaping.

1029. Acetylene must not be used at a pressure in excess of:

1) 5 psi(g)
2) 10 psi(g)
3) 15 psi(g)
4) 20 psi(g)

**Ans.3** Acetylene must not be used at a pressure in excess of 15 psi(g), or 30 psi(a). Above this pressure, acetylene may become unstable and explode.

1030. The most common cause of indoor air quality problems is:

1) Off-gassing from building materials/furnishings
2) Cigarette smoking
3) Processes using volatile chemicals
4) Poor ventilation

**Ans.4** Poor ventilation is the most common cause of indoor air quality problems because air contaminants from sources such as cigarette smoke, processes, and off-gassing build-up can cause hypersensitivity in people. The other answers contribute to the cause, but poor ventilation is usually the root of the problem.

1031. What is the common terminology for paint that does not hit the part in painting operations?

1) Overspray
2) Sludge
3) Dragout
4) Blowdown

**Ans.1** Overspray is the terminology for the wasted paint. Water curtain paint booths or dry filter paint booths are then necessary to capture the wasted paint solids, each resulting in a difficult to manage waste. Paint booth filters that are auto-ignitable are especially difficult and expensive to manage.

1032. The primary hazard operating gasoline-powered industrial trucks in an unventilated space is:

1) Human exposure to explosions

2) Human exposure to carbon dioxide ($CO_2$)
3) Human exposure to carbon monoxide (CO)
4) Human exposure to nitrogen oxides ($NO_X$)

**Ans.3** Carbon monoxide is the primary human exposure hazard when operating gasoline-powered trucks within an enclosed space.

1033. A sound absorption coefficient of 0.6 means that _____ of the sound energy in a wave will be reflected.

1) 40%
2) 60%
3) 10(log 0.6/0.4)%
4) (0.6 log 10)%

**Ans.1** A sound absorption coefficient of 0.6 means that 40% of the sound energy in the wave will be reflected and 60% absorbed. The key word in the question was reflected, NOT absorbed.

1034. If a material has a sound absorption coefficient of 0.8, that means that _____ percent of the sound energy in the wave will be:

1) 20, absorbed
2) 10 * log (0.20/0.80), reflected
3) 80, absorbed
4) 80, reflected

**Ans.3** A sound absorption coefficient of 0.8 means that 80 % of the sound energy in the wave will be absorbed and 20% will be reflected.

1035. In a work area the noise levels are 95 dBA for 2 hr, 90 dBA for 4 hr and 85 dBA for 2 hr. Does this exceed the OSHA limits?

1) Yes
2) No
3) Can't determine from information given
4) Borderline exposure

**Ans.1** Using OSHA Allowable 90 dBA for an 8 hour period:
Allowable exposure time, $T = 8/2^{(L-90)/5}$
Therefore:
For 95 dBA, T = 4
For 90 dBA, T = 8
For 85 dBA, T = 16

Noise dose (%), $D = 100 [\text{sum } C_1/T_1]$

$D = 100 [2/4 + 4/8 + 2/16] = 100 [1.125] = 112.5\%$
This exceeds the allowable daily exposure by more than 12%.

1036. The LEAST critical factor in the analysis of noise exposure is:

1) A-weighted sound pressure level
2) Frequency spectrum of the noise
3) Duration of noise exposure
4) C-weighted sound pressure level

**Ans.4** It is important to use the A-weighted sound level, frequency duration, distribution as well as the type of noise (impact vs. continuous) in the analysis of noise exposures. In addition, you must also know the duration of exposure.

1037. What is the overall noise level to be expected when two machines are operating simultaneously in a room if the SPL due to machine X is 95 dB and the SPL due to machine Y is 99 dB ?

1) 95 dB
2) 97 dB
3) 99 dB
4) 100 dB

**Ans.** $L_{Pt} = 10 \log [\Sigma 10^{L_{pi}/10}]$

(Total Sound Pressure = 10 times the log of the sum of the 10 to the power of the individual sound pressure divided by 10)

$L_{Pt} = 10 \log [10^{95/10} + 10^{99/10}]$

$= 10 \log [3.2 * 10^9 + 7.9 * 10^9]$

$= 10 \log [1.1106 * 10^{10}]$

$= 100.45$ dB

1038. What is the permissible noise exposure in duration per hour of a 97 dBA exposure?

1) 4 hours
2) 3 hours
3) 2 hours
4) 1.5 hours

**Ans.2** 3 hours is the permissible noise exposure in duration per hour of a 97 dBA exposure.

1039. The following noise exposures were recorded:
92 dBA - 3 hours
95 dBA - 3 hours
90 dBA - 2 hours
What is the total exposure?

1) 80 dBA
2) 85 dBA

3) 90 dBA
4) 93 dBA

**Ans.4** Add all the exposure times and divide by 8 hours. We are using units of dBA.
Ex = [(92 dB * 3 hr) + (95 dB * 3 hr) + (90 dB * 2 hr)] / 8
= (276 dB-hr + 285 dB-hr + 180 dB-hr) / 8 hr
= 92.8 dBA

1040. The following noise exposures were recorded:
92 dBA - 3 hours
95 dBA - 3 hours
90 dBA - 2 hours
Is there an over exposure to noise?

1) Yes
2) No
3) Just at level
4) Slightly below

**Ans.1** Add all the exposure times and divide by 8 hours. We are using units of dBA.
Ex = [(92 dB * 3 hr) + (95 dB * 3 hr) + (90 dB * 2 hr)] / 8
= (276 dB-hr + 285 dB-hr + 180 dB-hr) / 8 hr
= 92.8 dBA
The calculated exposure of 92.6 dBA is higher than the allowable 90 dBA.

1041. The following noise exposures were recorded:

| dBA | Time |
|---|---|
| 110 | 0.5 hours |
| 90 | 2.0 hours |
| 80 | 3.0 hours |
| 85 | 2.5 hours |

Was the permissible noise exposure exceeded for an 8 hour period?

1) No (just under the limit)
2) Yes (just over the limit)
3) Yes (over the limit by 50%)
4) At limit

**Ans.** Exposure = □ [(Measured dB * Exposure Time) / (Measured dB / Allowable Exposure Time)]
Exposure = [(110 db * 0.5 hr) / (110 dB * 0.5 hr)]
+ [(90 db * 2 hr) / (90 db * 8 hr)]
+ [(80 db * 3 hr) / (80 db * 32 hr)]
+ [(85 db * 2.5 hr) / (85 dB * 16 hr)]
= 1.496
The total allowable exposure should be kept at 1.0 or lower. This one exceeds the limit by nearly 50%.
Exposure Table from 29 CFR 1910.

| Hours | dBA |
|---|---|
| 32 | 80 |
| 16 | 85 |
| 8 | 90 |
| 6 | 92 |
| 4 | 95 |
| 3 | 97 |
| 2 | 100 |
| 1 | 105 |
| 0.50 | 110 |
| 0.25 | 115 |

1042. If the variations in sound level oscillate with an interval of 1 second or less, the sound is considered:

1) Impact noise
2) Non-continuous noise
3) Continuous noise
4) An overexposure

**Ans.3** Continuous noise involves sound levels oscillating with an interval of 1 second or less.

1043. Calculate the estimated near field power density on a microwave beam axis emitted from a large circular antenna with an effective diameter of 3 m and an average power output of 200 W.

1) 11.3 mW/cm $^2$
2) 42.5 mW/cm $^2$
3) 200 mW/cm $^2$
4) 200 W/cm $^2$

**Ans.1** To solve this problem, use the near field power density formula where W is power density, P is average power, and D is the antenna diameter in centimeters. The formula is:
W = (16 P) / ( D$^2$)
3 m = 300 cm
W = (16 * 200 W) / [ (300 cm)$^2$]
= 3,200 W / 282,743 cm$^2$
= 0.0113 W/cm$^2$
= 11.3 mW/cm$^2$

1044. Calculate the estimated far field power density 6 m from a 900 MHz microwave antenna with an effective diameter of 3 m and an average power output of 200 W?

1) 0.35 mW/cm$^2$
2) 1.25 mW/cm$^2$
3) 5.40 mW/cm$^2$
4) 11.3 mW/cm$^2$

**Ans.1** To solve this problem, use the far field equation. W = (A P) / ( $^2$ r$^2$), where r is the distance from the antenna, A is the antenna effective area, P is the power output of the antenna, and  is the wavelength.
First, calculate the wavelength using the following formula. [ = c / f ] f is the frequency and c is the speed of light
= (3 * 10$^8$/(900 * 10$^6$)
= 1/(300 * 10$^{-2}$)
= 1/3 m = 100/3 cm

A = r$^2$
= (150 cm)$^2$

= 70,686 cm $^2$
W = (70,686 $cm^2$ * 200 W)/[(100/3 $cm)^2$)(600 $cm)^2$)]
= 0.0353 $W/cm^2$
= 35.34 $mW/cm^2$

1045. Which type of ionizing radiation, sometimes used in manufacturing operations, is the least penetrating?

1) Alpha
2) Beta
3) Gamma
4) X-ray

**Ans.1** Of the five kinds of radioactivity that are of general concern (e.g., alpha, beta, gamma, x-ray, and neutrons), alpha particles are the least penetrating. They will not penetrate thin barriers. For example, paper, cellophane, and even skin, will shield alpha particles.

1046. Which kind of radioactivity is considered to be a critical internal hazard?

1) Alpha
2) Beta
3) Gamma
4) X-ray

**Ans.1** Alpha particles may be inadvertently ingested, inhaled, or absorbed into the body as a result of improper handling. Internally, alpha particles are acutely hazardous because of their ionizing ability through very short distances in soft tissue. Depending on the chemical and physical form of the radioactive material, alpha particles may deposit in specific organs and produce acute radiation doses in those organs.

1047. The preferred choice of shielding material for beta radiation is:

1) Lead
2) Concrete
3) Plastic
4) Paper

**Ans.3** Beta particles are readily stopped in relatively small thicknesses of materials. The preferred choice for beta sources is low density plastic materials such as plexiglass or lucite. High density materials, like lead, are less effective because of the production of secondary X-rays when the beta particles strike lead atoms.

1048. A glass container of radium iodate powder shatters when accidentally dropped on the quarry tile floor of an experimental laboratory. What is the first action that should be taken?

1) Call the spill team for assistance
2) Vacuum the powder up off the floor quickly, using a vacuum with a HEPA filter
3) Notify the NRC
4) Evacuate the laboratory

**Ans.4** The first action to be taken following the spill of a powdered radium substance is to evacuate the area. Radium sources emit alpha particles which are easy to spread over large areas. Airborne, they present a major hazard to personnel by route of inhalation, thereby necessitating rapid evacuation from the scene of a spill.

1049. In a single year, ACGIH recommends that the occupational exposure of radiation to the whole body of non-pregnant individuals not exceed:

1) 10 mSv per calendar quarter
2) 20 mSv per calendar quarter
3) 25 mSv per calendar year
4) 50 mSv per calendar year

**Ans.4** This is the effective dose averaged over the year. The average effective dose averaged over five years should not exceed 20 mSv per year.

ACGIH TLVs and BEIs (1999)

1050. If you were milling asbestos, what effect would this have on the fibers?

1) Nothing
2) Reduce diameter
3) Reduce diameter and shorten length
4) Increase length

**Ans.3** Milling will tend to break up the fibers, resulting in both shorter and narrower fibers.

1051. In the cyanide electroplating of brass, which of the following components is frequently used?

1) Hydrogen cyanide
2) MFG
3) Cyanide salts
4) Zn salts

**Ans.3** Cyanide salts are commonly used in electroplating.

1052. Which of the following is not a primary irritant?

1) $NH_3$

2) HCl
3) $H_2S0_4$
4) $H_2S$

**Ans.4** $H_2S$, or hydrogen sulfide, is a secondary irritant since it does produce irritation, but that effect is overshadowed by its systemic effect on the body. Other secondary irritants include: ketones, esters, alcohols, and ethers. A primary irritant's action is limited to irritation.

1053. The category A1 carcinogen is:

1) Suspected human carcinogen
2) Probable human carcinogen
3) Confirmed human carcinogen
4) Not suspected as a human carcinogen

**Ans.3** Carcinogen category A1 is a confirmed human carcinogen, which means the agent is carcinogenic to humans based on the weight of evidence from epidemiological studies of, or convincing clinical evidence in, exposed humans. Category A2 is suspected human carcinogen and A5 is not suspected as a human carcinogen. Probable human carcinogen is not a carcinogen category.

1054. Acute poisoning refers to:

1) Low dose exposure over many years
2) Rapid absorption in a short period
3) Exposure to highly toxic materials over a month or less
4) Low dose exposures over a short period

**Ans.2** Acute exposures are the result of a single rapid exposure to a high dose of a material over a very short time period. Low or high doses to a toxic material over a period of days or weeks is not an acute exposure, it is a chronic exposure. Low doses over a short period do not constitute an acute exposure. An acute exposure results in immediate effects.

1055. Pneumoconiosis is a disease of the lungs that is caused by inhalation of:

1) Dusts
2) Fumes
3) Gases and vapors
4) Smoke

**Ans.1** Pneumoconiosis are a group of diseases of the lung that are caused by inhalation of dusts, such as asbestos, cotton dust, coal dust, crystalline silica, and others. Smoke, fumes, gases, and vapors can cause other diseases and effects, depending on the toxicology of the substance.

1056. Byssinosis is associated with exposure to:

1) Asbestos
2) Cotton dust
3) Ferric oxide
4) Crystalline silica

**Ans.2** Byssinosis is caused by long term exposure to cotton dust. Asbestos exposure can cause asbestosis and potentially mesothelioma (a form of lung cancer). Exposure to ferric oxide can cause siderosis. Exposure to crystalline silica can cause silicosis.

1057. Chemicals with a "C" notation with their TLV's:

1) Are carcinogenic
2) Are listed in Appendix C of the MSDS
3) Are either suspect or confirmed carcinogens
4) Have ceiling TLV limits assigned to them

**Ans.** A "C" notation denotes that the TLV is the ceiling limit for the chemical listed. This means that the assigned TLV is a maximum level that should not be exceeded at any time during the workday.

1058. The minimum lethal dosage (MLD) is defined as:

1) The largest amount that will kill the most susceptible of a group of test animals
2) The largest amount that will kill the most susceptible of a group of test people
3) The smallest amount that will kill the most susceptible of a group of test animals
4) The smallest amount that will kill the most susceptible of a group of test people

**Ans.3** The MLD is the smallest amount that will kill the most susceptible of a group of test animals.

1059. Which one of the following statements is NOT true?

1) Fog - small gas-borne particles resulting from incomplete combustion and consisting predominantly of carbon and other combustible materials
2) Mist - a term loosely applied to the dispersion of liquid particles, many of which are large enough to be individually visible without visual aid
3) Smog - a term derived from "smoke" and "fog" and applied to extensive atmospheric contamination by aerosols arising from a combination of natural and man-made sources
4) Dust - solid particles predominantly larger than colloidal and capable of temporary suspension in air or other gases

**Ans.1** Fog is a term loosely applied to visible aerosols in which the dispersed phase is liquid. Formation by condensation is implied. The definition for fog in the answer choice applies to smoke.

1060. Which of the following statements is true?

1) Gas - a substance below its critical temperature
2) Vapor - the gaseous phase of a substance above its critical temperature
3) Aerosol - a dispersion of gaseous particles of microscopic size in a liquid medium
4) Mist - dispersion of liquid particles, many of which are large enough to be individually visible without visual aid

**Ans.4** The definition of "mist" is correct.
Gas - a substance above its critical temperature

Vapor - the gaseous phase of a substance below its critical temperature
Aerosol - a dispersion of solid or liquid particles of microscopic size in a gaseous medium for instance, smoke, fog, and mist

1061. Which one of the following statements is NOT true?

1) Smoke - small gas-borne particles resulting from incomplete combustion and consisting predominantly of carbon and other combustible materials
2) Fog - a term loosely applied to visible aerosols in which the dispersed phase is liquid
3) Fume - solid particles generated by condensation from the gaseous state, generally after volatilization from melted substances and often accompanied by a chemical reaction, such as oxidation
4) Smog - a term loosely applied to solid particles predominantly larger than colloidal and capable of temporary suspension in air or other gases

**Ans.4** Smog is a term derived from "smoke" and "fog" and is applied to extensive atmospheric contamination by aerosols arising from a combination of natural and man-made sources The definition given in the answer choice applies to dust.

1062. Liquid droplets or solid particles dispersed in air, that are fine enough particle size to remain so dispersed for a period of time, are called:

1) Smoke
2) Aerosols
3) Vapors
4) Dusts

**Ans.2** Aerosols are liquid droplets or solid particles dispersed in air, that are fine enough particle size to remain so dispersed for a period of time.

1063. Suspended liquid droplets generated by condensation from the gaseous to liquid state are called:

1) Mists
2) Aerosols
3) Dusts
4) Vapors

**Ans.1** Mists are suspended liquid droplets generated by condensation from the gaseous to liquid state.

1064. A material that causes visible destruction or irreversible alterations in living tissue by chemical action at the site of contact is defined as:

1) Acidophilic
2) Corrosive
3) Reactive
4) Toxic

**Ans.2** A corrosive chemical causes visible destruction or irreversible alterations in living tissue by chemical action at the site of contact.

1065. Toxic effects may be broken down into the following major classifications:

1) Light exposure
2) Moderate exposure
3) Acute, chronic, narcosis, discomfort
4) Acute or chronic

**Ans.3** Toxic effects may be broken down into the following major classifications: acute, chronic, narcosis, and discomfort.

1066. A worker is exposed to a level of a chemical above its TLV-TWA but lower than its STEL. What is the minimum period required before the next STEL level exposure if the TLV-TWA is not violated?

1) None required
2) None required but only 4 such exposures per day
3) 60 minutes
4) 4 STEL exposure per day with each exposure separated by 60 minutes

**Ans.4** Up to four STEL exposures may occur each workday with each STEL exposure separated by at least 60 minutes.
ACGIH TLVs and BEIs

1067. You are monitoring a worker's exposure to a substance with a TLV-TWA but no published TLV-STEL or TLV-C. What is the maximal exposure allowed even with no indication of TLV-TWA violation?

1) 5 times the TLV-TWA
2) 3 times the TLV-TWA
3) 2 times the TLV-TWA
4) 1.5 times the TLV-TWA

**Ans.1** Under no circumstances should exposures exceed 5 times the TLV-TWA, provided that the TLV-TWA is not exceeded.
ACGIH TLVs and BEIs

1068. What do the following chemicals have in common: acetone, carbon disulfide, cresol, gasoline, mercury-aryl, and inorganic compounds?

1) No TLV-TWA's
2) Designated as "A2" carcinogens

3) There is a skin notation
4) TLV-TWA is higher than either the OSHA PEL and/or the NIOSH Recommended Exposure Level (REL)

**Ans.4** All of the above chemicals have TLV-TWA's above either the OSHA PEL and/or the NIOSH REL. Apply the most conservative allowable exposure.

1069. The analog to the breathing air exposure level TLV-TWA, is the biological monitoring level:

1) BEI
2) TLV
3) PD
4) BIO

**Ans.1** The biological monitoring level utilized by the ACGIH to assess worker exposure is termed a Biological Exposure Index (BEI). ACGIH has established dozens of BEI's for various hazardous chemicals.

1070. An operator of a degreaser tank noticed that the tank solvent (trichloroethylene) was leaking onto a flame under the tank. As the plant safety professional, what is your main concern of exposure?

1) Trichloroethylene
2) Phosgene
3) Flammability/explosion
4) Chlorine

**Ans.2** Trichloroethylene upon exposure to a flame, decomposes to form phosgene, a highly toxic gas. Trichloroethylene is not flammable.

1071. What is the primary hazard(s) of ethylene oxide?

1) Toxicity
2) Fire/explosion hazard
3) Carcinogen
4) 1, 2, and possibly 3

**Ans.4** Ethylene oxide is toxic and has a low flash point. Additionally it is an IARC Group 2A carcinogen (i.e., it is probably carcinogenic to humans).

1072. Which of the following does NOT carry a TLV "skin" notation?

1) Toluidine
2) 2,4,6-Trinitrotoluene (TNT)
3) Hydrazine
4) Lead

**Ans.4** All of the above chemicals, with the exception of lead, carry the "skin" notation. ACGIH TLVs and BEIs

1073. Fatalities among workers performing quenching operations on metal drosses have been caused by:

1) Arsine
2) Nitrous fumes
3) Beryllium
4) Vanadium pentoxide

**Ans.1** Most cases of arsine poisoning occur after the accidental generation of the gas in the workplace. During recent years many incidents have involved a reaction between arsenic and aluminum, with the subsequent release of hydrogen in the presence of water to permit the

formation of arsine gas. Arsine usually evolves when nascent hydrogen is generated in the presence of arsenical compounds or by the hydrolysis of a metallic arsenide in contact with water. Invariably, there is an acid medium where metal is present (e.g., dross residues, galvanized implements, aluminum tanks or implements) thereby creating the key ingredients necessary for arsine formation. Arsine generation is a concern of metal pickling operations.

1074. Which form of silica has the lowest TLV?

1) Quartz
2) Cristobalite
3) Silica, fused
4) Chrysotile

**Ans.2** The TLV for quartz and fused silica is 0.1 $mg/m^3$. The TLV for cristobalite is 0.05 $mg/m^3$. Chrysotile is a form of asbestos.
ACGIH TLVs and BEIs

1075. Which of the following irritants is not considered a primary irritant?

1) HCl
2) $NO_2$
3) $SO_2$
4) $NH_3$

**Ans.2** $NO_2$ is a secondary irritant in that it produces toxic system effects upon absorption on the lung tissue. The other chemicals are primary irritants in that they form non-toxic end products and primarily cause irritation .

1076. Which of the following chemicals are sensitizers?

1) TDl
2) MDl
3) HCl
4) Both 1 and 2

**Ans.4** Both TDI and MDI are strong sensitizers. Sensitizers cause a strong allergic reaction in sensitized individuals after initial exposure. An individual who is sensitized responds to extremely small concentrations.

1077. Which of the following gases is not a simple asphyxiant?

1) Carbon monoxide
2) Nitrogen
3) Methane
4) Argon

**Ans.1** Carbon monoxide binds with the body's hemoglobin tightly so that oxygen cannot be transported. Because of this it is considered a chemical asphyxiant.

1078. Of the following, which is FALSE?

1) A "skin" notation means that the chemical causes dermatitis
2) A mixture calculation may be used when toxic effects are similar
3) The term TWA implies that the exposure limit may be exceeded for short periods
4) A ceiling notation means that the level may not be exceeded at any time

**Ans1.** A skin notation means that is substance in question has the ability to penetrate the skin thus making the skin an entry route.

1079. Where the "skin" notation follows a substance TLV, it refers to all of the following, EXCEPT:

1) Exposure via eyes
2) Exposure via intestinal absorption
3) Exposure via mucous membranes
4) Exposure via cutaneous route

**Ans.2** The "skin" notation refers to the potential significant contribution to exposure by the cutaneous route, including mucous membranes and the eyes. While some are capable of causing irritation, dermatitis, and sensitization, these properties are not considered relevant when assigning a skin notation.

1080. Glacial acetic acid should be stored with:

1) Nitric acid
2) Flammable materials
3) Acids and bases
4) Sodium hydroxide

**Ans.2** Acetic acid has a flash point of 109ºF. This makes it a combustible liquid. Therefore the most appropriate storage area is FLAMMABLE.

1081. Storing chemicals according to their compatibility is an important factor in reducing risks. Which of the following groups of chemicals cannot be stored safely together?

1) Methyl ethyl ketone, toluene
2) Nitric acid, oils
3) Sodium hydroxide, sodium cyanide
4) 1,1,1-trichloroethane, paint thinners

**Ans.2** Nitric acid will oxidize oils and potentially cause a fire.

1082. A basic safeguard involving the storage of hazardous chemicals is:

1) Automatic sprinkler protection
2) Segregation
3) Grounding and bonding
4) Class I, Division 2 wiring

**Ans.2** Non-compatible chemicals, such as those that may react violently on mixing (such as acids with organic solvents or oxidizers with organic solvents) should be separated by a solid partition in major storage areas, or by a distance of not less than 20 feet .

1083. A worker's hand has been exposed to liquid nitrogen. What is the correct first aid?

1) Immerse hand in ice water
2) Immerse hand in warm water
3) Immerse hand in hot water
4) Keep hand in ice

**Ans.2** Per the "Accident Prevention Manual for Industrial Operations", the hand should be placed in warm water less than 115ºF. The treatment is for frostbite.

1084. In preparing the battery for a powered lifter, one must always:

1) Add acid to water
2) Add water to acid
3) Ground the lifter
4) Check tire pressure

**Ans.1** To prepare a battery "AAA" - always add acid to water to prevent splattering. (Remember: "Do as you oughtta, add acid to watta.")

1085. Typically, if a compound is not very soluble in water, but is soluble in octanol, which of the following statements would be expected to be accurate?

1) The compound is very mobile in soil
2) The compound is not very mobile in soil
3) The compound should exhibit moderate to high toxicity
4) None of the above

**Ans.2** The octanol-water partition coefficient is sometimes used to approximate the degree to which a chemical, when released, will tend to migrate (partition) between soil and water.

1086. What classification level is the most hazardous for flammable aerosols?

1) Level 1
2) Level 2
3) Level 3
4) Level4

**Ans.3** Level 1 aerosols are defined as producing less than 8,600 Btu/lb of aerosol. Level 2 aerosols produce 8,600-13,000 Btu/lb and Level 3 aerosols produce more than 13,000 Btu/lb. Level 4 aerosols do not exist.

1087. In the identification of fire class occupancies, Class II indicates the presence of:

1) Combustible dusts
2) Flammable vapors
3) Ordinary combustibles
4) Ignitable fibers

**Ans.1** Class II occupancies are locations where combustible dust is, or may be, in suspension in the air continuously, intermittently, or periodically. Hazardous dusts are separated into

groups: Group E for metal dust, Group F for carbon black, coal or coke dust, and Group G for flour, starch, or grain dust. NFPA 70 Article 500-3 (National Electric Code)

1088. The exit design requirements in the NFPA 101 Life Safety Code are based on classification of the occupancy. These classifications are dependent on the burning characteristics of:

1) Contents of the building
2) Quantity of combustible materials
3) Type of building construction
4) The type of flooring materials

**Ans.1** The contents are evaluated to determine the rate of fire spread, development of products of combustion, and components that affect the time available for evacuating the building. NFPA 101 6.2 (Life Safety Code)

1089. A Class I, Division I hazard includes all of the following EXCEPT?

1) A closed-loop system containing a flammable solvent
2) An open vat containing a flammable solvent
3) A cleaning operation involving a flammable solvent
4) A degreasing tank with a flammable solvent

**Ans.1** Class I, Division I systems involve operations where flammable solvent vapors would generally be present. One would not expect flammable vapors to be normally present in a closed-loop system. NFPA 70 Article 500-5 (National Electric Code)

1090. A Division 2 location is one where:

1) Combustible materials are always present
2) Combustible materials are not normally present
3) Combustible materials are never present
4) Combustible materials are stored near an ignition source

**Ans.2** A Division 1 location is expected to have combustible materials present while a Division 2 location should only have such materials present in emergencies or other unusual circumstances. NFPA 70 Article 500-5 (National Electric Code)

1091. A flammable liquid with a flash point of 72ºF and a boiling point below 100ºF would be classified by the NFPA as a:

1) Class I liquid
2) Class IA liquid
3) Class IB liquid
4) Class II liquid

**Ans.2** A Class I liquid refers to any flammable with a flashpoint below 100ºF. A Class IA liquid has a flashpoint below 73ºF and a boiling point below 100ºF. A Class IB liquid has a flash point below 73ºF and a boiling point at or above 100ºF. A Class II liquid has a flash point at or above 100ºF and a boiling point below 140ºF. NFPA 30 Flammable and Combustible Liquids Code

1092. "Means of Egress" as defined in the NFPA Life Safety Code includes all of the following EXCEPT:

1) Exit access
2) Exit doorway
3) Exit discharge
4) Exit

**Ans.2** NFPA 101 (Life Safety Code) states that a means of egress involves the exit access, exit, and exit discharge. These components of the means of egress are further defined in NFPA's Life Safety Code Handbook. NFPA 101 3.3.121 (Life Safety Code)

1093. Which of the following is among NFPA's classifications of hazard of contents for a building?

1) Conventional hazard
2) Ordinary hazard
3) Nominal hazard
4) Minimal hazard

**Ans.2** Of these, NFPA defines "Ordinary Hazard" as one of the classifications of the hazard of contents in a building. The other two classifications are "Low Hazard" and "High Hazard." NFPA 101 6.2.2 (Life Safety Code)

1094. The National Electrical Code utilizes a Class I, II, and III hazard identification system. Class II hazards involve:

1) Flammable liquids
2) Combustible dusts
3) Combustible fibers
4) Combustible powders

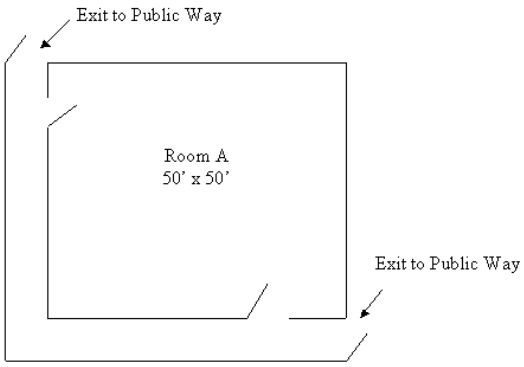
**Ans.2** The NEC Class I hazard category includes flammable liquids or gases, Class II includes combustible dusts, Class III involves fibers. NFPA 70 Article 500-6 (National Electric Code)

1095. Approved for hazardous (classified) location means?

1) Equipment is approved for the class, division, and group of location
2) There are four types of equipment specifically designed for hazardous (classified) locations
3) There are three types of equipment specifically designed for hazardous (classified) locations
4) Equipment need not be approved for the class, division and group of location

**Ans.1** Equipment must be approved for the class, division, and group of location. There are two types of equipment specifically designed for hazardous (classified) locations - explosionproof and dust-ignition-proof. NFPA 70 Article 500-2 (National Electric Code)

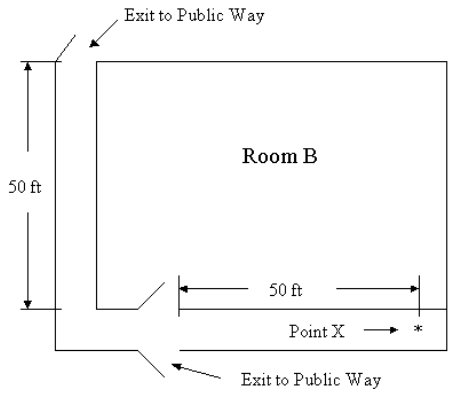
1096. Consider the following floor plan, and estimate the common path of travel from Room A (assume that Room A is essentially empty)?




1) 0 feet
2) 50 feet
3) 100 feet
4) 250 feet

**Ans.1** The common path of travel is the maximum distance one has to travel in order to have access to two or more exits. In this case, Room A already has two valid exits; therefore, the common path of travel is 0 feet.

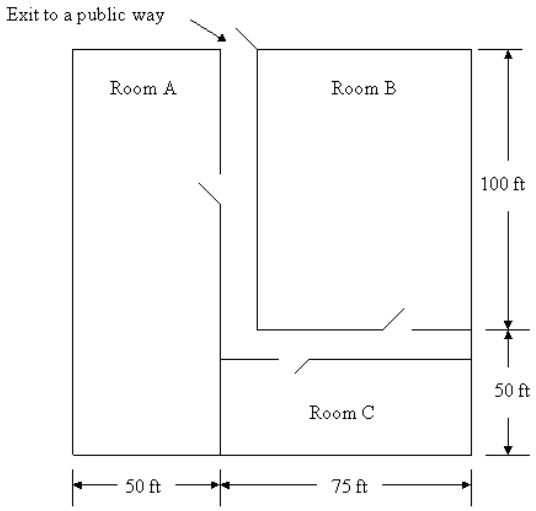
1097. Consider the following floor plan, and estimate the common path of travel from Point X.




1) 0 feet
2) 50 feet
3) 100 feet
4) 250 feet

**Ans.2** The common path of travel is the maximum distance one has to travel in order to have access to two or more exits. In this case, one has to travel the length of this dead-end corridor in order to have access to two different exits.

1098. Consider the following floor plan, and estimate the maximum travel distance in this floor plan (assume that all of the rooms are essentially empty).




1) 50 feet
2) 100 feet
3) 250 feet
4) 450 feet

**Ans.3** The maximum travel distance is the maximum distance one could travel to pass through the nearest exit discharging to a public way. In this case, the worst-case scenario is in the back of Room B. Assume that the furthest point is near the upper right corner of Room B. An estimate of the travel distance is 100 feet (to the door from Room B) plus about 50 feet to the corner of the corridor plus 100 feet to the exit.

1099. What is an example of an unfired pressure vessel?

1) Boiler
2) Steam jacketed kettle
3) Any vessel directly experiencing flame
4) Copper jacketed reactor

**Ans.2** An unfired pressure vessel does not have a direct flame impacting it. A steam jacketed kettle is one example.

1100. Which floor surface presents the most resistance to a slip/fall accident?

1) Steel
2) Granite
3) Creosote wood
4) Tile

**Ans.3** Creosote wood flooring would provide the most resistance to a slip/fall accident.

1101. Which natural fiber makes the safest rope?

1) Manila
2) Sisal
3) Cotton
4) Jute

**Ans.1** Per the "Accident Prevention Manual", manila rope is the best suited natural fiber rope for slings and capstans.

1102. Use the following data: 8 ft deep, unshored trench; floor of trench is 6 ft wide; Type A soil (sloping requirement for Type A soil is 3/4 horizontal to 1 vertical). What is the minimum acceptable width of the trench at grade level if the trench is properly sloped?

1) 12 ft
2) 18 ft
3) 21 ft
4) 24 ft

**Ans.2** Draw a sketch. For every foot in depth, the slope must be cut 3/4 ft horizontally. That means for 8 ft in depth, the slope must be cut 6 ft wide. Since this has to be done for both sides of the trench, the width must be 18 ft; (6 ft + 6 ft + 6 ft).

1103. What is not an acceptable method of controlling static discharge?

1) Grounding
2) Bonding
3) Conductive flooring
4) Open circuits

**Ans.4** Open circuits are not an acceptable method of controlling static discharge.

1104. If a wire is 200 ft long and has a resistance of 1.5 □ and a diameter of 2 mm, what will the resistance be if the length is 140 ft and the diameter is 2.5 mm?

1) 0.233 O
2) 0.323 O
3) 0.672 O
4) 0.840 O

**Ans.3** Use the formula: $R_2 = R_1(L_2 / L_1) (D_1^2 / D_2^2)$

$R_2 = R_1$ (140 ft/200 ft) $[(2 \text{ mm})^2/(2.5 \text{ mm})^2]$
$R_2 = 1.5$ □ * 0.7 * 0.64
$R_2 = 0.672$ □
Remember that the cross-sectional area of a round conductor changes with the square of the diameter.

1105. Which of the following substances is the best conductor?

1) Aluminum
2) Copper
3) Silver
4) Nickel

**Ans.3** Silver has the lowest resistivity at $1.47 \times 10^{-8}$ ohm-m.

1106. The resistance of a wire varies with all of the following EXCEPT:

1) Temperature
2) Length
3) Diameter
4) Nuclear orientation

**Ans.4** The resistance of a material varies with its temperature, length and diameter. Nuclear orientation has no effect on resistance.

1107. As the diameter of a wire increases, the resistance of the wire:

1) Increases
2) Decreases
3) Stays the same
4) Increases as the square of the cross-sectional area

**Ans.2** As the diameter of a wire increases, its resistance decreases.

1108. What size solid copper and aluminum conductors are used with a 110 V, 20 A electrical circuit?

1) 12 gauge for both
2) 14 gauge for both
3) 14 gauge for copper, aluminum cannot be used
4) 14 gauge for copper, 12 gauge for aluminum

**Ans.4** Aluminum conductors are not as efficient as copper so they must be larger. NFPA 70 Article 310-5 (National Electric Code)

1109. Class III, Division 1 electrical protection is provided in a plant processing:

1) Sugar
2) Cotton
3) Starch
4) Paint

**Ans.2** Class III, Division 1 locations are hazardous because of the presence of easily ignitable fibers or flyings in processing cotton. NFPA 70 Article 500-7 (National Electric Code)

1110. Any of the following are major cause(s) of electrical shock EXCEPT:

1) Ungrounded frames or inadequate grounding of frames on noncurrent-carrying parts of electrical machinery and tools
2) Inadequate protection of exposed current-carrying parts of electrical installations
3) Inadequate insulation of electrical conductors
4) GFCI devices

**Ans.4** GFCI stands for ground fault circuit interrupter. These devices prevent electrical shocks by opening the circuit upon detecting a fault in the circuit. The other answer choices are the three major causes of electrical shock as reported by OSHA in their student manual "Guide to Voluntary Compliance".

1111. What fundamental precaution should be taken during an electrical storm?

1) Stand perfectly still
2) Stand by a metal fence
3) Seek shelter within a grounded metal building
4) Seek shelter within a non-grounded non-metal building

**Ans.3** Precautions should be taken during an electrical storm, including seeking shelter within a grounded metal building.

1112. To control static discharge the following should be performed:

1) Lower relative humidity below 65%
2) Raise relative humidity above 65%
3) Lower relative humidity below 30%
4) Maintain relative humidity above between 10% and 60%

**Ans.2** To control static discharge one should raise relative humidity above 65%.

1113. Assume a fan operates at 81% of maximum efficiency and delivers 6,000 cfm of air against a total pressure of 2 in. $H_2O$. What horsepower is delivered?

1) 0.47 hp
2) 0.94 hp
3) 1.06 hp
4) 1.62 hp

**Ans.4** Q = Flow (cfm)
TPf = Total pressure (in. $H_2O$)
Max flow = 6,000 cfm / 0.81 = 7,407 cfm

AHP = Q (TPf/Max Flow)
= 6,000 cfm(2 in. $H_2O$ / 7,407 cfm)
= 1.62 hp

1114. In the application of principles of guarding, what does zero mechanical state (ZMS) mean?

1) All energy sources neutralized
2) All electrical energy off
3) Guards are not effective
4) All electrical energy locked and tagged out

**Ans.1** ZMS is defined in ANSI Z-241.1 as the state with all energy sources neutralized.

1115. Gas supply hoses for temporary heaters should have a burst pressure of:

1) 500 psi(g)
2) 750 psi(g)
3) 1,000 psi(g)
4) 1,250 psi(g)

**Ans.4** The burst pressure rating is 1,250 psi(g).

1116. The work rest on a grinding wheel should be no more than _____ away from the wheel.

1) 1/8 in
2) 1/2 in
3) 3/4 in
4) 1 in

**Ans.1** A work rest should be, at most, 1/8 in. away from wheel. This is necessary to prevent the object being ground from being caught between the wheel, possibly injuring the operator. 29 CFR 1919.215 (a)(4)

1117. Which of the following is not a type of compressed gas cylinder relief device?

1) Spring-loaded
2) Frangible disc
3) Fusible metal
4) Spring-loaded backed up by fusible metal

**Ans.4** All of the answer choices are allowed except, spring-loaded backed up by fusible metal. A frangible disc is the only relieving mechanism that is permitted to be backed up by a fusible metal device.

1118. Of the four gases shown below, which would cause the most problems with a cylinder valve assembly?

1) Nitrogen
2) Arsine
3) Hydrogen chloride
4) Air

**Ans.3** Hydrogen chloride (HCl) gas is highly corrosive and can cause problems with cylinder valves.

1119. How much horsepower is expended by a gasoline engine driving a winch used to raise a 2,200 lb fixture a distance of 20 ft in 10 sec?

1) 0.13 hp
2) 1.30 hp
3) 4.00 hp
4) 8.00 hp

**Ans.4** Horsepower is obtained from the formula:
Work = Force * Distance
W = 2,200 lb * 20 ft = 44,000 ft-lb
Power = Work/time
P = 44,000 ft-lb / 10 sec
P = 4,400 ft-lb/sec
Converting ft-lb/sec to hp:
P(hp) = (4,400 ft-lb/sec) / [550(ft-lb/sec)/hp]
P = 8.00 hp

1120. What effort must be exerted by a material handler who is required to push a 200 lb barrel up a 10 ft long inclined plank on to a dock that is 2 ft above the ground? (Neglect friction.)

1) 40 lb
2) 80 lb
3) 120 lb
4) 160 lb

**Ans.1** The work done in pushing a barrel up an inclined plane is equal to the resistive load multiplied by the height of the plane:
Length of Incline * Effort = Resistive Load * Height of Incline
10 ft * Effort = 200 lb * 2 ft
Effort = (200 lb * 2 ft) / 10 ft = 40 lb

1121. A wheelbarrow contains a load of 216 lb at an average distance of 1.50 ft from the axle. If the shaft of the wheelbarrow is 4 ft long, what force will be needed to raise the handles of the wheelbarrow?

1) 27 lb
2) 54 lb
3) 81 lb
4) 108 lb

**Ans.3** The moment of the 216 lb load is: 216 lb * 1.50 ft = 324 ft-lb
This force tends to turn the shaft in the clockwise direction.
The unknown force (f) is applied to the handles 4 ft from the fulcrum.
324 ft-lb = 4 ft * f
f = 324 ft-lb / 4 ft f = 81 lb

1122. Given a load of 2,000 lb and a two-leg sling making an angle of 30º with the horizontal, what is the tension of each leg in the sling?

1) 1,000 lb
2) 1,155 lb
3) 2,000 lb
4) 5,000 lb

**Ans.3** Divide the total load by 2 because this is a two-leg sling. Use the trigonometric identity for sine. The hypotenuse leg represents the tension in the actual sling leg.
sin 30º = 1,000 lb / hypotenuse leg length = 2,000 lb

1123. A box weighing 500 lb is to be pushed up a ramp with a 5% grade. If the coefficient of friction is 0.15, what is the force required to push the box up the ramp?

1) 100 lb
2) 250 lb
3) 500 lb
4) 1,000 lb

**Ans.1** 5% grade = $\tan^{-1}$ 0.05 = 2.86 º
cos 2.86º = resultant force / 500 lb
r = (0.9988) 500 lb
r = 499.4 lb
Calculate the friction force component:
friction force = f r
= (0.15)(499.4 lb)
= 75 lb
Calculate the dynamic force component:
sin 2.86º = d / 500 lb
d = 0.0499 * 500 lb
d = 25 lb
total force = dynamic force + friction force
total force = 25 lb + 75 lb
= 100 lb

1124. Hazardous chemical drums stacked 3-high limit response to a leak from one of the drums because of all of the following EXCEPT:

1) Leaks may not be accessible
2) Falling drums may aggravate the situation
3) Dispersion of sprinkler water to lower drums may be limited
4) Containment of spill on the ground will be hampered

**Ans.4** Containment will not be a factor since any leaks will fall to the ground where they can ultimately be contained. Containment at the source of the leak will be a problem in this situation.

1125. Wire rope is used widely instead of fiber rope because:

1) It has greater strength for the same diameter
2) Its strength is constant when wet or dry
3) It is not as durable
4) Both 1 and 2

**Ans.4** Wire rope is composed of wires, strands, and core. The wires are drawn to a predetermined size and laid together in various arrangements helically formed around the core. In general, the greater the number of wires in a strand and the greater the number of strands, the more flexible the rope.

1126. Of the following, which is the best source of power for forklifts used in a warehouse operation?

1) Electric
2) LPG
3) Gasoline
4) Either 2 or 3

**Ans.1** An electric forklift produces no emissions. This is an important consideration in an enclosed warehouse.

1127. You are asked to reduce the emissions of gasoline-powered forklifts used in a warehouse. What is your best option given the fact that you cannot afford to purchase electrically-powered trucks?

1) Convert the existing forklifts to diesel-power
2) Convert the existing forklifts to LP-gas-power
3) Provide warehouse employees with air-purifying respirators
4) Increase the warehouse dilution ventilation

**Ans.2** Of the above options, converting the gasoline-powered forklifts to LP-gas-power is the best. Conversions must be handled in accordance with the following standards: 29 CFR 1910.178 (d) 29 CFR 1910.178 (q)(12)

1128. Which of the following is not one of the common types of hitches used to secure a sling to a load?

1) Vertical
2) Horizontal
3) Choker
4) Basket

**Ans.2** Vertical, choker, and basket are types of hitches used to secure slings to loads lifted by hoists or cranes. There is no such thing as a horizontal hitch.

1129. A table of rated capacities for wire rope slings generally lists lifting capacities for different:

1) Sling lengths
2) Load angle factors
3) Sling ages
4) End fittings and hitch types

**Ans.4** Load rating tables for wire rope slings almost always list the weight which can be lifted for different diameter ropes by the end fittings and hitch type to be utilized. 29 CFR 1910.184 (f)

1130. Decreasing the angle between a sling and the load increases:

1) The weight which can be lifted
2) The effective weight of the load
3) The time it takes to make the load
4) The system overburden

**Ans.2** Decreasing the "load angle" increases the effective weight of the load, thereby increasing the capacity required to lift the load. For example, more weight can be lifted by a sling if the sling angle is 60º versus 45º. Refer to load angle factor tables , or perform the calculations yourself using trigonometry.

1131. If a uniformly distributed 5,000 lb load is to be lifted by a 2-leg bridle hitch with each leg forming an angle of 45º with the load, what is the load on each leg of the sling?

1) 1,768 lb
2) 2,500 lb
3) 3,535 lb
4) 5,000 lb

**Ans.3** First, divide the total weight to be lifted by the number of legs on the sling or hitch. If the load is not uniform, then the exact weight on the leg lifting more would have to be calculated. In this case the load is uniformly distributed so:
5,000 lb / 2 (since there are two legs) = 2,500 lb
Then calculate the effective weight on each leg by multiplying the weight on each leg by the load angle factor.
2,500 lb / sin 45º = 3,535 lb

1132. If an analytical procedure requires 4 mg of a substance having a TLV of 100 ppm and a molecular weight of 156, what is the minimum time required to collect one sample at the TLV if the sampling rate is 1.5 lpm?

1) 31.32 min
2) 10.59 min
3) 8.34 min
4) 4.17 min

**Ans.4** First determine $mg/m^3$:
$mg/m^3$ = (100 ppm * 156)/24.45
= 638 $mg/m^3$ or 0.638 mg/l
0.638 mg/l * 1.5 lpm = 0.96 mg/min
4 mg/0.96 mg/min = 4.17 min

1133. If 100 ml of methylene chloride is spilled into an unventilated room 4 m * 4 m * 10 m, what would the concentration be after it has all evaporated? (SG of methyl chloride is 1.336 and MW is 85)

1) 150 ppm
2) 100 ppm
3) 220 ppm
4) 240 ppm

**Ans.4** Total volume of room = 4 m * 4 m * 10 m = 160 $m^3$
100 ml * 1.336 g/ml = 133.6 g = 133,600 mg
133,600 mg/160 $m^3$ = 835 $mg/m^3$
Converting $mg/m^3$ to ppm:
(835 $mg/m^3$ * 24.45) / 85 = 240 ppm

1134. How much acetone would have to be evaporated into a room15 ft * 12 ft * 10 ft to give a final concentration of 10 ppm at STP? (SG of acetone is 0.79 and MW is 58)

1) 1.53 ml
2) 1.01 ml
3) 1.95 ml
4) 1.33 ml

**Ans.1** 15 ft * 12 ft * 10 ft = 1,800 $ft^3$
1,800 $ft^3$ / (35.3 $ft^3/m^3$) = 51 $m^3$
$mg/m^3$ = (ppm * MW) / 24.45
= (10 ppm * 58) / 24.45
= 23.7$mg/m^3$
23.7 $mg/m^3$ * 51 $m^3$ = 1208.7 mg = 1.21 g
1.21 g/0.79 g/ml = 1.53 ml

1135. Workplace air at 120ºF was sampled and 70 mg of methylchloroform was found in an 850 liter sample. What would the concentration have been at normal temperature and pressure (NTP)? (MW of methylchloroform is 133)

1) 16.3 ppm
2) 21.4 ppm
3) 25.4 ppm
4) 30.1 ppm

**Ans.1** 70 mg/0.85 $m^3$ = 82 $mg/m^3$
ppm = [(82 $mg/m^3$ * 24.45)/133] 322° K/298° K
= 16.3 ppm
Remember, for this conversion:
T = 25ºC = 298° K
T = 120ºF = 322° K

1136. Which of the following is not a chemical asphyxiant?

1) Aniline
2) Sodium nitrite
3) Hydrogen cyanide
4) Argon

**Ans.4** Argon is a simple asphyxiant. The other chemicals are all chemical asphyxiants.

1137. What would the resulting acetone vapor concentration be if 2 g of acetone were allowed to evaporate in a chamber 5 ft * 10 ft * 7 ft? (MW of acetone is 58)

1) 55 ppm
2) 65 ppm
3) 75 ppm
4) 85 ppm

**Ans.4** 5 ft * 10 ft * 7 ft = 350 $ft^3$
350 $ft^3$/35.3 $ft^3/m^3$ = 9.92 $m^3$
2 g = 2,000 mg
2,000 mg/9.92 $m^3$ = 20l.6 $mg/m^3$
Converting $mg/m^3$ to ppm
ppm = (201.6 $mg/m^3$ * 24.45) / 58
= 85 ppm

1138. What volume would 100 $cm^3$ of carbon monoxide occupy at 70ºF and 1 atm if the temperature and pressure were changed to 150ºF and 2 atm?

1) 100 $cm^3$
2) 65 $cm^3$
3) 57.7 $cm^3$
4) 52.9 $cm^3$

**Ans.3** This is the equation we will use.
$(P_1V_1) / T_1 = (P_2V_2) / T_2$
Perform these temperature conversions first.
70ºF = 21ºC = 294° K
150ºF = 66ºC = 339° K
Once the conversions are performed, rearrange the equation above to solve for $V_1$.
$V_1 = (P_2V_2T_1) / (P_1T_2)$
= (1 atm * 100 $cm^3$ * 339° K) / (2 atm * 294° K)
= 57.7 $cm^3$

1139. A worker is exposed to xylene during her 8-hr work day in the following concentrations: 2 hr at 100 ppm; 2 hr at 150 ppm; and 4 hr at 50 ppm. What is her time-weighted-average (TWA) exposure? (MW of xylene is 106)

1) 50 ppm
2) 56 ppm
3) 77 ppm
4) 88 ppm

**Ans.4** TWA = [(2 hr * 100 ppm) + (2 hr * 150 ppm) + (4 hr * 50 ppm)] / 8 hr
TWA = (200 hr-ppm + 300 hr-ppm + 200 hr-ppm) / 8 hr
TWA = 87.5 ppm

1140. What volume of phosphorus oxychloride would be needed to attain a concentration of 25 ppm in a 50 $ft^3$ container? (SG of phosphorus oxychloride is 1.675 and MW is 153)

1) 2.9 ml
2) 0.13 ml
3) 1.3 ml
4) 117 ml

**Ans.2** First convert $ft^3$ to $m^3$
50 $ft^3$/35.3 $ft^3/m^3$=1.4 $m^3$
Determine $mg/m^3$ by converting from ppm
$mg/m^3$ = (25 ppm * 153) / 24.45 = 156.4 $mg/m^3$
156.4 $mg/m^3$ * 1.4 $m^3$ = 219 mg = 0.219 g
Divide by specific gravity to get volume.
0.219 g / 1.675 g/ml = 0.13 ml

1141. What concentration of xylene would be produced using a permeation tube if the tube weighed 114.123 g at the start and 113.997 g after 5.8 hr if the flow rate over the tube was 92.3 lpm?

1) 3.92 mg/$m^3$
2) 30.9 mg/$m^3$
3) 100 mg/$m^3$
4) 4.5 mg/$m^3$

**Ans.1** Determine total grams collected:
114.123 g - 113.997 g = 0.126 g = 126 mg
Total sample:
5.8 hr * 92.3 lpm * 60 min/hr = 32,120 l or 32.1 $m^3$
126 mg/32.1 $m^3$ = 3.92 mg/$m^3$

1142. How much acetone would have to be evaporated in a room 25 ft x 25 ft x 25 ft to give a final concentration of 25 ppm at STP? (SG of acetone is 0.79 and MW is 58)

1) 25.6 ml
2) 27.9 ml
3) 33.2 ml
4) 45.7 ml

**Ans.3** First determine room volume:
25 ft * 25 ft * 25 ft = 15,625 $ft^3$
15,625 $ft^3$/35.3 $ft^3/m^3$ = 442.6 $m^3$
Determine mg/$m^3$ equivalent of 25 ppm
mg/$m^3$ = (25 ppm * 58) / 24.45 = 59.3 mg/$m^3$
59.3 mg/$m^3$ * 442.6 $m^3$ = 26,246 mg = 26.246 g
Divide by the specific gravity to obtain volume.
26.246 g / 0.79g/ml = 33.2 ml

1143. Assuming perfect mixing of dilution air with contaminated air, how long will it take to dilute 500 ppm of contaminant to 1 ppm assuming a dilution rate of 230 cfm and a room size of 1,000 $ft^3$?

1) 40 min
2) 36 min
3) 35 min
4) 27 min

**Ans.4** Using the equation:
$\ln (C_2/C_1) = -Q/V (t_2-t_1)$
then:
$t = -V/Q [\ln (C_2/C_1)]$
t = time to dilution (min), $C_2$= Concentration after t min (ppm), $C_1$= Initial concentration (ppm), Q = dilution rate (cfm), and V = Room volume ($ft^3$)

t = - 1000/230 [ln (1/500)]
t = -4.35 [ -6.21]
t = 27.02

1144. Assuming perfect mixing of dilution air with contaminated air, how long will it take to dilute 10,000 ppm of contaminated air to 1 ppm assuming a dilution rate of 230 cfm and a small office with dimensions of 10 ft x 10 ft x 10 ft?

1) 27 min
2) 33 min
3) 40 min
4) 47 min

**Ans.3** Using the equation t = [ln (Co/C)] / (Q/V)
where t = time to dilution (min), C = Concentration after t min (ppm), Co = Initial concentration (ppm), Q = dilution rate (cfm), and V = Room volume ($ft^3$)
t = [In (10,000 ppm/1 ppm]/(230 cfm/1000 $ft^3$)
t = 9.21/.23 min
t = 40 min

1145. What gas concentration results when 5 ml of dry phosphorus pentafluoride is injected by gas syringe into a 13 gallon glass carboy filled with air? (MW of dry phosphorus pentafluoride is 126).

1) 50 ppm
2) 100 ppm
3) 102 ppm
4) 107 ppm

**Ans.3** Use the gas laws. 126 g / 24.45 l = 5.15 g/l = 5.15 mg/ml
Concentration * volume = amount
5.15 mg/ml * 5 ml = 25.75 mg
Next convert gallons to $m^3$
13 gal * 3.785 l/gal = 49.205 l = 0.049205 $m^3$
Then convert mg/$m^3$ to ppm
(25.75 mg/0.049205 $m^3$) 24.45) / 126 = 102 ppm

1146. A welder was exposed to airborne lead at a concentration of 270 □g/$m^3$ for 75 min while burning bolts, 40 □g/$m^3$ for 3 hr 19 min when welding painted steel, and 600 □g/$m^3$ for 30 min when using an abrasive blaster to remove lead-containing paint. The balance of his 8-hour day was spent in an environment with an airborne lead concentration of 1.5 □g/$m^3$. What was his TWA exposure to lead?

1) 43.4 □g/m
2) 47.9 □g/m
3) 96.8 □g/m
4) 100.5 □g/m

**Ans.** Set up the following table:

| JOB | C * t = TWA |
|---|---|
| Burning | 270.0 * 75 = 20,250 |
| Welding | 40.0 * 199 =7,960 |
| Blasting | 600.00 * 30 = 18,000 |
| Other | 1.5 * 176 = 264 |
| Total | 46,474 |

8 hr = 480 min
TWA = (46,474 (□g/$m^3$)(min) / 480 min = 96.8 □g/$m^3$

1147. A gas mixture is 75% methane and 25% propane by volume. The respective LEL's are 5% and 2.1%. What is the LEL (in air) for the gas mixture?

1) 2.5%
2) 3.2%
3) 4.5%
4) 3.7%

**Ans.4** Use LeChatelier's Law. LEL(mixture) = 100 / [(75/5) + (25/2.1)] = 3.7%
Think of this as a weighted average of the LEL's.

1148. A gas mixture is 75% methane and 25% propane by volume. The respective UEL's are 15% and 9.5%. What is the UEL (in air) for the mixture?

1) 10.7%
2) 11.5%
3) 12.3%
4) 13.2%

**Ans.4** Use LeChatelier's Law. UEL(mixture) = 100 / [(75/15 + (25/9.5)] = 13.2%
Think of this as a weighted average of the UEL's.

1149. Most asbestos work involves which type of fiber?

1) Chrysotile
2) Amosite
3) Crocidolite
4) Cristobalite

**Ans.1** Chrysotile is the most common form of asbestos.

1150. In the scrubbing of a biosafety cabinet effluent, what is the most universal scrubbing technology utilized?

1) Entry of effluent into chemical scrubber

2) HEPA filtration
3) Incineration
4) Chemical sanitation

**Ans.2** HEPA filtration of biosafety cabinet effluent is the most common technology utilized. Chemical scrubbing may be required if chemicals are used in the cabinet. Incineration is used only for the most toxic pathogens.

1151. What is the recommended safety factor for manila rope in a block and tackle assembly?

1) 1
2) 5
3) 10
4) 20

**Ans.3** Per the "Accident Prevention Manual for Industrial Operation" the safety factor is 10 due to the potential for shock.

1152. A pistol grip power tool is best suited for:

1) Application on a vertical plane
2) Application on a horizontal plane
3) Assembly with screws
4) Assembly of small parts

**Ans.1** A pistol grip used on a vertical plane, such as a wall, allows the wrist to remain in the neutral position.

1153. Which of the following engineering controls would be the most appropriate for a worker exposed in a steel mill to air temperatures exceeding 105ºF?

1) Use of increased general ventilation
2) Use of cooling fans

3) Provision of cooling fans
4) Application of radiant heat shielding

**Ans.4** Although increasing general ventilation, using cooling fans, or providing spot cooling would result in some improvement, the application of radiant heat shielding is more appropriate. A steel mill (or forge shop, foundry, glass manufacturing plant) is a hot-dry industry, where the thermal load on the worker is increased by the sensible heat given off by the process equipment. The major heat source is radiant heat from molten metals and cherry red ingots, while that of the surrounding hot air is secondary. Radiant heat has no appreciable heating effect on the air it passes through, but its energy is absorbed by any object it strikes. Therefore, the application of a shield to either absorb or reflect the radiation from the heat source would be appropriate.

1154. When designing an ergonomically desirable work place, which of the following is a good design practice?

1) Allow room for employees to lean forward or to the side to reach objects
2) Design the work so that employees can move about and use different muscle groups
3) Place objects within close reach to discourage excessive movement
4) Design work which requires the arms to be held up or out

**Ans.2** Allowing employees freedom of movement and the use of different muscle groups helps prevent overuse of any one muscle group. Leaning, restricted movement, and extension of arms all can cause injury over the long-term.

1155. Work surface height for standing work places should be designed as follows for precision work:

1) Below wrist height with padded rests for elbows
2) Shoulder height with a back rest

3) Above elbow height with padded forearm rests
4) Standing work places should be discouraged

**Ans.3** The work should be placed above elbow height for best posture and provide padded forearm rests to allow resting of arms on surface. Work surfaces below elbow height may cause arm fatigue, and shoulder height work will cause back and shoulder fatigue. Standing work places can be well-designed to allow for low stress work with low fatigue. Adjustability will allow the employees to determine the best height for their needs.

1156. Manipulative tasks are highly repetitive and can cause an overload of muscles and joints. All of the following are good design guidelines for manipulative arm and hand work EXCEPT:

1) Keep movements of arms and hands well within the full range of motion of body joints
2) Prevent body joints from exercising the extremes of their ranges of motion
3) Keep elbows at an angle of 60º at or above mid-torso height
4) Keep wrists straight

**Ans.3** Elbows should be kept at 90º and below mid-torso height for maximum comfort and least strain on the muscles and tendons. Keeping body joints within the range of motion and away from extremes reduces potential for strains and sprains. Straight wrists will prevent repetitive motion injuries.

1157. Methods of minimizing rates of repetition in jobs include the following EXCEPT:

1) Dividing jobs to reduce the number of job elements
2) Allowing alternate use of limbs
3) Allowing self-pacing
4) Allowing frequent rotation between jobs which use different postures and muscles

**Ans.1** The goal is to increase the number of job elements to reduce the repetitiveness of the task. This can only be done by combining jobs, not dividing them. The other answers help to reduce repetition through allowing change and alternatives to using the same muscles continuously.

1158. Manual materials handling tasks can be designed to avoid injuries. Modifications to the object to be handled may include all of the following EXCEPT:

1) Designing objects so the arms can be easily extended around it
2) Installing handles or hand holds wherever possible
3) Avoiding lopsided weights and loose contents
4) Using hoists or other lift assists when object weight is excessive

**Ans.1** Extension of the arms around objects should be avoided. Alternatively, use handles, lifting devices, and even weighting to improve the task.

1159. Manual materials handling tasks can be designed to avoid injuries. Modifications to the task may include all of the following EXCEPT:

1) Keeping loads as close to the body as possible during lifting
2) Keeping loads off the floor by using platforms or stands
3) Keeping all repetitive lifts between knee and waist height
4) Avoiding twisting, stretching, or leaning with the load

**Ans.3** Repetitive lifts should be kept between knuckle and shoulder height to reduce bending and reaching for the load. Keeping the loads close to the body, off the floor, and avoiding twisting, stretching, or leaning will help avoid the musculoskeletal, crushing, or falling injuries frequently caused by manual materials handling.

1160. When attempting to reduce ergonomic injuries, which of the following actions is the most proactive?

1) Design reviews
2) Injury/illness data reviews
3) Medical surveillance
4) Audits/assessments

**Ans.1** Design reviews allow for ergonomic issues to be identified and eliminated before workers are exposed, which is the most proactive step. Reviewing injury/illness data, conducting medical surveillance, and performing audits are actions that find the problem once it has occurred or is occurring.

1161. All of the following are administrative controls that could be considered for the reduction or prevention of ergonomic injuries EXCEPT:

1) Job expansion
2) Tool changes
3) Job rotation
4) Rest periods

**Ans.2** Tool changes are an engineering control while job expansion, job rotation, and rest periods are administrative controls.

1162. Common job risk factors for the development of cumulative trauma disorder (CTD) include all of the following EXCEPT:

1) Excessive force
2) Abnormal postures
3) Mechanical compression
4) Simple repetition

**Ans.4** The accumulation of micro-injuries related to the risk factors (e.g., excessive force, abnormal postures, mechanical compression) eventually causes the injury. If the risk factors are not present, simple repetition will not contribute to the risk for developing CTD. On the other hand, if the risk factors ARE present, repetition will compound the effect and hasten the development of CTD.

1163. A construction worker operating a jackhammer would be most affected by which of the following hazards?

1) Vibrations of 1 to 125 Hz
2) Dust hazards
3) Sparks
4) Eye hazards

**Ans.1** The use of jackhammers without protective gloves can lead to "white finger disease", a deadening of the nerves due to the vibration from the jackhammer. The vibration frequency that is most hazardous is between 1 to 125 Hz.

1164. When safety nets are necessary, the safety nets are required to extend _____ ft beyond the edge of the work surface.

1) 4
2) 6
3) 8
4) 25

**Ans.3** Safety nets shall be provided when workplaces are more than 25 ft above the ground or other surfaces where the use of ladders, scaffolds, catch platforms, temporary floors, safety lines, or safety belts is impractical. Nets shall extend 8 ft beyond the edge of the work surface where employees are exposed and shall be installed as close under the work surface as practical, but in no case more than 25 ft below such work surface.
29 CFR 1926.105 (c)

1165. Quality of illumination pertains to the:

1) Distribution of brightness in the visual environment
2) Desirable number of footcandles that satisfies environmental needs
3) Illumination ratio between surfaces
4) Differences in the ratio between the light source and the reflected surface

**Ans.1** The factors to be considered for good illumination are quantity, the amount of illumination that produces brightness on the task and its surroundings, and quality, which pertains to the distribution of brightness in a visual environment and includes the color of light, its direction, diffusion, and degree of glare. In a positive sense, quality of illumination

implies that all brightnesses contribute favorably to visual performance, visual comfort, ease of seeing, safety, and esthetics for the specific task involved.

1166. Ventilation systems are designed for any of the following EXCEPT:

1) Supply oxygen
2) Remove contaminants
3) Provide comfort
4) Supply carbon dioxide

**Ans.4** The purpose of ventilation includes dilution of contaminants and supplying make up air.

1167. Canopy hoods can best be used when:

1) High toxicity materials are used
2) Thermal currents can be taken advantage of
3) Energy costs are a concern
4) Canopy hoods should never be used

**Ans.2** Canopy hoods are best used where heated materials give off contaminants that tend to rise which the canopy hood can capture and remove. The toxicity of the material is less of an issue if the canopy hood is able to keep the concentration below hazardous levels. Canopy hoods do tend to draw a high volume or air, which can lead to higher makeup air and energy costs.

1168. In determining the volume of dilution air needed in a room, you need to know:

1) TLV of the contaminant
2) Rate of generation of the contaminant
3) Size of room
4) Both 1 and 3

**Ans.4** All that is needed for determining the volume of dilution air needed is the specific gravity and molecular weight of the contaminant, the TLV, and generation rate. A "K" value

needs to be determined, based on the safety factor and mixing characteristics of the room. The size of the room is not a factor in this calculation.

1169. As a rule of thumb, what percentage of the exhaust face velocity of an unflanged open duct is present at 1 duct diameter from the duct?

1) 5%
2) 10%
3) 25%
4) 50%

**Ans.2** At one duct diameter from an unflanged open duct exhaust, there is only 10% of the initial face velocity available for capture of contaminants. Conversely, at 30 duct diameters from the source of a blowing airstream, there is still 10% of the velocity measured at the source.

1170. Of the following, which is the appropriate type of hood for a welding operation?

1) Side hood
2) Canopy
3) Downdraft bench booth
4) Wind tunnel

**Ans.3** Downdraft bench booth is an appropriate type of hood for a welding operation.

1171. The blowing range of a fan is about _____ times greater than the suction range.

1) 2
2) 30
3) 60
4) 10

**Ans.2** The blowing range is about 30 times greater. This is why capture velocity near the contaminant is very critical.

1172. A fan exhausting a paint spray booth is required to be nonferrous or non-sparking.

1) True
2) False
3) Depends on the occupancy of the building
4) Depends upon the training of the users

**Ans.1** The fan rotating element or the casing must be nonferrous or non-sparking. 29 CFR 1910.107 (c)(6)

1173. What is the acceptable ventilation rate for a flammable liquid storage room inside a building?

1) 2 air changes/hr
2) 6 air changes/hr
3) 10 air changes/hr
4) 25 air changes/hr

**Ans.2** OSHA requires 6 air changes/hr for an interior flammable liquid storage room. 29 CFR 1910.106 (d)(4)(iv)

1174. Baffles are put on the canopy of a shake out hood to:

1) Channel dust to the hopper
2) Prevent plugging of shaker grid
3) Provide better air distribution
4) Lower the noise level

**Ans.3** The function of the baffles is to provide for better air distribution and hence better control of the air contaminant from this foundry operation.

1175. Blast gates are used to:

1) Generate proper capture velocities
2) Balance the resistance to air flow
3) Properly vent sudden overpressures
4) Reduce energy loss when VP changes to SP

**Ans.2** Blast gates are used to balance the resistance to air flow in different ducts.

1176. Flexible ducting used for movable exhaust systems has the following advantage over conventional galvanized ducts:

1) Decreased resistance
2) Maneuverability of hood
3) Availability of hood shapes
4) Increased static pressure

**Ans.2** The principle advantage of flexible ducting is its maneuverability. The hood should be located as close as possible to the contaminant source. Once galvanized ducts are installed they are difficult and expensive to move.

1177. From an aerosol generation standpoint, which of the following housekeeping techniques is most preferable?

1) Dry sweeping
2) Wet scrubbing
3) Vacuuming
4) Air blowing

**Ans.3** All mechanical cleaning techniques generate significant amounts of aerosols. Vacuuming removes material from the area.

1178. A key disadvantage of canopy hoods is that:

1) More material is needed to construct the system
2) Larger fans are needed
3) Contaminants are drawn through the breathing zone
4) Contaminants are converted into aerosols

**Ans.3** Since canopies are above the worker, the contaminant must travel very close to, and most of the time through, the worker's breathing zone.

1179. Round ducts are better than square ducts because round ducts:

1) Can be made air tight easier
2) Use less material for same size
3) Are stronger
4) Have less friction

**Ans.3** In industrial applications, high velocities, and therefore high SP's are normally used. This condition results in duct collapses. Circular ducts are more able to withstand this stress.

1180. One of the problems with the old method of using cotton earplugs was the cerumen. Cerumen is also detrimental to some of the modern earplugs. What is cerumen?

1) Ear canal hair cells
2) Earwax
3) Fluids secreted by the cochlea
4) Secretions of the hypothalamus

**Ans.2** Cerumen is earwax. If cotton earplugs were used, the earwax would bind the cotton together, negating any protective effect.

1181. What is the most expedient method of protecting a worker from noise?

1) Ear protection
2) Limited exposure time
3) Engineering controls
4) Administrative controls

**Ans.1** The quickest way is to provide hearing protection. The most effective long term solution is to engineer out the source of the noise, if feasible.

1182. When controlling noise, the following are all proven techniques EXCEPT:

1) Substituting the process
2) Minimizing sound emitting areas
3) Providing adsorption panels
4) Eliminating leakage paths

**Ans.3** The panels need to absorb the sound waves, not adsorb them. Process substitution can provide alternative technologies that are less noisy. Reducing the areas that radiate noise will also reduce exposure.

1183. When hazards from operating machinery cannot be eliminated, _____ provides protection to the employee from inadvertent contact with a moving or dangerous part.

1) Training
2) Guarding
3) Personal protective equipment
4) Administrative controls

**Ans.2** Where hazards cannot be eliminated, guarding provides protection against potential failure of people to utilize equipment correctly. Both safe design and guarding minimize the effect of individual differences on accident frequency or severity.

1184. Which of the following is a manner by which sound may reach the inner ear even when hearing protectors are employed?

1) Use of foam rubber plugs
2) Hearing protective device vibration
3) Conduction through bone and tissue
4) Both 2 and 3

**Ans.4** The effectiveness of hearing protective devices depends on the manner in which the sound energy is transmitted through or around the device. Therefore, if the device does not fit properly there will be leakage. Sound can also travel by conduction.

1185. A condition which is caused by a very loud impulsive sound which can cause a ringing in the ear and an immediate loss of hearing is called:

1) Anacuses
2) Mastoidectomy
3) Lateral semnisci

4) Tinnitus

**Ans.4** Tinnitus is a condition which is caused by a very loud impulsive sound which can cause a ringing in the ear and an immediate loss of hearing.

1186. If an estimated near field power density of a microwave antenna is calculated as 14 $mW/cm^2$, what is your conclusion?

1) There is no hazard
2) There is a hazard
3) Further investigation is necessary to determine if there is a hazard
4) No conclusions may be made

**Ans.3** Since the result is greater than the OSHA exposure guide for microwaves of 10 $mW/cm^2$, further investigation is necessary. Consideration must be given to the distance from the antenna at which a worker may be exposed and the length of time they may be exposed. 29 CFR 1910.97 (a)(2)(i)

1187. When making a final assessment of the potential hazard from electromagnetic waves based on calculations, what other factors need to be considered?

1) The possibility of reflections from the ground or surfaces
2) The length of time of the exposure
3) Both 1 and 2
4) None of the above

**Ans.3** When making an assessment of the potential hazard of electromagnetic waves, the possibility of reflections and the length of exposure need to be considered.

1188. Which of the following is most vulnerable to damage/injury precipitated by exposure to electromagnetic radiation:

1) Eye

2) Liver
3) Lung
4) Kidney

**Ans.1** Of the above choices, the eye is the most vulnerable part of the body to electromagnetic radiation. The skin is also vulnerable - particularly to ultraviolet radiation exposure (sunburn). The susceptibility of the eye is partially due to its receiving radiations not entirely limited to the visible portion of the electromagnetic spectrum. The function of certain portions of the ocular apparatus may be most easily impaired by minimal or threshold damage because of the delicate functional balance of component parts.

1189. A Class II laser:

1) Does not have enough power to injure a person accidentally when viewed for less than 1/4 sec
2) Does not have enough power to injure person accidentally when viewed for less than 1/2 sec
3) Cannot produce accidental injury if viewed directly for more than 1/2 sec
4) Produces a hazardous direct or specularly reflected beam

**Ans.1** To provide guidance on the evolution and control of laser radiation hazards, a classification scheme was established (ANSI Z-136.l, Standard for the Safe Use of Lasers). This scheme is based only on the ability of the laser to produce injury, which in five classes, range from safe to very hazardous. Class II, or "low-power" devices, are those visible lasers which do not have enough power to injure a person accidentally, but which may produce retinal injury when viewed directly for more than 1/4 sec.

1190. A Class IIIb laser:

1) Does not have enough power to injure a person accidentally
2) Cannot induce injury when viewed within the beam with the unaided eye
3) Can produce accidental injury if viewed directly
4) Produces a hazardous direct or specularly reflected beam

**Ans.3** To provide guidance on the evaluation and control of laser radiation hazards, a classification scheme was established (ANSI Z-136.1, Standard for the Safe Use of Lasers). This scheme is based only on the ability of the laser to produce injury, which in five classes, range from safe to very hazardous. A Class IIIb consists of lasers which can produce accidental injury if viewed directly. Intrabeam viewing of either the direct beam or a mirror-like (specular) reflection of the beam should also be considered hazardous.

1191. Carbon dioxide lasers are particularly hazardous because:

1) They are Q-switched
2) They produce a continuous beam
3) They are invisible to the human eye
4) They are difficult to control

**Ans.3** Laser light formed in carbon dioxide lasers is invisible to the human eye.

1192. Current X-ray machines feature the greatest danger from:

1) Beta
2) Gamma
3) Radiofrequency
4) High voltage

**Ans.4** X-ray machines have numerous levels of safety design incorporated into them. The highest risk now is from the high voltages required to generate X-rays.

1193. If a curie represents $3.7 \times 10^{10}$ disintegrations per second, how many disintegrations per second in a microcurie?

1) $3.7 \times 10^{10}$

2) $3.7 \times 10^7$
3) $3.7 \times 10^4$
4) $3.7 \times 10^6$

**Ans.3** A microcurie has $10^{-6}$ activity of a curie
Conv = ($3.7 \times 10^{10}$ dps/curie) / $10^6$ microcurie/curie
= $3.7 \times 10^4$ dps/microcurie

1194. All of the following are non-ionizing radiation sources EXCEPT:

1) Radiofrequency
2) Ultraviolet
3) Infrared
4) X-ray

**Ans.4** All of the above radiation forms are considered non-ionizing except for X-rays.

1195. The dose rate at 2 m from a particular gamma source is 50 mrem/hr. At what distance will it give a dose rate of 2.0 mrem/hr?

1) 0.5 m
2) 5 m
3) 10 m
4) 20 m

**Ans.3** Use the inverse square law for radiation distance.
$D_1 R_1^2 = D_2 R_2^2$
$50 * 2^2 = 2R_2^2$
$R_2^2 = 100\ m^2$
$R_2 = 10$ m

1196. Radon produces particulate _____ products.

1) Decay
2) Industrial
3) Radioactive
4) Both 1 and 3

**Ans.4** Radon gas decays to form radioactive particulates which can be collected on a high volume sampler.

1197. Which laser forms are considered intrinsically the most dangerous?

1) Ruby
2) Visible
3) Invisible
4) Gas

**Ans.3** Invisible laser radiation requires the most safeguards as it cannot be seen by the human eye.

1198. Excessive exposure to ultraviolet (UV) radiation for a short period of time can result in:

1) Welder's flash
2) Stoker's syndrome
3) Hyperthermia
4) Pyrogenic response

**Ans.1** Acute UV exposure in a short time can result in welder's flash. Symptoms are: red eyes, foreign body sensation, gritty feeling in the eye, and excessive tears. The condition is usually temporary and rarely causes permanent damage.

1199. One problem associated with flux-cored arc welding of stainless steel is the production of aerosols containing:

1) Nickel and Chromium
2) Carbon and Nickel
3) Fluorides, Nickel and Chromium
4) Copper, Nickel and Acid gases

**Ans.3** Welding fumes can contain constituents of the material being welded and the consumable materials (e.g. flux). Stainless steel welding results in fumes containing nickel and chromium. Often flux is used which contain fluorides. These fluorides are also released into the air during the welding process.

Blunt, J., and Balchin, N.C., *Health and Safety in Welding and Allied Processes*, 5th ed.
Finch, R., *Welder's Handbook: A Complete Guide to MIG, TIG, ARC and Oxyacetylene Welding*

1200. Sources of ozone include any of the following EXCEPT:

1) Arc welders
2) Office photocopying machines
3) Ultraviolet quartz lamps
4) VDT monitors

**Ans.4** Ozone can be produced by the interaction of ultraviolet light, electric arcs, and oxygen. Sources of ultraviolet light include electric arcs such as those produced during arc welding, photocopy machines, laser printers, and ultraviolet lamps. VDT monitors do not produce ozone.

1201. The room designed to recharge and replace acid batteries for forklifts requires the following EXCEPT:

1) Eyewash/safety shower
2) Ventilation
3) Facilities to neutralize electrolytes

4) Blow-out construction

**Ans.4** All of the above except blow-out construction are required. 29 CFR 1910.178 (g)(2)

1202. Which of the following types of welding produces ozone?

1) Gas Metal Arc Welding
2) Sweating Carbon Rod
3) Brazing
4) Oxyacetylene welding

**Ans.1** Ozone is produced in greater quantities by gas metal arc welding (GMAW or short-arc), gas tungsten arc welding (GTAW or heli-arc), and plasma arc cutting. Arc welding produces intense ultraviolet light that can create ozone from oxygen.

Construction Safety and Health Outreach Program, U.S. Department of Labor OSHA Office of Training and Education, "Welding Health Hazards", May 1996.
Blunt, J., and Balchin, N.C., *Health and Safety in Welding and Allied Processes*, 5th ed.

1203. What horizontal force is necessary to slide a 50 lb wooden crate on a level wooden floor? (assume the coefficient of static friction for wood on wood is 0.40)

1) 20 lb
2) 30 lb
3) 50 lb
4) 125 lb

**Ans.** Ff = uFn, where Ff is final force, u is the coefficient of static friction and Fn is the weight of the object being pushed or pulled
Ff = 0.4 * 50 lb = 20 lb

1204. Solid particles generated by condensation from the gaseous state, generally after volatilization from molten metals, are called:

1) Smokes
2) Mists
3) Gases
4) Fumes

**Ans.4** Fumes are solid particles generated by condensation from the gaseous state, generally after volatilization from molten metals.

1205. Determine the wavelength of a low frequency sound of 100 Hz, which is traveling through steel at 16,400 ft/sec.

1) 200 ft
2) 200 in
3) 164 ft
4) 164 m

**Ans.3** Use the formula: C = □ f
C = Speed = 16,400 ft/sec
□ = Wavelength
f = Frequency in Hz = 100 Hz
C = □ f
□ = C / f
= (16,400 ft/sec) / 100 Hz
= 164 ft

1206. The decay rate of radium is 1.36 x $10^{-11}$ per sec. What is its half life?

1) 0.0059 x $10^{10}$ sec
2) 0.5095 x $10^{10}$ sec
3) 5.0955 x $10^{10}$ sec
4) 5.0955 sec

**Ans.3** Use the formula: Decay Rate = 0.693 / Half Life
Half life = 0.693 / 1.36 x $10^{-11}$ per sec
= 5.0955 x $10^{10}$ sec

1207. What is a "photon"?

1) Electromagnetic radiation
2) Alpha particle
3) Nucleon
4) Sound wave

**Ans.1** A photon is any item on the electromagnetic (EM) spectrum. An ionizing photon can be an x-ray, gamma, or cosmic ray. Do not confuse "phonon", a sound wave, with "photon." A nucleon is a particle in the nucleus, either a proton or a neutron. Non-ionizing electromagnetic (EM) radiation types would include infrared and visible light.

1208. Define non-ionizing radiation.

1) Radiation reduced in energy from ionizing
2) The visible portion of the electromagnetic spectrum
3) Radiofrequency (RF)
4) EM radiation less than 10 keV

**Ans.4** Particles are not considered in the discussion of non-ionizing radiation. The non-ionizing portion of the spectrum begins at 10 keV and below. This roughly corresponds to the area of soft x-rays and high energy ultraviolet light.

1209. Which is the LEAST likely definition of a neutron?

1) An effective neutral component of the nucleus
2) A nucleon
3) A particle with 1 amu
4) The opposite of proton

**Ans.4** Neutrons are not fundamental particles like electrons and protons. They will decay when separated from the nuclear force into a positive and negative component and, therefore,

are considered to have an effective charge of zero. Neutrons, like protons, have an amu about equal to 1 and are considered to be a nucleon.

1210. What is the general formula in standardized notation for Newton's second law?

1) $E = mc^2$
2) $F = ma$
3) $E = I R$
4) c = wavelength * frequency

**Ans.2** Each of the above formulae are fundamental. The first is Einstein's equivalent of mass and energy; the second, is Newton's second law, with force being equal to mass times acceleration; the third, is Ohm's Law, with voltage being equal to current times resistance. The last equation identifies that the speed of light is a constant.

1211. What is the mathematical relationship of the wavelength of a photon to its frequency?

1) Direct
2) Inverse
3) Independent
4) Depends on the energy

**Ans.2** The wave equation: speed of light = frequency * wavelength. The speed of light is a constant; therefore, the relationship is inverse. In other words, as the wavelength decreases the frequency will increase. Conversely, if the wavelength increases the frequency will decrease.

1212. What is the "half-life" of a radionuclide?

1) Amount of time to eliminate ½ the dosage
2) Average time that the average radionuclide exists
3) Time required for a radionuclide to decay to ½ activity
4) Amount of time for the body to eliminate ½ the dose

**Ans.3** The "half-life" is also known as the physical half-life. One half-life is the time it takes for a given amount of a radioactive material to decrease in activity by 50%.

1213. A carbon dioxide laser has a wavelength in the _____ region.

1) Green
2) Red
3) Ultraviolet
4) Infrared

**Ans.4** Wavelength = 10.6 □m. $CO_2$ lasers are invisible and can cause retinal damage. This wavelength is in the infrared region of the electromagnetic (EM) spectrum.

1214. Hot dry skin, dizziness, nausea, rapid weak pulse, and rapid shallow breathing are symptoms of:

1) Insulin shock
2) Myocardial infarction
3) Heat stroke
4) Stroke

**Ans.3** Heat stroke is a serious medical emergency caused by a failure of the central nervous system sweat control, thus leading to loss of evaporative cooling. This process leads to confusion, collapse, delirium, and coma. Cooling of the body must be started immediately if vital organ damage or death is to be averted.

1215. Evidence suggests that short term occupational exposure to vibration causes:

1) Severe physiological effects on the body
2) Moderate physiological effects on the body

3) Increased risk to the spine and the peripheral nervous system, the digestive system, and the reproductive system
4) Only small physiological effects which are reversible

**Ans.4** In most occupational settings, the damage from vibration is thought to occur from the micro-trauma induced by vibration exposure over a long period of time, usually years. In the initial stages, vibration injury is reversible if the worker is removed from the exposure or the exposure is eliminated from the job.

1216. The most important factor that causes carpal tunnel syndrome is:

1) Impact of the hand against blunt objects
2) Swelling of the transverse carpal ligament
3) Compression of the median nerve brought on by swelling of the tendons passing through the wrist
4) Swelling of the radial artery, which passes through the transverse carpal ligament

**Ans.3** Impact of the hand against blunt objects is considered incorrect although blunt trauma to the wrist could produce symptoms of CTS caused by swelling and edema.

1217. Tenosynovitis can best be described as:

1) Inflammation of the tendon and the synovial sheath
2) Inflammation of the tendons
3) A sore elbow
4) A torn ligament

**Ans.1** Tendons pass through synovial sheaths or membranes. Synovial membranes help lubricate tendons to facilitate movement. Repetitive use can irritate the sheath, the tendon, or both, causing a potential build-up of synovial fluid and possible inflammation.

1218. The minimum distance that combustibles should be moved away from welding and cutting operations is:

1) 10 ft

2) 25 ft
3) 35 ft
4) 50 ft

**Ans.3** The recommended distance of 35 ft is found in NFPA, FM, and most fire prevention codes.

1219. The minimum width of any exit access for new construction shall be no less than:

1) 28 in
2) 30 in
3) 32 in
4) 36 in

**Ans.4** The NFPA Life Safety Code (NFPA 101) states that the minimum width of any exit access shall be as specified for individual occupancies, but in no case shall such width be less than 36 inches (91 cm). In existing buildings, the minimum width shall not be less than 28 inches. NFPA 101 7.3.4.1 (Life Safety Code)

1220. An area where flammable solvents are transferred and stored should be wired as:

1) Class I, Division 1
2) Class I, Division 2
3) Class II, Division 1
4) Class II, Division 2

**Ans.1** A Class I, Division 1 rating is required for all electrical wiring (as well as electrical equipment) due to the expectation that a flammable atmosphere may be present at some time from operations. NFPA 70 Article 500-5 (National Electric Code)

1221. Fire safety relief devices are required on all compressed gas cylinders.

1) True
2) False
3) Depends upon building occupancy
4) Depends upon cylinder construction

**Ans.2** Cylinders containing toxic or poisonous gases are not required to have relief devices. Other classes of compressed gases have frangible disks, fusible disks, etc.

1222. A flammable liquid with a flash point of 63ºF and a boiling point of 110ºF would be classified by the NFPA as a:

1) Class I liquid
2) Class IA liquid
3) Class IB liquid
4) Class II liquid

**Ans.3** Class IB liquid is a flammable liquid with a flash point of 63ºF and a boiling point of 110ºF. NFPA 30 - Flammable and Combustible Liquids Code

1223. Design and layout requirements for emergency lighting in health care occupancies are outlined in:

1) NFPA 70 - National Electric Code
2) NFPA 72 - Fire Alarm Cods
3) NFPA 99 - Health Care Facilitiee
4) NFPA 101 - Life Safety Code

**Ans.3** NFPA 99, Chapter 8, specifies the design and layout of emergency lighting for health care facilities. NFPA 101 (Life Safety Code) specifies how they are powered.

1224. General purpose electrical equipment can cause explosions and fires in areas where flammable vapors, liquids, and gases, and combustible dusts or fibers are present. Which of the following is NOT an example of special purpose electrical equipment?

1) Non-sparking equipment
2) Motors and lighting fixtures
3) Intrinsically safe electrical equipment
4) Pressurized equipment

**Ans.2** Motors and lighting fixtures are examples of equipment that may heat up. These energy sources can produces temperatures high enough to cause ignition.

1225. Which of the following is NOT true?

1) Equipment and wiring approved as intrinsically safe is acceptable in any hazardous (classified) location for which it is designed
2) All interconnections between circuits must be evaluated to be sure that an unexpected source of ignition is not introduced through other non-intrinsically safe equipment
3) Intrinsically safe equipment is not capable of releasing sufficient electrical or thermal energy under normal or abnormal conditions to cause ignition of specific flammable or combustible atmospheric mixtures in their most easily ignitable concentrations
4) Separation of intrinsically safe and non-intrinsically safe wiring is not necessary to ensure that the circuits in hazardous (classified) locations remain safe

**Ans.4** Separation of intrinsically safe and non-intrinsically safe wiring IS necessary to ensure that the circuits in hazardous (classified) locations remain safe. NFPA 70 Article 504-30 (National Electric Code)

1226. Which one of the following statement is NOT true?

1) Generally, equipment installed in Class I locations need not be approved as explosion proof
2) Since it is impractical to keep flammable gases outside of enclosures, arcing equipment must be installed in enclosures that are designed to withstand an explosion
3) The equipment must not only be strong enough to withstand an internal explosion, but the enclosures must be designed to vent the resulting explosive gases
4) Venting must ensure that the gases are cooled to a temperature below the ignition temperature of the hazardous substance involved before being released into the hazardous atmosphere

**Ans.4** Generally, equipment installed in Class I locations must be approved as explosion proof. NFPA 70 Article 500-2 (National Electric Code)

1227. Which of the following organizations promulgates the primary safety standard for boilers?

1) ANSI
2) ASME
3) BOCA
4) NFPA

**Ans.2** The American Society of Mechanical Engineers (ASME) promulgates the "Boiler and Pressure Vessel Code."

1228. The TLV-STEL refers to a concentration:

1) That should never be exceeded
2) Average permitted for a 30 minute period
3) That is an emergency exposure level
4) Average permitted for a 15 minute period

**Ans.4** The threshold limit value-short term exposure limit (TLV-STEL) is an average concentration over a 15 minute period, that should not be exceeded at anytime during an 8 hour workday. The concentration may momentarily exceed the STEL, only if the 15 minute average is at or below the STEL.

1229. When a vacuum cleaner is used to clean up asbestos containing wastes, it must have _____ _____ exhaust ports.

1) Well maintained
2) HEPA filtered
3) Pressure demand
4) Demand only

**Ans.2** All vacuum cleaners used for asbestos-containing material (ACM), MUST have HEPA filtered exhausts.
29 CFR 1910.1001 (k)(4)

1230. During a power failure, the NFPA Life Safety Code (NFPA 101) requires that power-operated doors:

1) Operate automatically
2) Operate manually
3) Lock
4) Remain motionless

**Ans.2** The NFPA Life Safety Code requires that power-operated doors are capable of being operated manually during a power failure. NFPA 101 7.2.1.9.1 (Life Safety Code)

1231. The ACGIH classifies carcinogens into two categories. The category corresponding to Confirmed Human Carcinogens is:

1) A1
2) A2
3) A3
4) A4

**Ans.**

- A1 is the category for Confirmed Human Carcinogens.
- A2 is the category for Suspected Human Carcinogens.
- A3 is the category for Confirmed Animal Carcinogen with Unknown Relevance to Humans.
- A4 is the category for Not Classifiable as a Human Carcinogen.
- A5 is the category for Not Suspected as a Human Carcinogen.

Substances for which no human or experimental animal carcinogenic data have been reported are assigned no carcinogenicity designation.

1232. Shock mounts are responsible for reducing especially which of the following frequency ranges?

1) less than 500 Hz
2) less than 1,000 Hz
3) 1,000 - 2,000 Hz
4) 2,000 - 4,000 Hz

**Ans.1** Shock mounts and other vibration isolation techniques are useful in the low frequency ranges (i.e., less than 500 Hz).

1233. For the purpose of protecting workers in trenches, the type of soil is required to assess risk. A primary factor in differing types of soil is:

1) Depth
2) Cohesiveness
3) Width
4) Soil weight

**Ans.2** A soil's cohesiveness is critical to its stability in a trench or excavation.
29 CFR 1926 Subpart P, Appendix A

1234. In addition to shoring and shielding, what basic type of protective method is permitted to protect workers from a trench cave-in?

1) Sloping
2) Limiting the workers' exposure times in the trench
3) Posting a competent person as a safety observer outside the trench
4) Training

**Ans.1** Other accepted methods of protection include sloping or benching. Sloping and benching decrease the effective depth of the trench.
29 CFR 1926.652 (b)

1235. The point of operation for any machine tool is:

1) The place where a guard is usually placed
2) Where one moving part passes another
3) The place where work is done on material
4) Both 1 and 2

**Ans.3** Point of operation is the area on a machine where work is actually performed upon the material being processed. This is generally an area where injury can occur.
29 CFR 1910.212 (a)(3)

1236. The primary reason for enclosing a grinding wheel is:

1) Collect dust
2) Contain wheel breakage
3) Easier to use
4) Prevent wheel from breaking

**Ans.2** An enclosure will prevent wheel fragments from entering the work area if the grinding wheel breaks.

1237. The method of guarding by fixed physical barriers that are mounted on a machine to prevent access to the moving parts is called guarding by:

1) Enclosure
2) Interlock
3) Fixed guard
4) Barriers

**Ans.1** Enclosure is the method of guarding by fixed physical barriers that are mounted on a machine to prevent access to the moving parts.

1238. Which of the following is most appropriate in the routine occupational health examination of a machinist?

1) Exercise stress test
2) Serum protease inhibitor level
3) Allergy history assessment
4) Audiometric testing

**Ans.4** Machinists are likely to be exposed to noise that may produce hearing loss. The best answer is audiometric testing.

1239. Which of the following is NOT considered an engineering control method for noise?

1) Absorption devices installed inside the machine
2) Substituting a quieter machine for the loud machine
3) Enclosing the loud machine in a sound-dampening box
4) Donning ear muffs with inserts when using the machine

**Ans.4** Absorption, substitution, enclosure, and isolation are all considered engineering control measures for noise.

1240. Currently the term sievert (Sv) is becoming widely used in health physics. What is 1 sievert equivalent to?

1) 1 rem
2) 10 rems
3) 100 rems
4) 1,000 rems

**Ans.3** 1 sievert (Sv) = 100 rems

1241. At what dose of ionizing radiation does nausea, vomiting, and reduced white blood cell count occur?

1) 0-25 rad
2) 25-50 rad

3) 50-150 rad
4) 150-300 rad

**Ans.3** For exposure to ionizing radiation the following dose-response relationships exists:

| DOSE (rad) | EFFECT |
|---|---|
| 0 - 25 | No observable effect |
| 25 - 50 | Minor temporary blood changes |
| 50 - 150 | Nausea, vomiting, reduced white blood cell count |
| 150 - 300 | Increased severity of above symptoms plus diarrhea, malaise, loss of appetite - some death |

1242. Concentration of gases and vapors may be reported as:

1) Parts per million by weight
2) Particles per million by volume
3) Parts per million by volume
4) Particles per million by weight

**Ans.3** Gas and vapor concentrations in air are expressed in a volume/volume ratio. Based on toxicity, irritation, and other factors, the volume of gas or vapor is compared to one million parts of air. Particulates are used for solid contaminants. Weight/volume ratios are used for particulates.

1243. If two legs of a triangle have lengths of 15 feet and 25 feet, and the included angle is 135º, how long is the third side?

1) 29.2 ft

2) 30.0 ft
3) 34.6 ft
4) 37.2 ft

**Ans.4** Use the Law of Cosines.
$c^2 = a^2 + b^2 - 2 * a * b * \cos$
$c = (15\ ft)^2 + (25\ ft)^2 - 2\ ft * 15\ ft * 25\ ft * \cos$
$135 = 225\ ft^2 + 625\ ft^2 - 750\ ft^2 * -0.7071$
$= 1380.325\ ft^2$
$c = 37.15\ ft$

1244. Raynaud's Phenomenon can be caused by:

1) Repetitive motion stresses
2) Improper lifting
3) Improper posture
4) Hand-arm vibration

**Ans.4** Raynaud's Phenomenon (or Syndrome), also known as vibration white-finger disease, is caused by repeated exposure to hand-arm vibration in the 500 - 1,200 Hz range. Repetitive motion, improper lifting, and posture cause other types of ergonomic diseases.

1245. A unit of radiation reflecting the biological effects resulting from exposure of X or gamma radiation is called a:

1) Becquerel
2) Roentgen
3) Curie
4) Red

**Ans.2** A unit of radiation reflecting the biological effects resulting from exposure of X-ray or gamma radiation is called a roentgen.

1246. The primary red blood cell (erythrocyte, RBC) producing organ is the:

1) Bone marrow
2) Spleen
3) Liver
4) Thymus

**Ans.1** Up until the age of 4 or 5 years, the spleen and liver can produce RBC's in response to normal growth demands, but after this period, production of RBC's in extramedullary sites occurs only under pathologic conditions. Bone marrow is also the site for production of leukocytes (white blood cells) and thrombocytes (platelets).

1247. A cylinder contains 1,000.0 psi(g). What is its absolute pressure?

1) 1,014.7 psi(a)
2) 1,000.0 psi(a)
3) 985.3 psi(a)
4) 900.0 psi(a)

**Ans.1** Psi(g) (pounds per square inch gauge) = psi(a) (pounds per square inch absolute) - atmospheric pressure (14.7 psi)
psi(g) = psi(a) - 14.7 psi
1,000.0 psi = psi(a) - 14.7 psi
1,000.0 psi + 14.7 psi = psi(a) 1,014.7
psi = psi(a)

1248. A gas inside a sealed drum exerts 15.3 psi(g) at 25ºC. What is the pressure inside the drum when the temperature is 40ºC?

1) 16.1 psi(g)
2) 14.6 psi(g)
3) Molecular weight of material must be known
4) Volume of drum must be known

**Ans.1** $(P_1 V_1)/T_1 = (P_2 V_2)/T_2$
15.3 psi(g) / 298° K = $P_2$ / 313° K
$P_2$ = 16.1 psi(g)
Where:

$P_1$ = Initial pressure
$P_2$ = Final pressure
$T_1$ = Initial temperature (absolute)
$T_2$ = Final temperature (absolute)
$V_1$ = Initial volume
$V_2$ = Final volume
NOTE: Assume that the volume = 1. The volume does not change.

1249. A gas occupies a volume of 2.75 liters at 87ºC and 150 torr. What volume will the gas occupy at 25ºC and 745 torr?

1) 0 liters
2) 0.13 liters
3) 0.23 liters
4) 0.46 liters

**Ans.4** Use the combined Ideal Gas Law equation. $(P_1 V_1)/T_1 = (P_2 V_2)/T_2$
Remember to convert all temperatures to absolute.
(Ta = ºC + 273)
$V_2 = (P_1 V_1 T_2) / (P_2 T_1)$
= (150 torr * 2.75 l * 298° K) / (745 torr * 360° K)
= 0.46 liters
NOTE: 1 torr is also equal to 1 mm Hg

1250. A mixture of gases at 1 atm contains 66.0% $N_2$, 14% $O_2$ and 20% $CO_2$ by volume. What is the partial pressure of each gas in mm Hg?

1) $N_2$ = 501.6, $O_2$ = 106.4, $CO_2$ = 152.0
2) $N_2$ = 501.6, $O_2$ = 152.0, $CO_2$ = 106.4
3) $N_2$ = 217.1, $O_2$ = 152.0, $CO_2$ = 106.4
4) $N_2$ = 217.1, $O_2$ = 106.4, $CO_2$ = 152.0

**Ans.1** $P(total) = P(N_2) + P(O_2) + P(CO_2)$
= 1 atm = 760 torr = 760 mm Hg
$P(N_2)$ = 0.66 * 760 mm Hg = 501.6 mm Hg
$P(O_2)$ = 0.14 * 760 mm Hg = 106.4 mm Hg
$P(CO_2)$ = 0.20 * 760 mm Hg = 152.0 mm Hg

1251. When an atom absorbs energy from a source such as ultraviolet light, it often releases that energy in several smaller packets of energy. Which of the following is NOT correct?

1) The released energy is used for analytical purposes
2) The released energy is of a longer wavelength that the ultraviolet light
3) The released energy is a type of fluorescence
4) The released energy is usually X-rays

**Ans.4** The process of absorbing energy moves electrons to a higher energy. When the electrons revert back to a lower energy level, the excess energy may be released as light. This release usually takes place in two or three smaller releases, dividing the original energy into pieces, resulting in the longer wavelength. The science of spectrofluorometry is the analysis of this light.

1252. If a force acting on a surface is applied in a direction perpendicular to that surface then the force is called a:

1) Uni-directional force
2) Normal force
3) Perpendicular force
4) Friction force

**Ans.2** If a force acting on a surface is applied in a direction perpendicular to that surface then the force is called normal force.

1253. A system of forces is ____ if the lines of action of the forces have a common point of intersection.

1) Combined
2) Normal
3) Non-concurrent
4) Concurrent

**Ans.4** A system of forces is concurrent if the lines of action of the forces have a common point of intersection.

1254. A system of forces is said to be ____ if all the forces are acting on a two-dimensional surface.

1) Coplanar forces
2) Concurrent forces
3) Tangential force
4) Normal force

**Ans.1** A system of forces is said to be coplanar forces if all the forces are acting on a two-dimensional surface.

1255. A system of forces is ____ if all the forces have a common line of action.

1) Concurrent
2) Tangential
3) Coplanar
4) Collinear

**Ans.4** A system of forces is collinear if all the forces have a common line of action.

1256. A force applied on a body that will tend to elongate the body is called:

1) Applied
2) Tangential
3) Tensile
4) Normal

**Ans.3** A force applied on a body that will tend to elongate the body is called tensile.

1257. A force that will tend to shrink the body in the direction that it is applied is called:

1) Non-tensile
2) Tensile
3) Compressive
4) Coplanar

**Ans.3** A force that will tend to shrink the body in the direction that it is applied is called compressive.

1258. Ultrasonic noise cannot be heard by the human ear. What does the ear hear during ultrasonic operation?

1) Ultrasonic noise
2) Subharmonic noise
3) ELF noise
4) Nothing

**Ans.2** You actually hear subharmonics of the ultrasonic noise which are generated in your hearing range. An ultrasonic welder emitting noise at 20 KHz (human limit) will be heard at 10 kHz, the first subharmonic.

1259. In which part of the ear is mechanical energy translated to electrical energy?

1) Outer ear
2) Middle ear
3) Inner ear
4) Ear drum

**Ans.3** The inner ear contains the cochlea, which has tiny hair cells that vibrate when exposed to sound waves. These hairs then transmit electrical impulses to the brain along nerves attached to the hair cells.

1260. Most muscles are joined by hundreds or even thousands of nerve fibers of the efferent (motor) nervous system. The contact between the end point of the axon of one motor-neuron and the sarolemma of the muscle is called:

1) Reticulum switches
2) Cross bridges
3) Motor endplate
4) Motor muscle

**Ans.3** Most muscles are joined by hundreds or even thousands of nerve fibers of the efferent (motor) nervous system. The contact between the end point of the axon of the motor-neuron and the sarolemma of the muscle is called the motor endplate.

1261. The sacrum is located:

1) Along the upper spinal column
2) Below the lumbar vertebrae
3) Above the cervical vertebrae
4) Within the thoracic cavity

**Ans.2** The sacrum is located below the lumbar vertebrae.

1262. What factor LEAST influences the interchange of heat between man and his environment?

1) Moisture content of the air
2) Air velocity
3) Air density
4) Air temperature

**Ans.3** The density of the air is not one of the major influencing factors in heat exchange between man and the environment.

1263. What part of the body is most acutely damaged by ultraviolet radiation?

1) Cornea
2) Skin
3) Kidneys
4) Lung

**Ans.1** Ultraviolet light can cause the most acute damage to the eye's cornea. Factors which influence the damage include: exposure time, distance, wavelength, sensitivity of the individual, and presence of sensitizing agents.

1264. MIG (metal inert gas) welding is commonly used when working with:

1) Aluminum
2) Stainless Steel
3) Mild Steel
4) Titanium

**Ans.3** While MIG (metal inert gas) welding is capable of being used for all the listed metals, it is most effective for mild steels. TIG (tungsten inert gas) welding is more effective for aluminum, stainless steel and titanium. Fumes are a concern with both MIG and TIG welding, but MIG welding tends to produce more fumes. All forms of arc welding produce potentially hazardous Ultraviolet (UV) light.

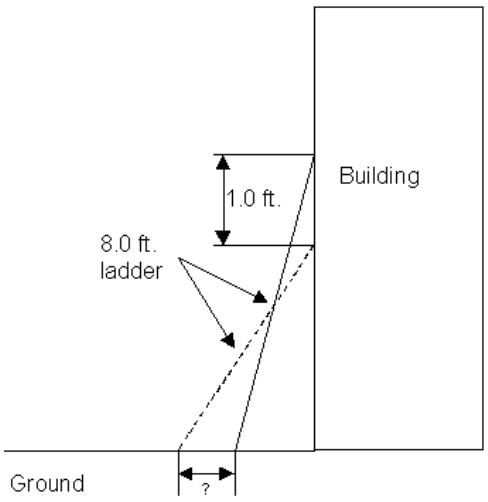
Blunt, J., and Balchin, N.C., *Health and Safety in Welding and Allied Processes*, 5th ed.
Finch, R., *Welder's Handbook: A Complete Guide to MIG, TIG, ARC and Oxyacetylene Welding*

1265. The primary machine shop tool used in the cold forming of metals is the:

1) Drill
2) Turning machine
3) Lathe
4) Power press

**Ans.4** The power press is a versatile tool used in a number of operations to cold form metals.

1266.



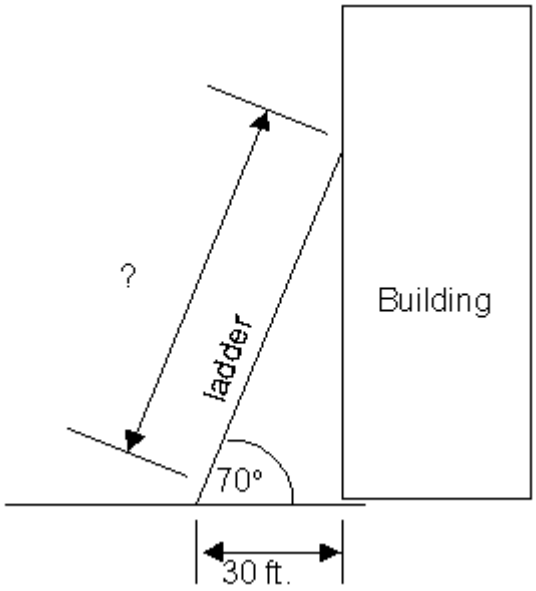

An 8.0 ft steel ladder is placed against a building. The top of the ladder is 7.6 ft up the building wall then slips 1.0 ft down. How far did the bottom of the ladder move?

1) 1.0 ft
2) 2.0 ft
3) 2.5 ft
4) 4.5 ft

**Ans.2** After the ladder slips 1 foot it is 6.6 ft up on the house. Using the Pythagorean Theorem for right triangles (i.e., $a^2 + b^2 = c^2$ where a and b are sides of a right triangle, and c is the hypotenuse.)
Solving for the new base: $(8.0\text{ ft})^2 - (6.6\text{ ft})^2 = (\text{new base})^2$
$64\text{ ft}^2 - 43.6\text{ ft}^2 = (\text{new base})^2$
$20.4\text{ ft}^2 = (\text{new base})^2$
$\sqrt{20.4}\text{ ft}^2 = \text{new base}$
New base = 4.5 ft

Solving for the original base: $(8.0 \text{ ft})^2 - (7.6 \text{ ft})^2 = (\text{original base})^2$
Original base = 2.5 ft. Therefore the base changed from 2.5 ft to 4.5 ft, a change of 2.0 ft.

1267.




A ladder leans against the side of a building and makes an angle of 70° with the ground. The foot of the ladder is 30 ft from the building. What is the length of the ladder?

1) 11 ft
2) 32 ft
3) 88 ft
4) 92 ft

**Ans.3** Use the cosine trigonometric identity.
Cosine of Angle = Adjacent Leg Length/Hypotenuse Leg Length
cos 70° = 30 ft / Hypotenuse Leg Length

Hypotenuse Leg Length = 30 ft / cos 70º
= 30 ft / 0.3420 = 88 ft

1268. If a 220 lb man runs up a 10 ft flight of stairs in 4 sec, how much horsepower does he develop?

1) 1.0 hp
2) 4.0 hp
3) 5.5 hp
4) 16 hp

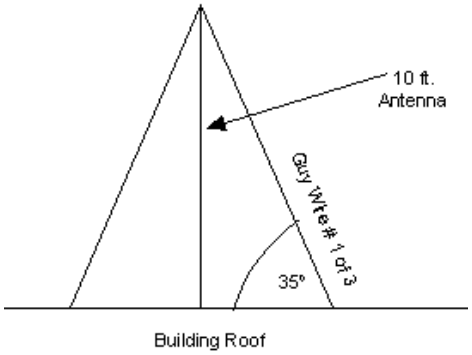
**Ans.1** 1 hp = 550 ft-lb/sec
Power = (10 ft * 220 lb) / 4 sec
= 2,200 ft-lb / 4 sec
= 550 ft-lb/sec = 1.0 hp

1269. What is the absolute pressure at the bottom of an open-air tank filled with sea water to a depth of 10 ft with an average density of 64 $lb/ft^3$?

1) 4.44 psi(a)
2) 19.3 psi(a)
3) 640 psi(a)
4) 655 psi(a)

**Ans.2** Use the formula: Pressure = Height * Density
P = 10 ft * 64 $lb/ft^3$
P = 640 $lb/ft^2$
There are 144 $in^2$ in a $ft^2$
640 $lb/ft^2$ / 144 $in^2/ft^2$ = 4.44 psi
Adding standard atmospheric pressure to the head pressure calculated above, the total absolute pressure at the bottom of the sea water tank is:
4.44 psi + 14.9 psi = 19.3 psi(a)

1270.

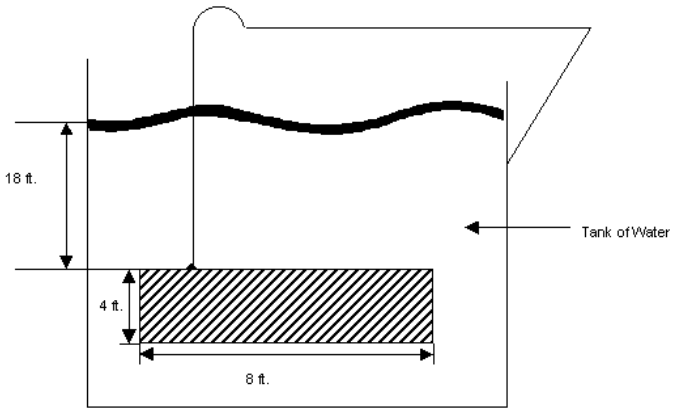



To secure a roof antenna three guy wires are required. The wires will make an angle of 35º with respect to the roof. How much guy wire is needed if the antenna is 10 ft high?

1) 13 ft
2) 18 ft
3) 37 ft
4) 52 ft

**Ans.4** Use the sine trigonometric identity. Sine of Angle = Opposite Leg Length / Hypotenuse Leg Length
sin 35º = 10 ft / Hypotenuse Leg Length
Hypotenuse Leg Length = 10 ft / sin 35º = 10 ft/ 0.5736 = 17.4 ft
Total for 3 guy wires = 3 * 17.4 ft = 52.2 ft.

1271.

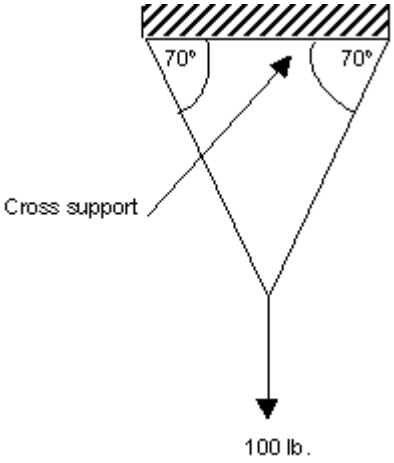



A 4 ft high by 8 ft wide panel is submerged such that the top of the panel is located 18 ft below the water line. What total force is experienced by the panel?

1) 100,000 lb
2) 80,000 lb
3) 60,000 lb
4) 40,000 lb

**Ans.4** Use the following formulas: Area = Length * Width = 8 ft * 4 ft = 32 $ft^2$
Pressure = Weight * depth at VERTICAL CENTER of panel = 62.4 $lb/ft^3$ * 20 ft = 1,248 $lb/ft^2$
Total Force = Area * Pressure = 32 $ft^2$ * 1,248 $lb/ft^2$ = 39,936 lb

1272.




What is the axial tensile stress on this cross support (neglecting the effects of beam weight)?

1) 18 lb
2) 36 lb
3) 48 lb
4) 96 lb

**Ans.2** Use the tangent trigonometric identity. Since the load is supported by 2 diagonal beams, divide the load by 2 and calculate the vertical component for each side. The vertical component is:
100 lb / 2 = 50 lb
tangent of angle = opposite length / adjacent length
tan 70º = 50 lb / adjacent length
= 50 lb / tan 70 º
= 50 lb / 2.75
= 18.2 lb

The other diagonal support beam is contributing the same stress.
Therefore, the total stress on the cross support is 18.2 lb + 18.2 lb = 36.4 lb

1273. A sled and passenger weigh 86 lb. The coefficient of friction of steel runners on snow is 0.07. What force is needed to get the sled moving?

1) 5.1 lb
2) 6.0 lb
3) 6.7 lb
4) 8.5 lb

**Ans.2** Force = f * weight
= 0.07 (86 lb)
= 6.02 lb

1274. If water weighs 8.34 lb / gal and 100 gal are moved up from the bottom to the top of a 100 ft tank, what work is done?

1) 385 ft-lb
2) 4,932 ft-lb
3) 8,340 ft-lb
4) 83,400 ft-lb

**Ans.4** Use the formula: W = F D (where W = work, F = force and D = distance)
weight of water = 8.34 lb / gal * 100 gal = 834 lb
W = 834 lb * 100 ft = 83,400 ft-lb

1275. What size of motor on a hoist rated at 2,000 lb is required if the motor were required to raise a rated load of 25 ft in no longer than 3 sec?

1) 20 hp
2) 25 hp
3) 30 hp
4) 35 hp

**Ans.3** Use the formula: Power = (F D) / t (where F = force, D = distance, t = time)
P = 2,000 lb * 25 ft / 3 sec = 16,667 ft-lb/sec
= (16,667 ft-lb/sec) / (550 ft-lb/sec per hp)
= 30.3 hp

1276. A three-leg sling is supported from a single point and attaches to a 1,000 lb steel cylindrical object at three equidistant points along the cylinder's circumference (i.e., the attachment points on the steel cylinder are 120º apart). The center of gravity of the cylinder runs through its center. When attached to this steel cylinder, each leg of the sling creates a 30º angle with respect to the load. What is the tension in each leg of the sling?

1) 333 lb
2) 385 lb
3) 577 lb
4) 667 lb

**Ans.4** Since the center of gravity of the load is through its center, and since each leg of the sling attaches to the load equidistant from its adjacent legs, each leg will carry exactly 1/3 of the load. The angle each leg of the sling makes with the load is measured as 30º. Therefore, the tension in each leg of the sling is equal to 1/3 of the total load (1,000 lb) divided the sine of 30º.
Tension (in one leg) = 1/3 of total load / sin 30º
Tension (one leg) = 1/3 of 1,000 lbs/sin 30º
Tension (one leg) = 333.3/0.5
Tension (one leg) = 667 lbs

When you have the angle of each leg of the sling with respect to the load (or with respect to the horizontal), divide the load by the sine of the angle.

1277. Determine the horsepower required to lift a 120 lb weight to a height of 40 ft in 65 sec.

1) 0.134 hp
2) 0.268 hp
3) 1.340 hp
4) 2.680 hp

**Ans.1** Use the the formula: Power = Work / Time (Note: Work = Force * Distance)
Power = Force * Distance / Time
= 120 lb * 40 ft / 65 sec
= 73.8 ft-lb/sec
1 hp = 550 ft-lb/sec
Power = (73.8 ft-lb/sec) / (550 ft-lb/sec) / hp
= 0.134 hp

1278. Which of the following correlates the parts of an overhead crane with the correct direction of travel?

1) Bridge: Horizontal, Trolley: Horizontal, Hoist: Vertical
2) Bridge: Vertical, Trolley: Horizontal, Hoist: Horizontal
3) Bridge: Horizontal, Trolley: Vertical, Hoist: Vertical
4) Bridge: Horizontal, Trolley: N/A, Hoist: Vertical

**Ans.1** The trolley and hoist move horizontally along the bridge rails. The trolley is connected to the bridge and moves at right angles to it. The hoist is connected to the trolley and does the lifting. Combined, the trolley and the bridge travel at right angles to each other and enable the crane to reach any point under the bridge rails, limited only by the travel limits.

1279. Which of the following best describes the type of health hazard produced by noise?

1) Hearing changes correctable by surgery
2) Permanent damage to the hair cells of inner ear
3) Reversible damage to the ear drum
4) Damage to the outer ear

**Ans.2** Damage to the hair cells of the inner ear may occur. Bald spots may form and hearing is impaired at the frequencies corresponding to the location of the bald spot.

1280. An employee has suffered a noise induced hearing loss. In which frequency range would you expect to see the loss on the employee's audiogram?

1) 500 Hz
2) 750 Hz
3) 1,000 Hz

4) 4,000 Hz

**Ans.4** Noise induced hearing loss affects the higher frequencies. This may be referred to as the 4,000 Hz notch in an audiogram.

1281. Hearing loss attributed to listening to loud music from portable music players is called:

1) Psychogenic
2) Sociocusis
3) Sensorineural
4) Conductive

**Ans.2** Hearing loss due to social habits is called sociocusis.

1282. The ossicles are located in the:

1) External ear
2) Middle ear
3) Inner ear
4) Ear drum

**Ans.2** The ossicles is located in the middle ear.

1283. Heat energy of the body is primarily lost by:

1) Radiation
2) Convection
3) Conduction
4) Evaporation

**Ans.4** Evaporation is the primary mechanism by which the body loses heat to the surrounding environment.

1284. A vehicle is traveling at a constant speed of 45 mph. On average, the vehicle accelerates at a rate of 1 ft/sec$^2$ four times per hour for 30 seconds to pass other vehicles. How far does the vehicle travel in one hour?

1) 46.1 miles
2) 49.0 miles
3) 46.4 miles
4) 47.3 miles

**Ans.3** In 58 minutes the vehicle goes 45 mph * (58 min / 60 min/hr) = 43.5 miles To determine the distance traveled in the remaining two minutes of acceleration, use $s = v_o t + at^2/2$
Convert 45 mph to 66 ft/sec
t = 120 seconds (2 minutes)
s = (66 ft/sec)(120 sec) + (1 ft/sec$^2$)(120 sec)$^2$) / 2
= 7,920 ft + 7200 ft
= 15,120 ft = 2.9 miles
43.5 miles + 2.9 miles = 46.4 miles

1285. If a drag racer has a terminal velocity of 200 mph after a ¼ mile race, what has been the average acceleration?

1) 50 ft/sec$^2$
2) 32.5 ft/sec$^2$
3) 18 ft/sec$^2$
4) 23.4 ft/sec$^2$

**Ans.2** 200 mph = 293 ft/sec ; ¼ mile = 1,320 ft
$v^2 = v_o^2 + 2as$, where v = ending velocity, $v_o$ = beginning velocity, a = acceleration, and s = distance.
(293 ft/sec)$^2$ = 0 + 2a( 1,320 ft )
a = 85,849 ft$^2$/sec$^2$ / 2,640 ft
= 32.5 ft/sec$^2$

1286. If a man applies a force in a horizontal direction to slide an 80 lb crate 10 ft, how much work did he perform? (Assume the coefficient of static friction is 0.40 between the crate and the floor.)

1) 320 ft-lb
2) 610 ft-lb
3) 800 ft-lb
4) 1,120 ft-lb

**Ans.1** Since the force is being applied in the horizontal direction, the only force to be overcome is the friction between the crate and the floor.
$F = \mu N$
$= (0.40)(80 \text{ lb}) = 32 \text{ lb}$
Work = Fs = (32 lb)(10 ft) = 320 ft-lb

1287. If a man lifts a 90 lb crate 3 ft, how much work did he perform?

1) 3 ft-lb
2) 30 ft-lb
3) 90 ft-lb
4) 270 ft-lb

**Ans.4** Since the force is being applied in the vertical direction, the only force to be overcome is the weight.
Work = Fs
Work = 90 lb * 3 ft = 270 ft-lb

1288. A man pushes a 30 lb object for 8 ft on a $15^{o}$ inclined plane. If the dynamic coefficient of friction is 0.40, how much work did he perform?

1) 155 ft-lb
2) 240 ft-lb
3) 92.8 ft-lb
4) 62.4 ft-lb

**Ans.1** Draw a picture. The 30 lb object exerts forces perpendicular to the plane and in the direction of the plane, forming a vector triangle. The force from the weight in the direction of the plane can be determined by:
cos 75 = Fp / 30 lb
Fp = 7.8 lb
The force perpendicular to the plane.
Fn = sin 75 * 30 lb = 28.9 lb
Dynamic friction force to be overcome is: Ff = 0.4 * 28.9 lb = 11.6 lb
The man has to overcome both forces.
Ft = 11.6 lb + 7.8 lb = 19.4 lb
Work performed = 19.4 lb * 8 ft = 155.2 ft-lb

1289. How much kinetic energy is required to bring a 3,000 lb automobile to a speed of 45 mph?

1) 3,000 ft-lb
2) 135,000 ft-lb
3) 188,000 ft-lb
4) 203,000 ft-lb

**Ans.4** Use the formula for calculating kinetic energy:
KE = $(mv^2) / 2$ m
= W/g = 3,000 lb / (32.2 $ft/sec^2$)
= 93.2 lb-$sec^2$/ft v
= 45 mph = 66 $ft/sec^2$
KE = [(93.2 lb-$sec^2$/ft)(66 $ft/sec)^2$] / 2
= 202,990 ft-lb

1290. How much kinetic energy is required to stop a 4,000 lb automobile travelling at 60 mph?

1) 353,000 ft-lb
2) 481,000 ft-lb
3) 4,000 ft-lb
4) 240,000 ft-lb

**Ans.2** KE = $mv^2/2$
m = W/g m

= 4,000 lb / 32.2 ft/sec$^2$
= 124.2 lb-sec$^2$/ft v
= 60 mph = 88 ft/sec$^2$
KE = [(124.2 lb-sec$^2$/ft)(88 ft/sec)$^2$] / 2
= 480,902 ft-lb

1291. If 300,000 ft-lb of kinetic energy is required to bring a moving automobile to a stop, and the total braking power transmitted to the road is 1,500 lb, how far did the vehicle go before it came to a stop?

1) 3,000 ft
2) 200 ft
3) 150 ft
4) 300 ft

**Ans.2** The difference in kinetic energy of two states is equivalent to the work performed.
300,000 ft-lb = 1,500 * d
d = 200 ft

1292. What is the potential energy with respect to gravity of a 10 lb steel ball at a height of 12 ft from the surface of the earth?

1) 120 ft-lb
2) 1.2 ft-lb
3) 0.875 ft-lb
4) 322 ft-lb

**Ans.1** PE = mgh
W = mg
PE = Wh
= (10 lb)(12 ft )
= 120 ft-lb

1293. The factor which is primarily responsible for dust gaining entrance to the alveolar regions of the lung is:

1) Composition of the dust
2) Concentration
3) Particle size
4) Condition of the mucociliary system

**Ans.3** Particle size is the most important factor in determining respirability. It is the ability of the smaller particles to negotiate the turns and twists of the respiratory tract that allows them to gain access to the lower portions of the respiratory system. The composition and concentration are important once the deposition of the particle has occurred; that may cause the biological response to the substance. The condition of the mucociliary system affects the clearance of particles from the lung, once deposited.

1294. The size particle which has the greatest retention in the lung is:

1) 0.05 to 1.0 micron
2) 0.5 to 0.75 microns
3) 1.0 to 5.0 microns
4) 5.0 to 10.0 microns

**Ans.3** Particles in the size range of 1.0 to 5.0 microns penetrate the lungs into the alveolar area where they have less of a chance of being cleared from the lung. Particles less than 1.0 micron tend to stay in the airstream and are breathed back out, while particles greater than 5.0 microns are captured in the upper respiratory tract and do not get to the lower portions of the lungs.

1295. An irreversible diseased lung condition in which the alveolar walls have lost their resiliency resulting in an excessive reduction in the lung capacity is called:

1) Eclema
2) Dipuria
3) Emphysema
4) Physema

**Ans.3** Emphysema is an irreversible lung condition in which the alveolar walls have lost their resiliency, resulting in an excessive reduction in the lung capacity.

1296. A dark-blue to purplish coloration of the skin and the mucous membranes caused by lack of oxygen utilization by the body is defined as a:

1) Toxemic effect
2) Canasine effect
3) Cyanotic effect
4) Cryogenic effect

**Ans.3** The cyanotic effect causes a dark-blue to purplish coloration of the skin and the mucous membranes caused by lack of oxygen utilization by the body.

1297. The effect of local or generalized grey-blue colored impregnation of the skin tissue with silver is called:

1) Asphyxia
2) Argyria
3) Atrophy
4) Diaphoresis

**Ans.2** Argyria is the effect of local or generalized grey-blue colored impregnation of the skin tissue with silver.

1298. What part of the eye is acutely damaged by the welders flash of ultraviolet rays?

1) Pupil
2) Sclera
3) Cornea
4) Retina

**Ans.3** The cornea is acutely damaged by the welders flash of ultraviolet rays.

1299. Activation-dependent carcinogens are those which require bioactivation of the parent molecule to an ultimate carcinogen. Synonyms for this term include:

1) Primary carcinogens
2) Procarcinogens
3) Direct-acting carcinogens
4) Cocarcinogens

**Ans.2** Procarcinogen, precarcinogen, secondary carcinogen, and indirect-acting carcinogens are all terms implying activation-dependence. Activation- independent, direct-acting, or primary carcinogens are those in which the parent molecule is the ultimate carcinogen.

1300. Carcinogens that interact with DNA are _____ carcinogens.

1) Activation-dependent
2) Genotoxic
3) Epigenetic
4) Unclassified

**Ans.2** Activation-dependent carcinogens are one type of genotoxic carcinogens. Epigenetic carcinogens are basically those for which no evidence of DNA reactivity has been found.

1301. Chemical carcinogens are defined operationally by their ability to cause neoplasms. Which of the following is NOT acceptable evidence for a carcinogenic effect?

1) Earlier development of tumors
2) Increase in benign tumors
3) Greater multiplicity of tumors

4) Presence of chemical DNA adducts

**Ans.4** The presence of chemical DNA adducts is indicative of a genetic lesion, not a tumor.

1302. The minimum amount of electric current passing through the body to produce a fatality is:

1) 7 A
2) 2 A
3) 500 - 750 mA
4) 70 - 100 mA

**Ans.4** The National Safety Council publicizes that 70 - 100 mA (milliamperes) is enough current to be fatal.

1303. When irritant smoke is used to test the fit of a respirator, the smoke is an aerosol of the following chemical.

1) Stannic chloride
2) Titanium tetrachloride
3) DEHP (Diethylhexylphthalate)
4) Either 1 or 2

**Ans.4** In the irritant smoke test, either stannic chloride or titanium tetrachloride can be used.

1304. Good ergonomic design for standing workstations would include all of the following EXCEPT:

1) Opportunity for frequent adjustments in posture
2) Anti-fatigue floor mats
3) Foot rests or a bar rail

4) Foot controls for adjustments of work

**Ans.4** Foot controls should be avoided for standing workstations, since this can cause shifting of weight to one leg and abnormal postures. Frequent adjustments, floor mats, and foot rests allow the employee to pick and choose the most comfortable positions as needed.

1305. Which is the intensity of illumination of a warehouse floor at a distance of 20 ft from a 1,000 candle power source?

1) 2.5 footcandle
2) 5.0 footcandle
3) 10 footcandle
4) 20 footcandle

**Ans.1** The intensity of illumination of a surface is inversely proportional to the square of its distance from the light source.
Expressed as a formula: Intensity of illumination = Power of source/distance$^2$ = 1,000 candle/(20 ft)$^2$ = 2.5 footcandle

1306. If the distance between a point source of light and a surface is doubled, the intensity of illumination on the surface will be:

1) The same
2) Reduced to 1/2 of the intensity
3) Reduced to 1/4 of the intensity
4) Reduced to 1/8 of the intensity

**Ans.3** The intensity of illumination from a point source varies inversely as the square of the distance. As the distance from the light source increases, the illumination decreases. When the distance is 2 times as great, the illumination is $(1/2)^2$ or 1/4 as much as it was before.

1307. What is the intensity of illumination of the top of a bench at a distance of 4 ft from a 150 candlepower light source?

1) 9.4 footcandle
2) 25 footcandle
3) 37 footcandle
4) 75 footcandle

**Ans.1** Intensity of illumination of a surface is inversely proportional to the square of the distance from a light source.
Expressed as a formula: Intensity of illumination = Power of source/distance$^2$
= 150 candle/(4 ft)$^2$ = 150 candle/16 ft$^2$ = 9.4 footcandle

1308. The inverse square law as applied to illumination states:

1) The light source as received varies inversely as the square of the power
2) The light intensity of the source varies inversely as the distance in lumens
3) Light intensity on a surface varies inversely with the square of the distance between source and surface
4) Surface illumination varies directly with the inverse square of the distance

**Ans.3** Light intensity on a surface varies inversely with the square of the distance between source and surface.

1309. Brightness ratio may be defined as:

1) Ratio of illumination between surfaces
2) Ratio of difference between light source and reflected surface
3) Ratio of square of distance between illumination surfaces
4) Ratio of distance between source and surface

**Ans.2** The brightness ratio of is the ratio of difference between light source and reflected surface.

1310. What would be the effect on illumination if a light source 1 ft from a worker is moved 10 ft further away?

1) The light illumination is halved

2) The light illumination is doubled
3) The light illumination is reduced by a factor of 10
4) The light illumination is reduced by a factor of 100

**Ans.4** Like other forms of radiation, light obeys the inverse square law. In other words, if you move away 10 ft, the light will decrease by a factor of $(10\ ft)^2$, or 100.

1311. Which one of the following is NOT correct?

1) Intensity - How much light a source gives off in a given direction
2) Lumen - Unit of light output from a light source
3) Footcandle - Quantity of light falling on a surface
4) Footlambert - Percentage of light that is reflected from a surface

**Ans.4** Footlambert is a unit of luminance. A surface emitting one lumen per square foot of surface has a luminance (brightness) of one footlambert. Reflectance describes how much light is reflected from a surface. A perfectly white surface has a reflectance of one or 100%.

1312. The unit used as the unit of illumination equivalent to a lumen/$ft^2$ is which of the following?

1) Footcandle
2) Lumen
3) Candela
4) LUX

**Ans.1** The classic unit of illumination is the footcandle, which is equivalent to one lumen/$ft^2$ .

1313. What is the intensity of illumination on top of an assembly bench at a distance of 2 ft from a 200 candlepower light?

1) 400 footcandle

2) 200 footcandle
3) 100 footcandle
4) 50 footcandle

**Ans.4** The intensity of illumination of a surface is inversely proportional to the square of its distance from the light source.
Expressed as a formula: Intensity of illumination = Power of source/distance$^2$ = 200 candle/(4 ft)$^2$ = 50 footcandle

1314. The proper slant for a portable straight ladder is _____ ft vertical to _____ ft horizontal.

1) 1, 4
2) 1, 5
3) 4, 1
4) 4, 4

**Ans.3** The proper slant is 4 ft vertical to 1 ft horizontal. 29 CFR 1910.25 (d)(2) 29 CFR 1910.26 (c)(3)

1315. An electrician assures you that the a standard duplex wall receptacle is wired correctly. She shows you on her multimeter (properly set for AC voltage) that the voltage across the brass-colored terminal and the terminal labeled with a "W" is 115 V. Do you agree with her assurance that the receptacle is wired correctly?

1) Yes
2) No. Ask her to test the voltage across the brass-colored terminal and the green-colored terminal. If it reads 115 V, then agree with her assurance
3) No. Ask her to test the voltage across the "W"-labeled terminal and the green-colored terminal. If it reads 115 V, then agree with her assurance
4) No. Ask her to test the voltage across the "W"-labeled terminal and the green-colored terminal. If it reads 0 V, then agree with her assurance

**Ans.4** There should be no voltage across the grounded conductor ("W"-labeled terminal) and the ground wire (green-colored terminal). If there is voltage, then the white wire and the black wire (hot wire) are reversed. NFPA 70 Article 200-10 (National Electric Code) describes the identification of terminals.

1316. The basic purpose of local exhaust ventilation is to:

1) Prevent re-entrance of air contaminants
2) Remove contaminants at their source
3) Provide dilution ventilation
4) Both 1 and n

**Ans.2** The purpose of local exhaust ventilation is to remove the air contaminants at the source, not to dilute them.

1317. The maximum permissible occupational exposure to incident electromagnetic energy of frequencies of 10 MHz to 100 GHz averaged over any possible 0.1 hr period is:

1) 0.5 $mW/cm^2$
2) 1.0 $mW/cm^2$
3) 10 $mW/cm^2$
4) 1.0 $W/cm^2$

**Ans.3** According to the OSHA radiation protection guide for normal environmental conditions for incident electromagnetic energy of frequencies from 10 MHz to 100 GHz, the maximum permissible exposure is 10 $mW/cm^2$ as averaged over any possible 0.1 hr period. This standard also applies to microwave cooking ovens which usually operate at 915 or 2,450 MHz. 29 CFR 1910.97 (a)(2)(i)

1318. Radon 222 has a half life of 3.8 days. What will the activity be of 1.0 microcurie of radon 222 after 15.2 days?

1) 0.05 microcurie
2) 0.50 microcurie
3) 0.62 microcurie

4) 0.062 microcurie

**Ans.4** One half life results in a decrease of 1/2 of the activity of the radon 222. How many half lives is 15.2 days?
15.2 / 3.8 = 4 half lives
4 half lives = $(1/2)^4$ = 1/16
1/16 of the original activity will remain.
1/16 x 1.0 microcurie = 0.062 microcurie

1319. Video display terminals have received much attention recently for their possible association with reproductive effects. The reproductive effects have been attributed to ionizing radiation and:

1) Low frequency pulsed magnetic fields
2) Thermal stress
3) Microwave emissions
4) Extremely low frequency (ELF) electric fields

**Ans.1** The main area of discussion has been the possible effects of low level pulsed magnetic fields. There have been several animal studies that have shown the waves to have potential teratogenic effects. The results have been difficult to replicate and some other exposures were not accounted for in the studies. This area will continue to be controversial.

1320. What is an alpha particle?

1) Helium nucleus component with a double positive charge
2) High-speed electron
3) X-rays from a nucleus
4) Unknown radiation component in a calibration

**Ans.1** Alpha rays are doubly ionized particles, not electromagnetic radiation. They are composed of two protons and two neutrons. An alpha particle is a helium nucleus.

1321. Which of the following is correct? Compressed air:

1) Must never be used for cleaning purposes
2) Must never be used to blow debris from a person
3) May be used for cleaning purposes when the pressure is reduced to below 50 psi
4) May be used to blow debris from a person when the pressure is reduced to below 30 psi

**Ans.2** Compressed air must never be used to blow debris from a person. In fact, compressed air shall not be used for cleaning purposes except where reduced to less than 30 psi, and then only with effective chip guarding and personal protective equipment.
29 CFR 1910.242 (b)

1322. The lenses in an arc welder's helmet are NOT required to:

1) Protect against metal spatter
2) Protect the eyes against ultraviolet radiation
3) Not degenerate from exposure to ultraviolet radiation
4) Protect the eyes against infrared radiation

**Ans.4** The welding helmet contains two lenses. The outer lens is clear glass. It functions to protect the welder from molten metal spatter. The inner lens is tinted and is capable of screening ultraviolet radiation. It will not degenerate from exposure to ultraviolet radiation. It is not designed to protect the eyes from heat (i.e., infrared radiation). ANSI Z87.1 29 CFR 1910.252 (b)(2)

1323. A metatarsal guard provides the following protection:

1) Toe
2) Toe to inseam
3) Toe to ankle
4) Toe to knee

**Ans.3** A metatarsal guard provides toe to ankle protection. The metatarsal guards are commonly used in foundries to protect employees from crushing injuries to the foot.

1324. Reactive metals are usually arc welded under a shield of inert gas such as argon. Metals that are welded in this manner include aluminum and titanium. The major pollutants from this operation is (are):

1) Phosgene
2) Ozone
3) Ultraviolet light
4) Both 2 and 3

**Ans4.** Metal inert gas (MIG) welding produces relatively little metal fumes. It does however, produce significant amounts of ultraviolet radiation and, as a secondary pollutant, ozone.

1325. An automobile traveling 55 mi/hr is slowing up at the rate of 5 mi/hr per second. How many seconds will it take to stop?

1) 0.09 sec
2) 0.40 sec
3) 6.00 sec
4) 11.0 sec

**Ans4.** Newton's Law: V (velocity) = a (acceleration) * t (time) applies.
Initial velocity is substituted for final velocity when a moving body is being negatively accelerated.
V = a t, or rearranging,
t = V/a t = 55 mi/hr / (5 mi/hr/sec)
= 11 sec

1326. A light meter used to measure the intensity of surface illumination provides readings measured in:

1) Lumens
2) Watts per foot
3) Footcandles
4) Candlepower

**Ans.3** A light meter is the instrument used to measure the intensity of surface illuminations. Readings are measured in footcandles. By definition, a footcandle is the illumination at a point of a surface which is one foot, and perpendicular to a uniform point source of one candle. More briefly, a footcandle is a unit of illumination.

1327. Which of the following types of electromagnetic radiation has the shortest wavelength?

1) Infrared light
2) Ultraviolet light
3) Visible light
4) Gamma waves

**Ans.4** Gamma rays have the shortest wavelength. This is followed by ultraviolet light, visible light, and infrared light.

1328. A 6 in. diameter grinding wheel rotating at 500 rpm has a surface speed of:

1) 556 ft/min
2) 690 ft/min
3) 785 ft/min
4) 970 ft/min

**Ans.3** You first must calculate the circumference of the wheel.
C = pd
= 3.14 * 0.5 ft
= 1.57 ft
Speed = C * rpm

= 1.57 ft * 500 rpm
= 785 ft/min

1329. One of the most fundamental relationships in the field of acoustics is the relationship between wavelength, velocity, and frequency. Wavelength (l) =

1) f/c
2) c/f
3) Force * mass
4) Force * g

**Ans.2** Wavelength is equal to the velocity divided by the frequency. This implies that wavelength is inversely proportional to frequency and directly proportional to velocity.

1330. Solve the following problem:(6! 6!) / (3! 2!)

1) 47
2) 4,320
3) 43,200
4) 432,000

**Ans.3** The factorial (!) notation indicates that you multiply all numbers less than or equal to the number, e.g.,
6! = 6 * 5 * 4 * 3 * 2 * 1 = 720
The problem then becomes: Answer = (720 * 720) / (6 * 2) = 43,200

1331. The time that it takes to complete one wave cycle (peak to peak) is called the:

1) Wavelength
2) Period
3) Frequency
4) Q factor

**Ans.2** The period is the TIME it takes to complete one wave cycle. It is measured in seconds. The distance between the peaks is the wavelength.

1332. Transmission loss is:

1) Frequency dependent
2) Amplitude dependent
3) Different for different substances
4) Both 1 and 3

**Ans.4** Transmission loss is frequency dependent and changes depending on the substance. For example, oak doors have a different transmission loss than do concrete blocks.

1333. Transmittance and absorbance in spectrophotometry are:

1) Inverses of one another
2) Always 180º out of phase to one another
3) Synonyms
4) Not related in any way

**Ans.1** Transmittance and absorbance are simple inverses of one another. The amount of light absorbed is absorbance and transmittance is the amount of light that is not absorbed.

1334. Which of the following is a root of the equation? $6x^2 + 16x + 8 = 0$

1) -2
2) 0
3) -1
4) 1

**Ans.1** Substituting x = -2 into the equation you find it yields 0, therefore x = -2 is a root.

1335. A 10 lb steel ball is dropped from a height of 12 ft. What are the potential and kinetic energies of the ball when it reaches a height of 8 ft?

1) PE = 80 ft-lb, KE = 40 ft-lb
2) PE = 80 ft-lb, KE = 80 ft-lb
3) PE = 40 ft-lb, KE = 80 ft-lb
4) PE = 40 ft-lb, KE = 40 ft-lb

**Ans.1** PE = m * g * h (at 8 ft)
PE = (10 lb / 32.2 ft/sec$^2$) 32.2 ft/sec$^2$ * 8 ft = 80 ft-lb
KE represents the work performed from 12 ft to 8 ft
KE = 10 lb (12 ft - 8 ft) = 40 ft-lb

1336. A 10 lb steel ball is dropped from a height of 12 ft. What is the speed of the ball after it has fallen 4 ft?

1) 16 ft/sec
2) 18 ft/sec
3) 20 ft/sec
4) 22 ft/sec

**Ans.1** KE represents the work performed during the 4 ft part of the fall.
KE = 10 lb * 4 ft = 40 ft-lb
KE also is equal to ½ * m * v$^2$
m = 10 lb / 32.2 ft/sec$^2$ = 0.3106 lb-sec$^2$/ft 40 ft-lb
= (0.3106 lb-sec$^2$/ft * v$^2$) / 2 v
= 16 ft/sec

1337. What is the potential energy with respect to the elastic force of a ball pulled 6 in. in a horizontal direction from the undeformed position?(Assume a spring constant of 8 lb/in.)

1) 196 in.-lb
2) 144 in.-lb
3) 48 in.-lb
4) 6 in.-lb

**Ans.2** PE = $ks^2 / 2$ = 8 lb/in.$(6\ in)^2$) / 2 = 144 in.-lb

1338. At what frequency are the A, B, and C acoustic scales equal?

1) 250 Hz
2) 1,000 Hz
3) 500 Hz
4) 2,000 Hz

**Ans.2** The response curves for the three scales listed all cross at 1,000 Hz.

1339. Relative humidity is determined by:

1) Wet bulb temperature alone
2) Vane anemometer
3) Impinger sample
4) WBGT and DBT with psychrometric chart

**Ans.4** Relative humidity is the ratio of the amount of water the air has to the amount of water it can hold.

1340. Blood is carried to the heart by the:

1) Arteries
2) Capillaries
3) Spleen

4) Veins

**Ans.4** The veins carry blood to the heart. Arteries carry blood away. (The pulmonary artery and pulmonary vein are exceptions to this).

1341. Light passes through the crystalline lens in the eye and focuses on the:

1) Cornea
2) Iris
3) Pupil
4) Retina

**Ans.4** Light enters the eye through the pupil, travels through the transparent crystalline lens, then travels through the vitreous humor (eyeball), and finally focuses on the retina.

1342. Carbon monoxide is particularly dangerous because:

1) It inhibits blood cells from carrying oxygen
2) It is not easily transferred in cellular respiration
3) It is a systemic poison that inhibits enzyme generation
4) It paralyzes the respiratory center of the brain

**Ans.1** Of all the gases that have poisonous effects upon man and animals, carbon monoxide is the most widely encountered. It exerts its effects by combining with the hemoglobin of the blood and interrupting the normal oxygen supply to the body tissues.

1343. Fats, oils, and waxes are identified with:

1) Carbohydrates
2) Proteins

3) Lipids
4) Glucose

**Ans.3** Lipids are one of the four main classes of biological molecules that all living cells contain. Lipids include fat, waxes, and oils. Their primary functions include storing energy and making up a major part of cell membranes.

1344. There are four main classes of complex biological molecules that are contained in all living cells. Which of the following is NOT one of them?

1) Carbohydrates
2) Proteins
3) Hormones
4) Lipids

**Ans.3** Hormones are not one of the main classes of complex biological molecules. The four main classes of complex biological molecules that all living cells contain include carbohydrates, proteins, lipids, and nucleic acid.

1345. AIDS is caused by a:

1) Protozoa
2) Rickettsia
3) Bacteria
4) Virus

**Ans.4** The human immunodeficiency virus (HIV) is the infectious etiologic agent causing AIDS.

1346. The genetic code is best described as:

1) Uncomplicated

2) Detached
3) Degenerate
4) Variable between species

**Ans.3** The genetic code is degenerate. That is, an amino acid may be coded by more than one codon. (A codon is a sequence of three adjacent nucleotides that specifies the insertion of an amino acid in a specific structural position during protein synthesis).

1347. Nature tends to maintain a balanced and symmetrical state for all organisms to exist. In a normal healthy person this is called:

1) State of balance
2) Anthropometry
3) Positive feed-back mechanism
4) Homeostasis

**Ans.4** Homeostasis is a state of physiological equilibrium produced by a balance of functions and chemical composition within an organism.

1348. The three parts of a human cell consist of the nucleus and:

1) Protoplasm and cytoplasm
2) Membrane and cytoplasm
3) Chromosomes and membrane
4) DNA and RNA

**Ans.2** A cell is a structural unit of which tissues are made. The cell is the basic building block of life. It is a living thing which grows and reproduces itself. If has three basic parts: the nucleus, cytoplasm, and membrane.

1349. The basic material from which all tissue is made is called:

1) Plasma
2) Cytoplasm
3) Protoplasm
4) Nucleus

**Ans.3** Protoplasm is the basic material from which all living tissue is made. Physically, it is a viscous, translucent, semifluid colloid composed mainly of proteins, carbohydrates, fats, salts, and water.

1350. All hoistway openings are required to be protected by double doors or gates that:

1) Are interlocked with the controls
2) Can be easily opened
3) Cannot be opened when the car is not at the landing
4) Both 1 and 3

**Ans.4** "All hoistway openings shall be protected by double doors or gates that are interlocked with controls, so that the car cannot be started until all gates or doors are closed, and so that gates or doors cannot be opened when the car is not at the landing." ANSI Z-17.2

1351. ANSI recommends the lighting level for corridors in industrial facilities have a minimum of _____ footcandles.

1) 10
2) 20
3) 30
4) 40

**Ans.2** The American National Standards Institute (ANSI) Standard Practice for Industrial Lighting (ANSI A22.2-1984) provides minimum lighting levels for safety purposes. The minimum they recommend for corridors in industrial facilities is 20 footcandles. The Illuminating Engineering Society (IES) publishes a Lighting Handbook which recommends higher illumination levels to maximize productivity.

1352. Aisles that are utilized by both powered industrial trucks and pedestrians should be at least:

1) 2 ft wider than the widest load
2) 3 ft wider than the widest load
3) 4 ft wider than the widest load
4) 5 ft wider than the widest load

**Ans.2** The aisle width requirement is 3 ft wider than the widest load of the powered industrial truck.

1353. The building type least likely to survive an earthquake is:

1) Wood
2) Steel
3) Masonry
4) Both 2 and 3

**Ans.3** Masonry, due to its characteristics, is most adversely affected by earthquakes.

1354. ESFR sprinkler technology is most often used to protect:

1) Residential buildings
2) High rise buildings
3) Heat treating processes
4) Rack storage

**Ans.4** ESFR (early suppression fast response) sprinklers are designed to fight high challenge, fast developing fires, often found in rack storage systems. ESFR sprinklers are also used in flammable liquid storage areas. The ESFR sprinkler is designed to extinguish the fire, rather than just control a fire, like other sprinkler heads.

1355. A sprinkler system employing automatic sprinklers attached to a piping system containing air and a supplemental detection system installed in the same areas as the sprinklers; where the actuation of the detection system opens a valve that permits water flow into the sprinkler piping system and to be discharged from any sprinklers that are open is called a(n):

1) Dry system
2) Pre-action system
3) Deluge system
4) ESFR system

**Ans.2** Pre-action and deluge systems are similar in design except that a pre-action system has closed sprinkler heads, whereas a deluge system has open heads. A pre-action system is often used to protect computer rooms or other areas with high value contents.

1356. Open sprinklers are used in what type of sprinkler system?

1) Dry system
2) Pre-action system
3) Deluge system
4) Wet system

**Ans.3** Deluge systems are designed to discharge water through every head when activated for rapid spreading fires, such as flammable liquid storage areas and aircraft hangars with jet fuel.

1357. Water storage tanks can be protected from internal corrosion on wetted surfaces by painting or by:

1) Anodic protection
2) Cathodic protection
3) Static protection
4) Galvanization

**Ans.2** Cathodic protection using anodes of aluminum are common. Anodes must be renewed annually.

1358. Which technique is employed by many hydraulically designed sprinkler systems to reduce friction loss and improve system performance?

1) Double check valves
2) Booster pumps
3) Grid or looped piping
4) Dual risers

**Ans.3** Grid or looped piping systems reduce friction loss by putting the water supply from the riser closer to an operating sprinkler head and by bringing multiple sources of water to each head.

1359. A paint spray operation will be located on the assembly floor. What precautions, with regard to ignition sources, must be exercised?

1) None are needed
2) All ignition sources must be eliminated
3) All flammable and combustible material must be removed from the area
4) All ignition sources should be prohibited within 20 ft of the spraying area

**Ans.4** There shall be no open flame or spark producing equipment in any spraying area within 20 ft, unless separated by a partition. 29 CFR 1910.107 (c)(2)

1360. A gas furnace installed in a ceiling that is used for heating a building is vented through the attic of the building. It is vented with the following type of vent:

1) Single wall metal pipe
2) Type L vent
3) Type BW gas vent
4) Type B gas vent

**Ans.4** A Type B vent is appropriate for a gas fired furnace with a draft hood. A BW vent is used for a gas fired wall furnace. Type L vents are for oil fired furnaces. Single wall pipe cannot be used when the flue passes through an attic or concealed space.

1361. Whenever flammable or combustible liquids are transferred from one container to another:

1) Both containers shall be effectively bonded and grounded
2) Only the dispensing container shall be bonded and the other container shall be grounded
3) Both containers need only be grounded
4) Both containers need only be bonded

**Ans.1** To prevent discharge sparks of static electricity, it is necessary to both ground and bond each container.

1362. A room 50 ft x 20 ft x 10 ft contains 100 ppm of $CCl_4$. How much time is required to lower the concentration to 25 ppm if a blower generating 300 cfm is used to clear the room?

1) 7.5 min
2) 11.1 min
3) 46 min
4) 54 min

**Ans.3** $t = \log_{10}(C/Co)\ (-2.303)\ (P/Q)$
Where:
C = end concentration
Co = beginning concentration
P = room volume
Q = flow of air
Substituting we get:
$t = \log_{10}(25/100)\ (-2.303)\ (10{,}000\ ft^3/300\ cfm) = 46$ min

1363. A 50 ft x 190 ft x 20 ft building is supplied with 8,000 cfm. What is the number of air changes per hour in the building?

1) 1.4 air changes/hr
2) 2.5 air changes/hr
3) 3.6 air changes/hr
4) 4.7 air changes/hr

**Ans.2** Determine total building volume:
50 ft * 190 ft * 20 ft = 190,000 $ft^3$
Convert cfm to $ft^3$/hr:
8,000 cfm * 60 min/hr = 480,000 $ft^3$/hr
480,000 $ft^3$/hr / 190,000 $ft^3$ = 2.5 air changes/hr

1364. A large office building is designed for 5.3 air changes per hour. With this design, how many minutes will it take for one air change?

1) 9.1 min
2) 10.2 min
3) 11.3 min
4) 12.8 min

**Ans.3** Simply divide 60 min/hr by 5.3 air changes/hr
Minutes per Air Change = (60 min/hr) / (5.3 air changes/hr) = 11.3 min/air change

1365. A large office building currently houses 50 people. If the building HVAC system supplies 8,000 cfm and 90% of the air is recirculated, what is the outdoor air ventilation rate per person?

1) 0 cfm/person
2) 16 cfm/person
3) 25 cfm/person
4) 800 cfm/person

**Ans.2** First determine the amount of outdoor air:
8,000 cfm * 0.1 = 800 cfm
Then determine cfm/person:
800 cfm/50 people = 16 cfm/person

1366. Canopy hoods are rarely used due to the often poor performance of this hood type. One equation used, however, to size a canopy hood is:

1) Q = VA
2) Q = Ce * SPh * V
3) Q = 1.4 PDV
4) Q = (LWH/2)V

**Ans.3** The equation used to size canopy hoods is
Q = (1.4 * P * D * V)
Where:
P = tank perimeter (ft), D = vertical distance between tank and edge of hood, and V = design capture velocity.

1367. If 500 cfm are needed to capture air contaminants 6 in. from a hood, what volume of air is needed to capture contaminants 12 in. from the hood face?

1) 550 cfm
2) 750 cfm
3) 2,000 cfm
4) 4,000 cfm

**Ans.** Required flow rate varies as the square of the distance from the contaminant source. A source twice as far away will require 4 times the original flow rate.

1368. An exhaust system operates at 25,000 cfm. A hood is added to the system which requires a total system capacity of 30,000 cfm. How much should the fan speed be increased to handle the extra exhaust volume?

1) 20%
2) 25%
3) 30%

4) 40%

**Ans.1** Utilizing the fan laws where cfm varies directly with fan rpm:
$Q_1/Q_2 = rpm_1/rpm_2 = 30{,}000$ cfm/25,000 cfm = 1.2 or 120%
Increase the fan rpm by 20% and you can accommodate hood. (Assuming maximum fan rpm and tip speed are not exceeded).

1369. An exhaust system operating at 25,000 cfm is increased to 30,000 cfm. What is the required increase in the fan horsepower to handle the increased volume?

1) 50%
2) 66%
3) 73%
4) 79%

**Ans.3** Fan horsepower varies as the cube of the fan speed and cube of the fan volume. Therefore, the required ratio of increase in the fan speed is:
$(30{,}000 \text{ CFM}/25{,}000 \text{ CFM})^3 = 1.73$
The required increase in the fan horsepower is 73%. You must ensure that the fan motor horsepower rating is ample.

1370. What is the approximate capture velocity 12 in. away from a 1 ft diameter duct exhausting air at a duct velocity of 4,000 ft/min?

1) 30 ft/min
2) 40 ft/min
3) 300 ft/min
4) 400 ft/min

**Ans.4** The rule of thumb is that the velocity one duct diameter away from an exhaust opening is about 10% of the duct velocity, or 400 ft/min.

1371. In order to ensure uniform capture and air distribution which would be applicable to most hoods, a maximum plenum velocity should be no more than _____% of the slot velocity.

1)
2)
3)
4)

**Ans.3** The rule of thumb is that the maximum plenum velocity should be 50% of the slot velocity. For a 2,000 fpm slot your maximum plenum velocity should not exceed 1,000 fpm.

1372. A duct velocity of 4,005 ft/min is equivalent to what VP?

1) 1.0 in
2) 2.0 in
3) 3.0 in
4) 0.5 in

**Ans.1** Using the equation V = 4,005 $\sqrt{VP}$
V/4,005 = $\sqrt{VP}$
4,005/4,005 = $\sqrt{VP}$
1.0 = $\sqrt{VP}$
1.0 in = VP

1373. An exhaust duct static pressure is -1.50 in.. Total pressure is -0.75 in.. What is the standard air velocity pressure?

1) -0.25 in
2) -0.50 in
3) -0.75 in
4) 0.75 in

**Ans.4** Using the equation SP + VP = TP
TP = -0.75 inch

SP = -1.50 inch
Substituting and rearranging we get:
VP = -0.75 - (-1.50) = 0.75 in

1374. An exhaust duct has a velocity pressure of 0.75 in.. What is the velocity inside of this duct?

1) 3,000 ft/min
2) 3,400 ft/min
3) 3,500 ft/min
4) 3,600 ft/min

**Ans.3** To solve this problem use the equation $V = 4{,}005\sqrt{VP}$
= 4,005 * 0.866
= 3,468 ft/min

1375. A 12 in. diameter duct has an average duct velocity of 3,200 ft/min. What is Q?

1) 1,600 cfm
2) 2,500 cfm
3) 2,600 cfm
4) 2,700 cfm

**Ans.2** Q = VA
Q = [□(6 in)$^2$]/[12 in/1 ft)$^2$)] 3,200 ft/min
= 2,512 cfm

1376. Two impingers are attached in series. What is the collection efficiency of the first impinger if it contains 95.6 □g, and the second impinger contains 7.5 □g of the same air contaminant?

1) 91.7%
2) 91.9%
3) 92.2%
4) 92.5%

**Ans.3** To calculate collection efficiency utilize the equation:
% efficiency = 100% [1 - ($c_2/c_1$)]
= 100% [1 - (7.5 □g/95.6 □g)]
= 100% [0.922]
= 92.2%

1377. A rotameter calibrated at 25ºC and 760 mm Hg gives a rate indicated as 1.45 l/min at 40ºC and 680 mm Hg. What is the corrected standard air flow rate?

1) 1.24 l/min
2) 1.36 l/min
3) 1.41 l/min
4) 1.45 l/min

**Ans.1** Utilize the correction equation:
Q = 1.45 l/min(680 mm Hg/760 mm Hg)(25ºC + 273ºC)/(40ºC + 273ºC)
= 1.24 l/min
The colder and denser air at standard conditions has greater buoyancy for the rotameter float ball.

1378. Calculate the density of air in units of $mg/cm^3$ under Standard Industrial Hygiene Condition:

1) 1.18 $mg/cm^3$
2) 1.25 $mg/cm^3$
3) 1.27 $mg/cm^3$
4) 1.31 $mg/cm^3$

**Ans.1** Density of air = Pressure air * Mass air / (R * Temperature air)
= 1 atm * 28.96 g/mole / (0.08205 l-atm/mole K * 298°K)
= 1.1838 mg/$cm^3$

1379. What is the density of dry air in units of lb/$ft^3$ at 70ºF, the industrial ventilation design standard?

1) 0.075 lb/$ft^3$
2) 0.15 lb/$ft^3$
3) 1.0 lb/$ft^3$
4) 1.5 lb/$ft^3$

**Ans.1** Density air = Pressure air * Mass air / (R * Temperature air)
Use Rankine for absolute temperature (70ºF = 529.6 R)
□ = [14.7 lb/$in^2$ * 28.96 lb/lb-mol / [(10.73 lb/$in^2$-$ft^3$/lb-mol-R) 529.6 R]
= 0.07496 lb/$ft^3$

1380. Calculate the density of standard air at 200ºC and at 0.7 atm:

1) 0.10 g/l
2) 0.52 g/l
3) 0.80 g/l
4) 0.95 g/l

**Ans.2** Remember to convert degrees C to Kelvin
Density is defined as mass per unit volume (m/V)
PV = nRT
PV = mRT/M
Remember that n = m/M where "m" is the mass and "M" is the molecular weight
We want to find the density; therefore, we need to rearrange the equation to solve for density (m/V)
PV = mRT/M
PVM = mRT
PVM/RT = m
PM/RT = m/V = □
□ = (0.7 atm * 28.96 g/mol) / (0.08205 l-atm/mol-K * 473° K)
= 0.522 g/l

1381. Static pressure on the exhaust side of a fan is normally:

1) Negative
2) Positive
3) Neither negative or positive
4) Greater than 100 MPa

**Ans.2** The static pressure on the exhaust side of a fan is normally positive in relation to atmospheric pressure. The static pressure on the suction side of an exhaust fan is negative compared with atmospheric pressure.

1382. A building 100 ft x 20 ft x 10 ft experiences 12 air changes per hour. What is the flow rate in the building?

1) 400 cfm
2) 1,667 cfm
3) 4,000 cfm
4) 240,000 cfm

**Ans.3** Volume of the room is 100 * 20 * 10 = 20,000 $ft^3$
Flow rate cfm = 20,000 $ft^3$ * 12 changes/hour / 60 min/hr
= 4,000 cfm

1383. Which of the following is true regarding a critical orifice?

1) Used only in the laboratory
2) Used ahead of the sampling device
3) Constant flow occurs when downstream pressure is more than 53% of upstream pressure
4) Each can only be used for a specific flow rate

**Ans.4** A critical orifice achieves a specific constant flow rate when downstream pressure is less than 53% of upstream pressure. It is used in the field when constant flow at a specific flow rate is important and a strong enough pump to achieve high static pressure is available. The orifice must be used downstream of the sampling device.

1384. Which of the following is NOT one of the three fan laws?

1) Velocity varies directly as the square of the RPM
2) Power varies directly as the cube of the RPM
3) Flow rate varies directly with the RPM
4) Static pressure varies directly as the square of the RPM

**Ans.1** Velocity is directly related to flow rate and the diameter of the ductwork. Therefore, the correlation between velocity and RPM would be a direct relationship if the diameter of the duct is constant. The other answers are the three fan laws.

1385. Velocity pressure is:

1) The velocity of air divided by 4005 times the square root of the Sp
2) The difference between total pressure and static pressure
3) Independent of velocity
4) Proportional to the square root of velocity

**Ans.2** The total pressure in a ventilation system is equal to the static pressure plus the velocity pressure. The velocity pressure, therefore, is the difference between the total pressure and the static pressure. The CFM is equal to the area of the duct multiplied by 4005 times the square root of the velocity pressure. The velocity pressure is directly related to the square of velocity of air in the duct.

1386. The flow through a 10 ft * 10 ft opening to a spray booth was calculated to be 20,000 cfm. What is the velocity of air at the spray booth face?

1) 2 ft/min
2) 20 ft/min
3) 200 ft/min
4) 2,000 ft/min

**Ans.3** The area of the spray booth face is 10 ft * 10 ft = 100 $ft^2$. To determine the velocity of air at the face of the booth, divide 20,000 cfm by 100 $ft^2$ = 200 ft/min.

1387. If a spray booth has a frontal opening of 4 ft * 6 ft and requires a flow of 120 ft/min, how many cubic feet of air would go through the opening every minute?

1) 1,620 cfm
2) 1,880 cfm
3) 2,280 cfm
4) 2,880 cfm

**Ans.4** Use the formula Q = AV
Q = 120 ft/min (4 ft * 6 ft)
= 120 ft/min * 24 $ft^2$
= 2,880 $ft^3$/min

1388. What is capture velocity?

1) Linear flow rate through a hood
2) The velocity required to move particles through a duct
3) The velocity required in front of a hood to capture contaminants
4) The velocity of air in a slot

**Ans.3** Capture velocity is the air velocity in front of a hood required to capture the contaminant and draw it into the hood.

1389. What is the coefficient of entry?

1) The ratio of actual to theoretical flow of a hood
2) Linear flow rate at the hood opening
3) Linear flow rate in the hood
4) The velocity required in front of a hood to capture contaminants

**Ans.1** The ratio of actual to theoretical flow caused by a given hood static pressure. The coefficient of entry reflects losses due to turbulence.

1390. What is static pressure?

1) The kinetic pressure exerted in the direction of flow
2) The potential pressure exerted in all directions
3) The pressure exerted by a vapor

4) The pressure due to weight of the atmosphere

**Ans.2** Static pressure is the potential pressure exerted in all directions. Static pressure is measured perpendicular to the direction of air flow.

1391. What is velocity pressure?

1) The kinetic pressure exerted in the direction of flw
2) The potential pressure exerted in all directions
3) The pressure exerted by a vapor
4) The pressure due to weight of the atmosphere

**Ans.1** Velocity pressure is always exerted in the direction of air flow.

1392. If the velocity pressure in a duct is 2 in. of water, what is the air velocity (at STP)?

1) 2,215 ft/min
2) 4,229 ft/min
3) 5,281 ft/min
4) 5,664 ft/min

**Ans.4** $V = 4{,}005 \sqrt{VP}$

$= 4{,}005 \sqrt{2}$

= 5,664 ft/min

1393. If the velocity of air in a duct is known to be 1,250 ft/min, what is the velocity pressure (assume STP)?

1) 0.097 in. of water
2) 0.31 in. of water
3) 0.097 mm Hg
4) 0.31 mm Hg

**Ans.1** $V = 4{,}005\sqrt{VP}$

$1{,}250 = 4{,}005\sqrt{VP}$

$1{,}250/4{,}005 = \sqrt{VP}$

VP = 0.097 inches of water

1394. When a fan is used to draw air through a duct, which of the following is true?

1) The total pressure is always equal to zero
2) The static pressure is always negative
3) The static pressure is always positive
4) The velocity pressure is always negative

**Ans.2** When a fan is used to draw air through a duct, the static pressure is always negative.

1395. If a hood's static pressure is measured at -1.25 in. of water and the coefficient of entry is know to be 0.82, what is the air velocity immediately downstream from the hood?

1) 1,187 ft/min
2) 2,889 ft/min
3) 3,261 ft/min
4) 3,672 ft/min

**Ans.4** $V = 4{,}005 * Ce\sqrt{SPh}$

$= 4{,}005 * 0.82\sqrt{1.25}$

= 3,672 ft/min

1396. Pressure drop occurs in a ventilation system due to all of the following EXCEPT:

1) Friction losses within the system
2) The number and type of bends in the system
3) Turbulence

4) Chemical composition of contaminant

**Ans.4** Pressure drop in a system can occur for a number of reason including friction losses, the number and types of bends, and turbulence. The answer least associated with pressure drop is chemical composition of contaminant.

1397. Which type of hood would you recommend be designed for operations which release sudden surges of hot gases and vapors?

1) Down draft
2) Enclosed
3) Canopy
4) Slotted

**Ans.3** Canopy hoods are effective for the control of hot processes and for those operations which release sudden surges of hot gases and vapors such as those where the thermal currents caused by rising hot air funnel the contaminant into the exhaust system.

1398. How is pressure normally stated when taken in exhaust ducts?

1) mm Hg
2) in. of water
3) psi
4) in. of Hg

**Ans.2** Normally pressure measurements are reported as inches of water.

1399. In general, which of the following operations would require the greatest capture velocity to achieve effective exposure control?

1) Evaporation from open surface tanks
2) Spray booths
3) Welding

4) Grinding

**Ans.4** Grinding operations release contaminants at high initial velocities. The other operations release contaminants with low velocities. Higher capture velocities would be required to entrain the releases from the grinding operations.

1400. Which respirator is an air-purifying type?

1) Abrasive blasting helmet
2) Particulate filter
3) Gas mask
4) Both 2 and 3

**Ans.4** Both the gas mask and the particulate filter respirators are air-purifying.

1401. Which glove would offer the best protection against halogenated hydrocarbons?

1) PVC
2) Neoprene
3) Viton
4) Natural rubber

**Ans.3** Out of all of the gloves tested, Viton is the one that offers protection against halogented hydrocarbons.

1402. A worker is using trichloroethylene as a degreaser in a tank. On occasion her hands may come in contact with the solvent. Which type of glove would you recommend?

1) Nitrile
2) Butyl

3) Poly-vinyl acetate
4) Viton

**Ans.4** Viton is the type of glove material that offers protection against chlorinated solvents

1403. In a machine shop, protective eye wear should be worn by:

1) Equipment operators
2) Shop supervisors
3) Casuals walking through the shop
4) All of the above

**Ans.4** If there is a flying object hazard, all who may be exposed should wear eye protection.

1404. Safety spectacles would be least effective in protecting an employee's eyes in which of the following situations?

1) Chipping rock
2) Pouring molten metals
3) Working in a high concentration of gaseous ammonia
4) Sharpening tools at a stand grinder

**Ans.3** Ammonia ($NH_3$) is a gas; spectacles would not provide protection.

1405. Which respirator has an assigned protection factor of 10,000?

1) SCBA operating in pressure demand mode
2) SAR in pressure demand mode with auxiliary SCBA in positive mode
3) SCBA operating in demand mode
4) Both 1 and 2

**Ans.4** The protection factors were determined by conducting quantitative fit testing on human volunteers. The higher the factor, the more protection. Therefore, a SAR with an escape SCBA offers the same protection as an SCBA. Both must be in the supply mode. The demand mode may introduce contaminants into the face during strenuous circumstances.

1406. The static magnetic field TLV for an 8-hour TWA whole body dose should not exceed:

1) 600 Gauss
2) 800 Gauss
3) 1,000 Gauss
4) 1,200 Gauss

**Ans.1** The TLV for static magnetic fields is 600 Gauss or 60 milliTeslas (mT) for a whole body dose. Extremity doses can be 6,000 Gauss (600 mT). Static magnetic field can pose a severe danger to workers with pacemakers, and their exposure should not exceed 5 Gauss (0.5 mT).

ACGIH TLVs and BEIs

1407. What minimum level of airflow is required for supplied air respirators?

1) 1 cfm
2) 2 cfm
3) 3 cfm
4) 4 cfm

**Ans.4** Supplied air respirators must have a minimum airflow of 4 cfm to the face piece.

1408. The vapor pressure of a solvent in air is $7 \times 10^{-3}$ mm Hg. What is the concentration in ppm?

1) 920 ppm
2) 92 ppm
3) 9.2 ppm
4) 0.092 ppm

**Ans.3** Cppm = (Pvapor / Ptotal) x $10^6$ ppm
= (7 x $10^{-3}$ mm Hg / 760 mm Hg) x $10^6$ ppm
= 9.2 ppm

1409. Biological exposure indices data is important because they include type(s) of exposure in addition to inhalation. Which of the following exposures are least likely to be assessed by a BEI?

1) Skin absorption
2) Injection
3) Ingestion
4) Transplacental

**Ans.4** BEI's measure certain biological functions to determine the "total" exposure from all routes. BEI's do not measure transplacental exposures. They may provide some useful information however on exposure of the fetus to workplace chemicals.

1410. A device that protects equipment from an overload by opening the circuit automatically is commonly called a:

1) Circuit breaker
2) Fuse
3) Ground fault circuit interrupter (GFCI)
4) Both 1 and 2

**Ans.4** Circuit breakers and fuses are overcurrent devices that open a circuit automatically when there is excessive current flow from an accidental ground, short circuit, or overload. The circuit breaker trips open when its mechanical mechanism is activated; it can be reset. The fuse contains a strip of wire or fusible metal link which melts when overloaded. GFCI devices do not protect equipment, per se. They are designed to protect people by opening a circuit rapidly upon detecting a fault.

1411. In choosing a noise control barrier, which represents the best choice of design?

1) Baffles
2) Partial enclosure
3) Air-tight enclosure
4) Non-reflective surfaces

**Ans.3** Air tight enclosures yield the best noise protection. Small leaks can cause significant noise leakage. It is best to design enclosures to keep leaks to an absolute minimum.

1412. In the design of a noise reducing enclosure, all of the following characteristics are desired EXCEPT:

1) Acoustical-absorbing material inside enclosure
2) Acoustical-absorbing material outside enclosure
3) Enclosure made of solid non-porous material
4) Air-tight enclosure

**Ans.2** Acoustical-absorbing material on the outside of an enclosure is much less effective than having the sound-absorbing material on the inside.

1413. When a respirator has a HEPA filter, the filter is 99.97 % effective at capturing monodisperse particle _____ microns or larger in diameter.

1) 1.0
2) 0.5
3) 0.3
4) 0.1

**Ans.3** HEPA filters are 99.97% efficient at collecting monodisperse particles 0.3 microns or larger.

1414. When a respirator is worn by more than one person it must go through an additional step of cleaning. This step is called:

1) Valve replacement
2) Disinfection
3) Quantitative fit testing
4) Qualitative fit testing

**Ans.2** Any time a respirator is worn by more than one person it must be disinfected. If possible, it is preferable to have each person have their own respirator. This is difficult, however, in the case of emergency response respirators such as SCBA's. The employer's Respiratory Protection Program must stipulate in writing how the disinfection process will occur.

1415. Which of the following items does not affect the fit of a respirator?

1) Strap tension
2) Sideburns
3) Dentures
4) Exhalation valve cover

**Ans.4** Dentures, sideburns, and the adjustment straps all effect how well a respirator fits. The exhalation valve, although very important to respirator function, does not effect the fit.

1416. Which of the following respirators has the highest protection factor?

1) Air purifying
2) Pressure demand full face piece
3) Demand full face piece
4) Continuous flow full face piece

**Ans.2** The pressure demand unit with full face piece has a protection factor of 10,000. The other listed devices all have lower protection factors.

1417. There are no established safety specifications for:

1) Bump caps
2) Safety glasses
3) Respirators
4) Boilers

**Ans.1** Bump caps were originally developed by the airline industry for work inside fuselages. They have no established specifications.

1418. The two most common techniques employed for the control of bioagents are:

1) Ventilation and housekeeping
2) Housekeeping and disinfection
3) Disinfection and sterilization
4) Sterilization and housekeeping

**Ans.3** Disinfection and sterilization are the two most common techniques used to control bioagents. It is important to note disinfection and sterilization are not synonymous operations. To disinfect means to get rid of some pathogenic microorganisms that cause infection, but does not imply complete destruction of all that are harmful, whereas, sterilization means the absolute destruction of all living organisms.

1419. Early hearing loss caused by noise:

1) Seriously affects speech reception
2) Makes a person's speech sound distorted
3) Affects telephone usage
4) Is not generally noticeable to the subject

**Ans.4** Most people do not notice the hearing loss until there is significant detriment to their hearing.

1420. What type of guarding should be installed at the approach zone to a fixed robotized workstation?

1) Chain link fence
2) Pressure sensitive mats
3) Fixed barrier with interlocked gate
4) Kill switch

**Ans.3** A robotized work station is divided into a robot movement zone (work envelope) and the zone of approach to it. The boundaries of the zone can be defined so that the limits within which protection is provided are known. Fixed barriers with interlocked gates permit workzone entry into an area, free from robot movement, which are monitored by the interlock. More detail on safety requirements for industrial robots can be found in ANSI/RIA R 15.06 - 1986 and NIOSH publication No. 88-108.

1421. Major types of machine guards include any of the following EXCEPT:

1) Push-button
2) Interlocked
3) Automatic
4) Fixed

**Ans.1** Machine guards include fixed, interlocked, and automatic.

1422. An uncolored sprinkler head is suitable for use in which temperature range?

1) 135ºF - 170ºF
2) 175ºF - 225ºF
3) 250ºF - 300ºF
4) 325ºF - 375ºF

**Ans.1** The NFPA has a color code for sprinkler heads to indicate temperature rating.

Uncolored: 135ºF - 170ºF
White: 175ºF - 225ºF

Blue: 250ºF - 300ºF
Red: 325ºF - 375ºF
Green: 400ºF - 475ºF
Orange: 500ºF - 650ºF
NFPA 13 2-2.3.2 - Standard for the Installation of Sprinkler Systems

1423. When walking through a chemical mixing area, you noticed that the automatic sprinkler heads faced the roof and were color-coded blue. What should this tell you?

1) It was an intermediate temperature pendent sprinkler head with an operating temperature of 175ºF- 225ºF
2) It was a high temperature upright sprinkler head with an operating temperature of 250ºF - 300ºF
3) It was an extra high temperature upright sprinkler head with an operating temperature of 325ºF - 375ºF
4) It was a high temperature pendent sprinkler head with an operating temperature of 250ºF - 300ºF

**Ans.2** Sprinkler heads facing up are called "upright". When color coded blue, it signifies a high temperature rating with a 250ºF - 300ºF operating temperature. NFPA 13 2-2.3.2 - Standard for the Installation of Sprinkler Systems

1424. The warning symbol for radiofrequency (RF) radiation is a:

1) Circle with a red background and black letters
2) Diamond with a red and black background with aluminum letters
3) Diamond with a yellow background and white letters
4) Yellow circle with a microwave antenna

**Ans.2** A sign with a diamond with a red and black background and aluminum letters denotes RF hazards.
29 CFR 1910.97 (a)(3)